Hierarchical Structural Realism and Universality Classes of Embedded Observers

Author: Alex Maybaum
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Abstract

This paper articulates the structural commitments of the Incompleteness of Observation (OI) framework beyond its specific cubic-lattice substratum, and develops the framework's relationship to broader unification programs in fundamental physics. The framework's structural realism operates on a two-dimensional hierarchy: an observation axis (Level A: observation occurs; Level B: observer-admission; Level C: universality classes; Level D: OI's specific representative) and an orthogonal gauge axis (Level G1: D-gauge; Level G2: SM gauge group; Level G3: substratum gauge group $\mathcal{G}_{\text{sub}}$; Level G4: universality-class equivalence). Different empirical predictions live at different intersections, with falsifiability stratified accordingly. *Part I* addresses the technical question of whether SM-reproducing string compactifications are gauge-equivalent representatives of OI's substratum equivalence class — a reading that would dissolve the landscape multiplicity problem. Through detailed analysis of finite-dimensional matrix-model formulations (BFSS, IKKT) we establish that OI's structural conditions A2 (determinism) and A5 (linearity) fail in ways that reflect structural commitments of string theory rather than artifacts of discretization. The substratum-level equivalence reading does not hold. *Part II* develops a broader structural framework — universality classes of embedded observers — within which OI and string theory can be precisely related. We articulate the algebraic formalization of "embedded observer" via the algebra-channel pair $(\mathcal{A}_V, \Phi)$, define universality classes by partial-trace observational equivalence, and establish that OI's $\mathcal{G}_{\text{sub}}$ is a sub-equivalence of universality-class equivalence. *Part III* applies a four-feature audit (Born rule, channel-level unitarity, P-indivisibility, commutant gauge-invariance) to characterize OI-string universality-class membership: three features transport directly between frameworks, with holographic effective field theory providing a direct analog of OI's P-indivisibility, and AdS/CFT bulk reconstruction providing a direct analog of OI's Stinespring lift. The fourth feature transports as a pattern but not as a specific output. The paper also formalizes an *observer-admission* distinction: substrata admit embedded observers when their bijection structure satisfies C1–C4, with most substrata being observer-lacking. Under this framing, the string landscape's 10^{500} vacua decompose structurally — most compactifications produce observer-lacking substrata,

with the SM-reproducing subset being structurally smaller. The OI-string relationship is reframed from "competing programs" to complementary roles within a universality-class structure.

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1. Introduction

1.1 Scope and structure

The framework developed in [Main, SM, GR, Substratum] establishes — under the reconstruction theorem's stated conditions — a uniquely fixed **local lattice/gauge residue** modulo the substratum gauge group $\mathcal{G}_{\text{sub}}$ within OI's specific structural class of finite deterministic bijection systems. H-Bell is separately required for a Bell-inclusive completion to coexist with that local sector, and the Bell-inclusive completion is not proved unique. This paper addresses two layers of structural commitments that lie beyond this core technical content.

The first layer (§2) articulates the *hierarchical structure* of the framework's structural realism. The framework operates on a two-dimensional hierarchy with an observation axis (foundational axiom → observer-admission → universality classes → OI's specific representative) and an orthogonal gauge axis (D-gauge → SM gauge → substratum gauge → universality-class equivalence). Different empirical predictions live at different intersections of these axes; falsifiability is stratified accordingly.

The second layer (Parts I, II, III) develops the framework's relationship to broader unification programs through two related but distinct questions:

Question 1: Substratum-level equivalence. Are SM-reproducing string compactifications gauge-equivalent representatives of OI's substratum equivalence class? An affirmative answer would dissolve the landscape multiplicity problem. Part I (§§3–7) develops this through technical analysis of matrix-model formulations, with negative outcome — OI's structural conditions A2 (determinism) and A5 (linearity) fail.

Question 2: Cross-framework structural realism. If substratum-level equivalence does not hold, is there a broader structural framework within which OI and string theory can be precisely related? Part II (§§8–13) develops this through formalization of universality classes of embedded observers — equivalence classes of substratum-with-observer systems under partial-trace observational equivalence, broader than $\mathcal{G}_{\text{sub}}$.

Part III (§14) synthesizes both questions through a four-feature audit (Born rule, channel-level unitarity, P-indivisibility, commutant gauge-invariance) characterizing the OI-string universality-class relationship.

The two parts have different epistemic statuses. Part I produces a *definitive structural result* (negative — matrix-model formulations fail OI's A2 and A5). Part II is *programmatic* — it articulates the universality-class proposal precisely,

formalizes it via definitions and propositions in §9, and identifies open technical questions whose resolution would establish or refute the framing. Definitive identification of specific universality classes — including the OI-string cross-class — requires substantial follow-up work along the lines flagged in §§11.2 and §13.1.

1.2 The string landscape as a multiplicity problem

For four decades, string theory has aspired to derive the Standard Model from compactification of higher-dimensional spacetime. The state of the art is summarized by Marchesano, Shiu, and Weigand [MSW2024]: many compactifications produce gauge groups that include $SU(3) \times SU(2) \times U(1)$ as a subgroup; some produce three generations of chiral matter; a smaller subset produces the observed hypercharge structure; a still smaller subset produces approximately the observed Yukawa pattern. No compactification has been demonstrated to reproduce the Standard Model uniquely with all parameters matching observation, and no general scheme is known to reproduce the flavor hierarchy [MSW2024, §2.3].

The conventional response to this state of affairs has been the landscape picture: string theory has approximately 10^{500} metastable vacua, each corresponding to a different low-energy effective theory, and our universe corresponds to whichever vacuum produces the SM with the observed parameters. The vacuum-selection problem — why our universe is in this particular vacuum — has been variously addressed by the anthropic principle, by eternal inflation populating all vacua, or by sharper conjectures (the swampland program) attempting to delineate which vacua are consistent with quantum gravity [Vafa2005, Palti2019, OoguriVafa2007].

The swampland program in particular has produced significant progress in constraining the landscape from above [Lehnert2025, MSW2024]. Conjectures such as the Weak Gravity Conjecture, the de Sitter Conjecture, the AdS Distance Conjecture, and the Cobordism Conjecture aim to identify which low-energy effective theories admit consistent UV completion in quantum gravity. These conjectures shrink the apparent multiplicity of vacua but do not, in their current form, single out a unique SM-reproducing vacuum.

1.3 The substratum picture

In a separate program [Main, SM, GR, Substratum], the Incompleteness of Observation (OI) framework approaches fundamental physics from a different direction. The starting point is the observation that quantum mechanics is the marginal-statistics description used by an embedded observer with finite distinguishable states; the trace-out over the hidden sector generates the reduced observable law — universally admitting the quantum representation — from a deterministic classical substratum. The framework develops two technical results:

Reconstruction theorem (Theorem 23, [Substratum §3]). Given E1–E7 together with A1–A6, M1-T, the hidden-sector mixing hypothesis for the C2 necessity direction, and Lemma 24.1's semigroup-transfer/completeness step, the **local propagating lattice/gauge sector** is fixed up to the stated substratum-gauge freedoms: $d = 3$, $K = 6$, the $(3, 2, 1)$ decomposition, $SU(3)\times SU(2)\times U(1)$ and the associated local representation data, with the qualifications stated in [SM], and $\hbar = c^3\epsilon^2/(4G\sigma)$, σ the entropy per boundary mode [GR §3.1]. M1-B is a separate Bell input. A single measurement-independent deterministic completion containing both that local sector and the Bell-violating composite exists only conditional on H-Bell, now the curvature-stability requirement for preparation-indexed state-dependent adjacency. Neither H-Bell nor Lemma 24.1 establishes uniqueness of the Bell-inclusive full completion.

Substratum gauge group (Theorem 24, [Substratum §4]). The equivalence class of substrata producing the same emergent physics is parametrized by a four-generator gauge group:

(i) *State relabeling.* Permutations of hidden-sector configurations.
(ii) *Alphabet change.* The local state-space size q in $\mathbb{Z}/q\mathbb{Z}$ is gauge under the alphabet-size hypothesis ([SM §2.7]).
(iii) *Deep-sector enlargement/reduction.* The hidden sector beyond the boundary layer admits arbitrary cardinality.
(iv) *Graph isomorphism (up to statistical isotropy).* Coupling graphs sharing dimension, spectral dimension, and statistical isotropy are gauge-equivalent.

The framework's structural-realism stance is that the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ — not any particular substratum representative — is what physics determines. Twenty-two quantitative SM predictions, including Cabibbo angle $1/(\pi\sqrt{2})$ matching observation at 0.05σ, Koide ratio $2/3$ matching at 0.02%, and the Bekenstein-Hawking $1/4$ coefficient (conditional on the horizon conditions of [GR §2, §8.5]; GW250114 confirms the classical area theorem, the coefficient untested directly), follow as consequences of the equivalence class under the conditions named at each step.

1.4 Question 1: the equivalence-class proposal

The two pictures describe the same observational target — SM physics — from very different directions. String theory begins from quantum gravity in $d = 10$ or $d = 11$ and seeks to reach the SM through compactification. OI begins from the embedded-observer ontology and reaches the SM through reconstruction. Question 1 asks whether the two descriptions are independent, complementary, or equivalent.

The core proposal investigated in Part I is:

Proposal (equivalence-class reading). Any string compactification that reproduces the observed Standard Model — gauge group $SU(3) \times SU(2) \times U(1)$, three chiral generations, anomaly-free hypercharges, the observed cosmologi-

cal constant — is a gauge-equivalent representative of OI's equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ under the four-generator structure of Theorem 24. Compactification moduli, Calabi-Yau topology, and flux choices play the role of hidden-sector relabeling (generator (i)) and deep-sector enlargement (generator (iii)).

This proposal has three substantive consequences if defensible:

(A) *Dissolution of the multiplicity problem.* The $\sim 10^{500}$ vacua are a redundancy of representation, not a multiplicity of physically distinct theories. Different compactifications producing the same observable physics are gauge-equivalent under $\mathcal{G}_{\text{sub}}$. The swampland program's task is reframed: rather than identifying which vacuum is "ours," the task is to verify that the SM-reproducing subset of the landscape forms a single equivalence class.

(B) *Cross-attribution of predictions.* OI's quantitative SM predictions become predictions of any SM-reproducing string compactification, since all such compactifications are gauge-equivalent. Conversely, string-theoretic computations within SM-reproducing compactifications become tests of OI's structural derivations.

(C) *Computational gateway.* OI's cubic-lattice substratum admits direct lattice-Monte-Carlo and lattice-perturbation-theory computation. String compactifications are mathematically rich but computationally demanding. Gauge-equivalence under the proposal would let OI's lattice serve as a computational tool for string-theoretic questions, with results lifted to string-theory predictions through the equivalence.

Pre-conditions for the proposal. The proposal is suggestive but requires three pre-conditions:

Pre-condition 1: Generator (ii) extension to continuum. OI's generator (ii) operates on the alphabet $\mathbb{Z}/q\mathbb{Z}$, a discrete structure. String theory uses continuous degrees of freedom — string positions on Calabi-Yau, complex-structure and Kähler moduli, RR flux quanta. The technical question is whether OI's alphabet-size hypothesis [SM §2.7] extends to continuum theories, or whether OI's discrete alphabet is essential to the reconstruction. If discrete nature is essential, the bridge between OI and string theory passes through finite-dimensional matrix-model formulations of strings (BFSS, IKKT), not the continuum string description directly.

Pre-condition 2: Existence of SM-reproducing compactifications. Despite forty years of effort, no string compactification has been demonstrated to uniquely reproduce the SM with all parameters matching observation [MSW2024]. The "SM-reproducing subset" of the landscape is conjectured to exist but has not been explicitly exhibited.

Pre-condition 3: A1-A6 compatibility under matrix-model bridge. Even with matrix-model finite-dimensional formulations as intermediaries, OI's structural conditions A1 (finiteness), A2 (deterministic bijection), A3 (bounded coupling),

A5 (linearity) may be violated by string-theoretic structure. Whether matrix models satisfy A1-A6 is not obvious and requires technical verification.

§§3-6 of Part I address these pre-conditions; §7 considers the consequences both if they hold and if they fail.

1.5 Question 2: why universality classes are substantively new

The negative answer to Question 1 (Part I) does not exhaust the structural relationship between OI and string theory. OI's $\mathcal{G}_{\text{sub}}$ classifies observationally-equivalent substrata *within OI's structural class* — substrata satisfying A1-A6. The reconstruction theorem establishes uniqueness of the local reconstructed residue within this class, not uniqueness of the Bell-inclusive completion and not uniqueness across structural classes.

Question 2 asks whether observational features alone — without committing to specific structural conditions — generate equivalence classes across all candidate substratum descriptions. If substantive *universality classes* exist, they would represent a stronger structural-realism claim than OI's $\mathcal{G}_{\text{sub}}$: not just "substrata are gauge-equivalent within OI's class" but "embedded-observer systems are gauge-equivalent across structural classes."

Part I's findings make this question concrete. The structural features of partial-trace observation (Born rule, channel-level unitarity, P-indivisibility, commutant gauge-invariance pattern — established in §14's four-feature audit) transport across the OI / string-theory boundary even when substratum-level structure does not. This provides a candidate universality class: the class characterized by these four features. The question is whether this class is well-defined, what its members are, and what additional features distinguish sub-classes.

Part II develops the formal framework for this question. §8 articulates three candidate formalizations of "embedded observer" (categorical, operational, algebraic), with the algebraic formalization adopted as principal. §9 develops the algebraic formalization, stating the key definitions (embedded observer, substratum-with-observer system, observational equivalence, universality class), proving propositions on the relation to OI's gauge group (Proposition 9.5) and the Stinespring uniqueness structure (Propositions 9.7a and 9.7b), and developing the observer-admission distinction (Definition 9.9 + Proposition 9.10 + Corollary 9.11) that operates at a level prior to universality-class membership. §10 uses OI's framework as a worked example. §11 examines the OI-string universality class. §12 develops implications for structural realism. §13 identifies open questions and limits, with §§13.1.1-13.1.3 developing three technical questions (algebra-channel *-isomorphism, prediction transport, partial-trace features sufficiency) in detail.

The two questions are not independent. Question 1's negative answer motivates Question 2's broader framework: if substrata cannot be gauge-equivalent at the substratum level, the next question is what equivalence does hold. Question

2's universality-class framework then provides the language for Part III's cross-framework synthesis.

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2. The framework's hierarchical structure

The framework's structural-realism stance operates on a two-dimensional hierarchy. An *observation axis* runs vertically — from the framework's foundational axiom to its specific empirical content. A *gauge axis* runs horizontally — from narrow gauge equivalences within the emergent QM to broad equivalences across structural classes. The two axes are orthogonal: depth on the observation axis is about scope (which substrata are within the framework's domain); breadth on the gauge axis is about transformations (which transformations are observable-preserving).

This section articulates the two axes, their orthogonality, and the multi-level structural realism that follows. The articulation is canonical for the framework — it locates where each technical result lives and stratifies the framework's predictions by hierarchy intersection. Subsequent sections of this paper develop content at specific levels (Part I at Level D × Level G3; Part II at Level C × Level G4; Part III at the partial-trace observational features sub-class within Level C).

2.1 The observation hierarchy

The observation hierarchy nests by inclusion: each deeper level is a more specific commitment.

Level A — The observation axiom. The framework begins with the fact that *observation occurs* [Main §1]. An observer records distinguishable outcomes of interactions with a system not wholly under the observer's control. Level A commits to: there is something to observe; some part of reality is structured to record outcomes (observers are embedded in reality, not external to it); observation has structural consequences. This level is shared with mainstream observer-essentiality programs since 2022 (CLPW, Maldacena, HUZ, DEHK, Slagle-Preskill); the framework's distinguishing feature is producing the consequences from a finite deterministic substratum without extra dimensions or stochasticity.

Level B — Observer admission. Not every substratum admits observation. The framework identifies four conditions on a substratum's bijection φ that are necessary for an embedded observer to exist [Main §3]:

(C1) **Coupling.** The interaction between the observer's degrees of freedom and the rest must be substantial.

(C2) **Record persistence.** The written record must survive to readback — in our universe realized by a slow bath, the hidden-sector correlation time much

longer than the observer's measurement timescale (one sufficient route).

(C3) **Sufficient capacity.** The hidden sector must have information capacity enough to store the interaction history.

(C4) **History readback.** The interaction must read back what was written — the load-bearing memory condition.

A substratum (S, φ) is *observer-admitting* if there exists a partition $V \subset S$ such that (S, φ, V) satisfies C1–C4, and *observer-lacking* otherwise (formal Definition 9.9 in §9.6). Level B commits to: the space of substrata is structurally partitioned into observer-admitting and observer-lacking subsets; most substrata are observer-lacking (random permutations on finite sets typically lack the slow-mixing, sufficient-capacity, coupled-subsystem structure required); the framework's content applies only to observer-admitting substrata.

This has structural-anthropic content. Standard anthropic reasoning treats parameters as contingent and explained by selection: we observe a universe with parameters compatible with us because we couldn't observe a universe without them. The observer-admission reading sharpens this: observer-lacking substrata are not candidates for "our universe" at all — they are mathematical objects of a different kind, without observer-extraction structure. Within observer-admitting substrata, anthropic reasoning operates normally.

Level C — Universality classes. Within observer-admitting substrata, equivalence classes are determined by partial-trace observational features. Two substratum-with-observer systems are *observationally equivalent* if their algebra-channel pairs $(\mathcal{A}_V, \Phi)$ are *-isomorphic with channel intertwining (Definition 9.3 in §9.1). A* universality class* is an equivalence class under observational equivalence (Definition 9.4).

Universality classes have internal sub-structure (§9.5). Three levels of refinement:

- *Partial-trace features sub-class:* characterized by structural features any partial-trace operation must produce (Born rule, unitarity, non-Markovian marginal, commutant restriction pattern). Most universal level.
- *Gauge-group sub-class:* characterized by which gauge group emerges. SM-reproducing classes share $SU(3) \times SU(2) \times U(1)$.
- *Algebra-channel sub-class:* characterized by specific algebra of observables and specific channel dynamics.

Level C commits to: structural realism at this level is class-level (the universality class is real, specific representatives are gauge); the class is broader than OI's $\mathcal{G}_{\text{sub}}$-equivalence (which operates within OI's specific structural class); multi-level structural realism is forced (different content at different sub-class refinements).

Level D — OI's specific representative. OI's framework selects a specific universality class (and a specific substratum representative within that class)

by adding structural conditions A1–A6 [Substratum §3]:

(A1) Finiteness; (A2) Determinism (bijection); (A3) Bounded coupling degree; (A4) Center independence; (A5) Linearity; (A6) Background independence.

Combined with the empirical inputs and modeling assumptions stated in Theorem 23, these determine — conditional on C2-mixing and Lemma 24.1's completeness — the **local lattice/gauge residue** up to $\mathcal{G}_{\text{sub}}$. H-Bell separately conditions Bell-inclusive existence, whose uniqueness is open (Theorem 23 in [Substratum]).

A1–A6 are not the same as observer-admission C1–C4; they operate at a different hierarchy level. C1–C4 are conditions for *any* embedded observer to exist (Level B); A1–A6 are conditions for membership in *OI's specific* universality class (Level D). Some relations: A1 (finiteness) is related to but distinct from C3 (capacity); A2 (determinism) is a Level D commitment that selects deterministic-substratum members of a universality class — observers can in principle exist on stochastic dynamics. Other A's (A3–A6) are pure class-specific: required for OI's specific gauge-structure derivation, not necessary for observer-admission.

Level D commits to OI's specific structural realization: $d = 3$ cubic lattice with $K = 6$ internal components, linear wave equation, $SU(3) \times SU(2) \times U(1)$ from cubic-group commutant, specific empirical predictions (Cabibbo $1/(\pi\sqrt{2})$, Koide $2/3$, Bekenstein-Hawking $A/4$ — twenty-two structural retrodictions in [SM §7] and [GR §7]). The framework's strongest realism claim — ontic structural realism applied to $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ — operates at Level D.

The hierarchy nests: Level A $\supset$ Level B $\supset$ Level C $\supset$ Level D. Each deeper level is a more specific commitment.

2.2 The gauge hierarchy

The gauge hierarchy is the breadth axis: which transformations are observable-preserving, with broader levels absorbing more degrees of freedom as gauge.

Level G1 — D-gauge on the emergent Hamiltonian. The diagonal-unitary equivalence on the emergent Hamiltonian: $H \to DHD^\dagger$ with D diagonal-unitary in the eigenbasis of measurement outcomes [GR §3.3]. This is an exactly invisible freedom of the transition data. Energy shifts and the antiunitary reflection are also invisible; their completeness under the stated gap and overlap hypotheses is conditional on H-BG [GR §3.3]. Two Hamiltonians H and $DHD^\dagger$ produce identical $T_{ij}(t) = |\langle j|e^{-iHt/\hbar}|i\rangle|^2$ in the measurement basis. G1 is the smallest gauge group; it operates within a single Hamiltonian / single QM realization.

Level G2 — SM gauge group on the emergent QFT. The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges as the commutant of the coupling matrix M on the emergent QFT [SM §4.4-§4.6]. Cubic-group decomposition of the $K = 6$ internal components produces multiplicities $(3, 2, 1)$; the commu-

tant of the cubic-group action on the $K = 6$ space is $SU(3) \times SU(2) \times U(1)$. Background independence (A6) promotes this global commutant symmetry to local gauge invariance. G2 includes G1 as a sub-symmetry. Level G2 operates within OI's specific universality class — the *uniqueness* of $SU(3) \times SU(2) \times U(1)$ depends on A3-A6 + E4. Other universality classes producing the SM at low energies would reach the same gauge group through different machinery (§14.4 on string-theoretic compactifications).

Level G3 — Substratum gauge group $\mathcal{G}_{\text{sub}}$. The group of transformations on (S, φ) preserving all observables — identical transition probabilities, identical emergent Hamiltonian (up to G1), identical $\hbar$ across all partitions of OI's structural class [Substratum §4 Theorem 24]. Four explicit families of generators: (i) hidden-sector relabeling, (ii) alphabet change (under the [SM §2.7] alphabet-size hypothesis), (iii) deep-sector enlargement, (iv) graph statistical-class equivalence. Stinespring uniqueness [Main §3.2], extended per the semigroup-transfer step ([Substratum] Lemma 24.1), conditionally establishes that these four exhaust the freedom. G3 includes G1 and G2: the SM gauge group is the visible-sector shadow of $\mathcal{G}_{\text{sub}}$ under the trace-out [Substratum §4]. G3 is an *equivalence-class symmetry*, not a Lagrangian symmetry — no Lagrangian exists at the substratum level. It plays the same structural role at the substratum level that Lagrangian gauge plays at the field-theory level.

Level G4 — Universality-class equivalence. The equivalence relation on substratum-with-observer systems by partial-trace observational features (Definition 9.4 in §9.1). Two systems are universality-class equivalent if their algebra-channel pairs are *-isomorphic with channel intertwining. G4 is broader than $\mathcal{G}_{\text{sub}}$: substrata related by $\mathcal{G}_{\text{sub}}$ are universality-class equivalent (since they produce the same algebra-channel pair), but substrata not related by $\mathcal{G}_{\text{sub}}$ may still be universality-class equivalent if they produce the same algebra-channel pair through structurally different machinery. G4 operates within the observer-admitting subset (Level B) — observer-lacking substrata are outside the universality-class structure entirely.

The hierarchy nests: G1 $\subset$ G2 $\subset$ G3 $\subset$ G4. Each broader level absorbs more degrees of freedom. The trace-out projects higher levels onto lower: trace-out of $\mathcal{G}_{\text{sub}}$ produces the SM gauge group as image; restriction to Hamiltonian projects G2 onto G1.

2.3 Orthogonality and multi-level structural realism

The two hierarchies are orthogonal. The observation axis is about scope; the gauge axis is about transformations. Levels G1, G2, G3 sit within Level D (OI's specific universality class with its specific gauge structure); Level G4 operates at Level C (universality-class equivalence, broader than OI's specific class). The orthogonality is what makes the framework's structural-realism stance multi-level.

What is real differs at each level:

- *Level G4 × Level C:* the algebra-channel structure of partial-trace observation — Born rule, channel-level unitarity, non-Markovian marginal, commutant gauge-invariance pattern. Universal across observer-admitting substrata.
- *Level G3 × Level D:* the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ — OI's specific substratum modulo the four-generator gauge group. Class-specific to OI.
- *Level G2 × Level D:* the emergent QFT structure — SM gauge group, matter content, gauge invariances. Forced uniquely in OI; shared with any SM-reproducing universality class.
- *Level G1 × Level D:* the emergent Hamiltonian's transition-probability content — dynamical structure modulo D-gauge.

These are not competing realisms but realism at different levels. The framework commits to all of them simultaneously, with content at each level being more class-specific (lower levels) or more class-universal (higher levels).

2.4 Predictions stratified by hierarchy level

The framework's empirical predictions live at specific intersections:

- *Level G4 × Level C predictions* — Born rule, channel-level unitarity, non-Markovian marginal, commutant restriction pattern. Universal across observer-admitting systems; trivially "predicted" because they are structural features of partial-trace observation. Not falsifiable within the framework.
- *Level G2 × Level D predictions* — SM gauge group, the $1 \oplus 3$ candidate-sector count (three physical generations under H-spin'), anomaly-free hypercharges. Forced uniquely in OI through cubic-group commutant; shared with any SM-reproducing universality class.
- *Level G3 × Level D predictions* — Cabibbo angle $1/(\pi\sqrt{2})$, Koide ratio $2/3$, Bekenstein-Hawking $A/4$, dark-sector phenomenology. Distinguishes OI from other universality-class members; forced by OI's specific $\mathcal{G}_{\text{sub}}$-equivalence class.

The framework's most distinguishing predictions live at Level G3 × Level D — they are class-specific to OI. Level G2 × Level D predictions are shared with any SM-reproducing class. Level G4 × Level C predictions are essentially structural.

Falsification is stratified accordingly: falsifying a Level G4 × Level C prediction is impossible without contradicting QM itself; falsifying a Level G2 × Level D prediction would refute the entire SM-reproducing universality class; falsifying a Level G3 × Level D prediction would refute OI's specific representative without necessarily refuting the broader universality-class framework. the JUNO match to $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ is a Level G3 × Level D result (retrodictive); the $A/4$ coefficient is a Level G2/G3 × Level D derivation with no direct empirical test (GW250114 constrains the classical area theorem, not the coefficient).

2.5 Where each level's content lives

The framework's papers carry content at different levels of the two-dimensional structure:

- *[Main]:* Levels A and B (observation axiom; embedded-observer characterization via C1–C4)
- *[SM]:* Level D × Levels G1–G2 (Standard Model structure on OI's specific cubic lattice)
- *[GR]:* Level D × Levels G1–G2 (gravitational thermodynamics on OI's specific cosmological-horizon structure)
- *[Substratum]:* Level D × Level G3 (reconstruction theorem; substratum gauge group $\mathcal{G}_{\text{sub}}$)
- *This paper:* Levels B and C × Level G4 (universality-class formalization, observer-admission distinction, OI-string relationship at universality-class level, landscape reframing)

Different theorems and structural conditions do the load-bearing work at each level: Level A is not a theorem (structural fact); Level B is C1–C4 ([Main §3]) plus Definition 9.9 + Proposition 9.10 + Corollary 9.11 (§9.6); Level C is Definitions 9.1–9.4 + Propositions 9.5, 9.7a and 9.7b (§9) plus the four-feature audit (§14); Level D is Theorem 23 (reconstruction) and Theorem 24 (substratum gauge group) ([Substratum §§3-4]) plus A1–A6 + E1–E4. On the gauge axis: G1 is the residual diagonal-unitary equivalence ([GR §3.3]); G2 is the cubic-group commutant theorem ([SM Theorem 7]) plus background independence; G3 is the four-generator characterization (Theorem 24) plus Stinespring uniqueness; G4 is Definition 9.4 plus the algebra-channel structure as the invariant.

The framework's empirical content (twenty-two classified quantitative predictions — seven parameter-free structural, the remainder layered, chained, or fitted [SM §7.6] — the specific cubic-lattice substratum, and the specific gauge group derivation) is concentrated at Level D × Level G3. The broader levels are where the framework connects to other unification programs and where universality-class structural realism applies — the content of Parts I, II, and III of this paper.

Part I: Substratum-Level Equivalence (Question 1)

3. Per-generator analysis applied to string compactifications

OI's gauge group $\mathcal{G}_{\text{sub}}$ has four explicit generators: (i) hidden-sector relabeling, (ii) alphabet change (under the alphabet-size hypothesis of [SM §2.7]), (iii) deep-sector enlargement/reduction, (iv) graph statistical-class equivalence. Each has

a natural but inexact analog in string compactifications. *Generator (i)* corresponds to relabeling internal-space coordinates (Calabi-Yau complex-structure parameters, Kähler-class moduli not affecting four-dimensional physics) — discrete in OI, continuous in strings. *Generator (ii)* corresponds to the matrix-rank parameter N in matrix models, or possibly the worldsheet central charge; the detailed analysis (§4) shows no straightforward extension of the alphabet-size hypothesis to string-theoretic continuum structure has been established. *Generator (iii)* corresponds to the choice of compactification (which Calabi-Yau, which flux, which D-brane arrangement) — structurally productive as deep-sector content, but contingent on the underlying structure satisfying A1-A6 (which §6 establishes it does not for matrix models). *Generator (iv)* corresponds to lattice discretizations of compactified manifolds with the same effective dimension and spectral properties — straightforward where it applies, but operates within OI's structural class on graphs sharing OI's specific properties; cross-class application requires shared statistical properties that are not automatic.

The natural identifications are suggestive but the structural incompatibilities documented in §6 prevent direct equivalence-class identification. Per §14.4, the *pattern* of gauge invariance via commutant restriction transports across frameworks, but the specific output (which gauge group, which generators) is forced through framework-specific machinery in each case. This section is preparatory to §6's substratum-level analysis and §14's observational-level analysis; readers who have internalized those sections may treat §3 as supporting documentation.

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4. Generator (ii) and the discrete-to-continuum bridge

This section examines the technical bridge between OI's discrete substratum (alphabet $\mathbb{Z}/q\mathbb{Z}$) and string theory's continuum structures (real-valued matrix entries in BFSS, complex moduli in compactifications, continuous fields on worldsheets). The bridge question is whether OI's alphabet-size hypothesis [SM §2.7] extends to continuum theories, or whether OI's discrete nature is essential to the reconstruction.

4.1 The alphabet-size hypothesis [SM §2.7]

OI's substratum has local state-space $\mathbb{Z}/q\mathbb{Z}$ for some integer $q \geq 2$. That the alphabet size q is a gauge parameter — that no observable depends on the choice of q — is the alphabet-size hypothesis of [SM §2.7], posed for linear dynamics. [SM §2.7] lists the results proved for every q and those that carry q; no proof covers every observable.

This hypothesis bears on OI's claim that A5 (linearity) is forced rather than assumed, conditionally: the forcing runs through *amplitude-scale* gauge (the unobservability of the absolute field scale), which [Substratum §4, Theorem 24, *Status*] classifies as an adjoined operational input rather than a theorem ([SM §2.7]) — its independent warrant by the observer architecture being an open,

referee-grade question. Linearity is therefore forced *given* that operational principle, not unconditionally. Non-linear dynamics over $\mathbb{Z}/q\mathbb{Z}$ generically produce q-dependent observable physics (x^2 mod 3 and x^2 mod 5 have different algebraic structure), violating the gauge equivalence. The hypothesis is stated within the discrete alphabet structure; it does not address whether $q \to \infty$ limit produces continuum dynamics or what role real-valued degrees of freedom play.

4.2 Matrix models as candidate finite-dimensional representatives

String-theoretic matrix models — BFSS [BFSS1996], IKKT [IKKT1996] — are finite-dimensional formulations of M-theory and type IIB string theory respectively. BFSS at finite matrix rank N has $9N^2$ continuous bosonic matrix entries plus fermionic counterparts; IKKT has a similar structure with one fewer dimension. These are finite-dimensional in the sense of bounded matrix rank; they are not finite-cardinality in the sense of OI's $|S| < \infty$ requirement.

Whether matrix models can be candidate representatives of OI's equivalence class through the bridge requires a discrete-to-continuum identification. Two natural candidates: (a) the matrix rank N playing the role of OI's alphabet size q, with $N \to \infty$ corresponding to $q \to \infty$; (b) lattice discretization of matrix-model dynamics, producing a finite-cardinality system that approximates the continuum matrix-model in an appropriate limit.

Candidate (a) is structurally suggestive — both q and N parametrize the size of the basic representation space. The disanalogy is that OI's alphabet-size hypothesis asserts q-independence of observables; the analogous claim for N would be N-independence of matrix-model observables, which is *not* a feature of matrix models. Different N values give different theories (BFSS at $N = 2$ differs from BFSS at $N = 10^9$), and the conjectured M-theory limit is specifically $N \to \infty$ at fixed coupling. The N-dependence of matrix-model dynamics is structural, not gauge.

Candidate (b) is technically tractable through lattice-discretized BFSS [BFSS-Lattice2015]. At fixed lattice spacing and matrix rank, the configuration space is finite-cardinality and OI's A1 holds. The remaining structural conditions A2 (deterministic bijection) and A5 (linearity) still fail per §6's analysis — lattice Monte Carlo is stochastic, and the Yang-Mills commutator-squared interaction remains nonlinear at the lattice level.

4.3 Possible bridges and their status

Three structural possibilities for the discrete-to-continuum bridge:

(a) $q \to \infty$ *limit gives continuous matrices.* Under this reading, OI's substratum at very large q approximates a continuum theory, with continuous matrix entries arising as the limit. This is the most direct continuum extension. The challenge is that the alphabet-size hypothesis is posed for

linear dynamics, which the continuum matrix-model violates. The bridge fails at A5.

(b) *Matrix rank N as analog of q.* Under this reading, the matrix-model parameter N plays the same gauge role as OI's q, with N-independence of observables required. This fails empirically — matrix-model dynamics is genuinely N-dependent — and the conjectured M-theory limit is at fixed $N \to \infty$, not at variable N.

(c) *Discrete-and-continuum genuinely incompatible.* Under this reading, OI's discrete structure is essential to its reconstruction theorem, and string theory's continuum structure is essential to its quantum-gravity content. The two frameworks describe physics through structurally distinct mathematical apparatus, with no natural bridge between them at the substratum level. This is the conclusion supported by §6's analysis.

The provisional conclusion is (c). The substratum-level bridge does not exist in the form the equivalence-class proposal of §1.3 required. The bridge does exist at the partial-trace observation level (per §14), but this is a different kind of bridge — it relates structural features of how observation extracts information, not the underlying substratum descriptions. The discrete-to-continuum question at the substratum level is orthogonal to the partial-trace-features question at the observational level.

This connects directly to the framing of §14.5: structural features of observation transport across frameworks (Born rule, unitarity, non-Markovian marginal); substratum-level structure does not (discrete vs continuum, bijection vs path-integral, linear vs nonlinear). The discrete-to-continuum bridge is part of the substratum-level layer that does not transport.

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5. SM-reproducing compactifications: a catalogue

Three principal classes of string compactifications produce the SM gauge group through structurally different mechanisms. *Heterotic* ($E_8 \times E_8$ or $SO(32)$ on Calabi-Yau threefolds): E_8 broken by CY holonomy and bundle data; three generations from CY topology (Euler character / 2); hypercharge from bundle structure. Specific realizations include the Schoen Calabi-Yau (Bouchard-Donagi 2006) and heterotic line-bundle models (Anderson-Gray-Lukas-Palti 2014). *Type II / D-brane* (IIA or IIB on Calabi-Yau with D-brane configurations): gauge fields from open strings on D-brane stacks producing $U(N)$ factors; chiral matter from D-brane intersections or magnetization; SM gauge group typically as a subgroup of larger Pati-Salam or left-right-symmetric structure. *F-theory* (on elliptically-fibered Calabi-Yau fourfolds): gauge group from singularity structure of the discriminant locus; $SU(5)$ GUT models from A_4 singularities with hypercharge from flux configuration breaking $SU(5)$ to the SM. None of the three classes uniquely determines all SM parameters

[MSW2024].

The pattern shared across the three classes — and shared with OI per §14.4 — is gauge invariance via commutant restriction on a substratum-level symmetry. The specific output (the SM gauge group precisely) is forced by framework-specific machinery in each case: CY holonomy and bundles for heterotic; D-brane configurations for Type II; singularity structure with flux for F-theory. OI's framework reaches the same target through a structurally simpler route: the cubic-group commutant on $K = 6$ internal components, with no compactification choices required. The paper's overall finding (per §6 + §14) places this catalogue in context: these compactifications are framework-specific realizations of an observational target (SM gauge group + three generations + hypercharges), not gauge-equivalent representatives of OI's substratum equivalence class. They share with OI the structural features of partial-trace observation (per §14.1-14.3 and the gauge-invariance pattern of §14.4) but realize the specific gauge content through structurally different machinery.

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6. Structural compatibility: A1–A6 under matrix-model bridges

The equivalence-class reading proposed in §1.3 requires that string compactifications producing the observed Standard Model be gauge-equivalent representatives of $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$. As discussed in §1.4 (Pre-condition 3), the natural bridge between OI's discrete substratum and string theory's continuum description passes through finite-dimensional matrix-model formulations — BFSS [BFSS1996] for type IIA / M-theory, IKKT [IKKT1996] for type IIB, and their close variants. This section examines whether matrix-model formulations satisfy OI's structural conditions A1–A6.

The structural conditions, as established in [Substratum §3.1] following the V1 reformulation [Substratum §3.1], are: A1 (finiteness), A2 (determinism / bijection), A3 (bounded coupling degree), A4 (center independence), A5 (linearity), A6 (background independence). All six are forced by the empirical inputs E1–E4 combined with the framework's gauge structure ([SM §2.7] alphabet-size hypothesis, [Substratum §4] Theorem 24 classical-substratum descent). For matrix models to be candidate representatives of OI's equivalence class, the same conditions must hold for the matrix-model description.

We address each condition in turn, taking the BFSS matrix model as the principal example. The conclusions extend to IKKT and to BFSS variants with appropriate modifications.

6.1 A1 (finiteness)

Condition. The configuration space S is finite.

BFSS status. The BFSS matrix model at finite N has nine bosonic matrices X^i (with $i \in \{1, \ldots, 9\}$) and sixteen fermionic matrices ψ_α, each of size $N \times N$ with continuous (real-valued bosonic; Grassmann-valued fermionic) entries [BFSS1996, see also [HigherPointBFSS2025]]. The configuration space is $\mathbb{R}^{9N^2}$ for the bosonic sector at fixed N, augmented by a Grassmann algebra for fermions. This is not finite.

Possible repair: lattice discretization. Lattice-discretized BFSS [BFSS-Lattice2015] places the matrix model on a discrete time lattice, and additionally discretizes the matrix entries through standard lattice gauge theory techniques. At fixed N and fixed lattice spacing, with appropriate UV cutoff on matrix-entry values, the configuration space becomes finite. The lattice discretization is the standard tool for nonperturbative numerical study of matrix models and is well-developed in the literature.

Status. A1 fails for continuum BFSS. A1 holds for lattice-discretized BFSS at fixed N and lattice spacing, with the proviso that the continuum and large-N limits — which are believed to recover string-theoretic content — produce $|S| \to \infty$ in a manner that may or may not preserve the equivalence-class structure.

6.2 A2 (determinism / bijection)

Condition. The substratum dynamics $\varphi : S \to S$ is a deterministic bijection.

BFSS status. BFSS dynamics is quantum, defined through the path integral over matrix configurations. At the operational level, lattice-discretized BFSS is studied through Monte Carlo simulation, which samples from the path-integral measure stochastically. Neither the continuum quantum dynamics nor the lattice Monte Carlo is a deterministic bijection on the configuration space.

The classical limit of BFSS — at $\hbar \to 0$, equivalent to taking large coupling λ at fixed N — gives Hamiltonian equations of motion on continuous phase space. These are deterministic but not bijective on a finite configuration space (since the configuration space is continuous). One could discretize phase space and time-step the equations of motion as a deterministic update, but this discretization is artificial and does not respect the underlying continuum dynamics.

Implications for the trace-out construction. OI's emergent-QM derivation [Main §3.2] uses the Stinespring dilation to lift visible-sector CPTP channels to unitaries on $\ell^2(S)$, with the descent restriction selecting permutation matrices and hence bijections. If the substratum dynamics is stochastic (path-integral) rather than deterministic, the trace-out gives a CPTP channel with two sources of randomness — substratum-intrinsic (from the path-integral) and observer-ignorance (from the trace-out) — rather than the observer-ignorance-only structure that OI's framework requires. This breaks the framework's identification of substratum bijection with observable QM unitarity.

Status. A2 fails for matrix models in their standard quantum formulation. The

failure is not a discretization artifact but a structural feature: matrix models are quantum theories, and their dynamics is quantum-mechanical (path-integral) rather than classical-deterministic. Repair would require reformulating the matrix model as a classical cellular-automaton dynamics with bijection structure, which is not the standard BFSS conjecture and would represent a substantial re-foundation rather than a discretization.

6.3 A3 (bounded coupling degree)

Condition. Each site is coupled to a bounded number of neighbors in φ's action.

Status. Reframing-dependent. Treating individual matrix entries as sites, BFSS's Yang-Mills interaction $-\frac{1}{4}\mathrm{Tr}[X^i, X^j]^2$ couples each entry to $\sim N^2$ others (unbounded in the large-N limit). Treating each matrix as a single site (9 bosonic sites, 16 fermionic in BFSS), coupling is bounded. The matrix-as-site reframing is in tension with OI's combinatorial substratum (where "site" is a single configuration value); without independent argument that this reframing is correct rather than a definitional convenience, A3 should be regarded as failing on the natural reading.

6.4 A4 (center independence)

Status. Holds. BFSS has manifest invariance under matrix translations $X^i \to X^i + c^i \mathbb{1}$ (the BFSS analog of spatial translation), and matrix indices a, b have no preferred values; the action is symmetric under index permutations. Both features are manifest in the BFSS action.

6.5 A5 (linearity)

Condition. The wave equation governing φ is linear in the configuration variables.

BFSS status. The BFSS action contains the commutator-squared interaction $-\frac{1}{4}\mathrm{Tr}[X^i, X^j]^2$, which is *quartic* in the matrix entries. The equations of motion derived from this action are nonlinear. This is not an accident or simplification: the nonlinear matrix interaction is what produces the gauge structure of M-theory, the emergence of D-brane dynamics, and the noncommutative-geometry interpretation of matrix theory.

OI's [SM §4.1] necessity argument for linearity rests on amplitude-scale gauge (§4.1 above; [Substratum §4]), an adjoined operational input that the alphabet-size hypothesis of [SM §2.7] does not entail. The hypothesis adds a consistency statement: nonlinear dynamics over $\mathbb{Z}/q\mathbb{Z}$ generically have q-dependent emergent physics (x^2 mod 3 vs x^2 mod 5 are algebraically distinct), violating it.

In matrix models, the analog of q is the matrix size N. Different N values give algebraically distinct theories — the BFSS coupling structure depends on N,

and the $N \to \infty$ limit (which is believed to recover the full M-theoretic content) is structurally distinct from finite N. Whether this N-dependence corresponds to OI's q-dependence — or whether N is more analogous to OI's deep-sector cardinality (generator (iii)) — is not obvious.

If N is analogous to q, then the N-dependence of matrix-model dynamics directly violates the OI requirement. If N is analogous to deep-sector cardinality, then N-independence at fixed observable physics is required for N to be gauge under generator (iii); this is the content of the BFSS conjecture (large-N matrix model = M-theory) but is not rigorously established and remains a conjecture.

Status. A5 fails directly: BFSS has nonlinear matrix dynamics, in contrast to OI's linear wave equation. The deeper question — whether the nonlinearity is fundamental or whether some reformulation of BFSS as a linear theory in a different basis is possible — is open but does not have a known answer in the affirmative direction. The most natural reading is that matrix models genuinely have nonlinear substratum dynamics, in structural disagreement with OI's reconstruction.

6.6 A6 (background independence)

Status. Holds in spirit, conditional on the matrix-spatial bridge. BFSS has manifest $U(N)$ gauge invariance under local matrix transformations $X^i \to UX^iU^{-1}$, which is the precursor of the spacetime gauge theory in the long-distance limit. OI's A6 requires covariance under spatially varying internal-index transformations, with the link coupling transformed alongside, but BFSS has no spatial structure beyond the matrix index; the BFSS analog is matrix-internal $U(N)$ gauge invariance. Whether matrix-internal gauge invariance plays the role A6 requires depends on the matrix-spatial bridge, established only in specific limits (e.g., the diagonal-matrix limit where eigenvalues correspond to D0-brane positions).

6.7 Summary of A1–A6 status

The tally across A1–A6:

Condition	BFSS status	Comment
A1 (finiteness)	Fails (continuum) / holds (lattice)	Repairable through lattice discretization
A2 (determinism / bijection)	Fails	Path-integral is stochastic, not bijection; structural mismatch
A3 (bounded coupling)	Fails (entry-as-site) / holds (matrix-as-site)	Reframing-dependent
A4 (center independence)	Holds	Manifest in BFSS

Condition	BFSS status	Comment
A5 (linearity)	Fails	Commutator-squared interaction is nonlinear; structural mismatch
A6 (background independence)	Holds (matrix-gauge) / conditional (spatial)	Depends on matrix-spatial bridge

The outcome is mixed. Three of six conditions (A4, A6 in spirit, A1 with discretization) are compatible with matrix-model structure. Three of six (A2, A3 on natural reading, A5) fail in ways that are not simple discretization issues but reflect structural commitments of matrix-model dynamics that diverge from OI's substratum.

The key failure is A5 (linearity). OI's reconstruction theorem derives, conditionally on H-link, the Standard Model gauge group through cubic-group decomposition of the $K = 6$ internal components, with the wave-equation linearity playing an essential role: it is the linearity that the alphabet-size hypothesis [SM §2.7] presupposes, and it is the wave-equation structure that produces the cubic-lattice spectrum from which the gauge group is derived. Matrix models reach the gauge group through a different route — emergent gauge fields from open strings ending on D-branes, or matrix-internal $U(N)$ symmetries enhanced in specific limits — and this route is genuinely nonlinear.

The implication for the equivalence-class proposal. For matrix models to be representatives of OI's equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$, they must satisfy the same A1–A6 that OI's reconstruction requires. The analysis above indicates this is not the case: at minimum A2 and A5 fail in ways that are not artifacts of the bridge but reflect genuine structural commitments. The equivalence-class reading therefore does not hold for matrix models as currently formulated.

This conclusion is structural rather than technical. Matrix models are the most direct finite-dimensional formulations of string theory, and their structural divergence from OI's substratum at A2 and A5 is informative about the relationship between the two programs. OI and string theory are not gauge-equivalent in the sense the equivalence-class proposal asks.

6.8 What can be salvaged

The negative conclusion in §7.8 admits five partial recoveries, ordered from most to least plausible.

(i) Different bridge. Matrix models are not the only finite-dimensional formulation of string theory. Alternative bridges include: (a) the AdS/CFT correspondence, where the boundary CFT lives on a finite-dimensional spacetime that may be discretized while preserving the bulk string content; (b) tensor-network reformulations of matrix models [TensorNetworkMatrix2024], which translate

continuous matrix dynamics into discrete tensor structures; (c) causal dynamical triangulations [Loll2019], a discretization scheme for quantum gravity with manifestly finite configuration space and combinatorial dynamics. Each of these admits independent investigation of A1–A6 compatibility, and the equivalence-class reading might hold for one of them even if it fails for matrix models. This paper does not pursue these alternatives in detail; they remain open directions.

(ii) Emergent string-theoretic features from OI substratum. A reformulation of the relationship asks not whether string theory's substratum-level structures (matrix models) are equivalent to OI's substratum, but whether OI's substratum produces string-theoretic features at appropriate emergent scales. OI's GR §5 already derives the Bekenstein-Hawking $A/4$ entropy through a thermal-self-consistency argument that is structurally distinct from string theory's microscopic D-brane count [StromingerVafa1996], yet produces the same numerical coefficient. If OI's substratum also produces, at appropriate emergent scales, holographic duality structure analogous to AdS/CFT [Maldacena1997], graviton modes from the long-distance limit of the substratum dynamics, and dualities analogous to T-duality and S-duality, then string theory's distinctive technical achievements would be recovered as features of OI's emergent physics rather than as independent substratum-level commitments. The structural barrier identified in §6 (linearity and determinism at the substratum level vs.~Yang-Mills nonlinearity and path-integral structure) does not preclude this recovery: effective field theories with nonlinear interactions and stochastic descriptions are routinely derived from underlying linear-deterministic dynamics through coarse-graining and emergent-scale phenomena. Whether OI specifically produces the right emergent-scale features to recover string-theoretic content is an open technical question requiring direct investigation; this paper does not pursue the analysis but flags it as the most consequential research direction left open by the negative result.

(iii) Approximate equivalence. The equivalence-class reading might hold *approximately* — in the sense that SM-reproducing matrix-model compactifications agree with OI's predictions to within experimental precision, even if the underlying structural conditions differ. This would be an empirical equivalence rather than a structural one and would be testable by computing OI-style predictions (Cabibbo angle, Koide ratio) within specific compactifications and comparing to OI's results. This is an empirical research direction not pursued here.

(iv) Reformulation of A2/A5. OI's A2 (determinism) and A5 (linearity) might admit reformulations that accommodate matrix-model structure. For A2: a path-integral over deterministic bijection trajectories produces ensemble-stochastic behavior at the coarse-grained level; if BFSS can be reformulated as such an ensemble, A2 might hold at the trajectory level. For A5: the commutator-squared interaction is linear in the *combination* $[X^i, X^j]$, with the nonlinearity arising from the commutator structure rather than higher-order terms; if "linearity" is read at the combination level, A5 might hold. Both

readings are speculative and have not been worked out in detail. The current state of OI's framework treats A2 and A5 in the standard sense (bijection at the configuration level; linear in single-variable expansion), and any reformulation must be argued explicitly.

(v) Shared observational invariants. A reformulation of the relationship asks not whether the substrata are equivalent (recoveries (i) and (iv)) or whether one produces the other (recovery (ii)) or whether predictions match empirically (recovery (iii)), but whether OI and string theory agree at the *observable* level because both must respect shared invariants of observation. Under this reading, the substrata of OI and string theory are genuinely different and not gauge-equivalent under any natural transformation — consistent with the structural divergence established in §6 — but the observations they produce are forced into agreement by structural features of the observation process itself. The intersection between the two frameworks lives at the observable level, not at the substratum level. OI's gauge group $\mathcal{G}_{\text{sub}}$ classifies observationally-equivalent substrata within OI's structural class; the proposal here is that a larger equivalence relation operates across structural classes, with observational agreement enforced by invariants that both OI's substratum and string theory's compactifications must respect. The constructive content of this recovery would be the identification of these observational invariants — features of the observation process that are independent of substratum description and that force any substratum producing the observed Standard Model to agree at the observable level. This is a stronger structural-realism claim than OI's reconstruction theorem alone establishes: the reconstruction theorem identifies invariants within OI's class; the present recovery would identify invariants across all classes producing the SM. The technical content of this recovery is open. It is a research direction that reframes the OI-string relationship from "two substratum descriptions of physics" to "two realizations of observational invariants that physics requires."

6.9 Summary

Matrix-model formulations of string theory do not satisfy OI's structural conditions A1–A6 in their standard formulation. The failures at A2 (determinism / bijection) and A5 (linearity) are deep — they reflect structural commitments of string theory that diverge from OI's substratum, not artifacts of the discretization bridge. The equivalence-class proposal of §1.3 therefore does not hold for matrix models as currently understood.

This negative result is the central technical finding of this paper. Section 6 develops its implications: that OI and string theory are independent programs with the same observational target, and that the conventional approach to combining them — string theory as a UV completion of the SM, with OI as a foundational reformulation — is structurally distinct from the equivalence-class reading that this paper investigated.

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7. Implications and discussion

The structural analysis of §6 establishes that matrix-model formulations of string theory do not satisfy OI's structural conditions A1–A6 in their standard formulation, with the failures at A2 (determinism / bijection) and A5 (linearity) reflecting structural commitments rather than discretization artifacts. The equivalence-class proposal of §1.3 — that SM-reproducing string compactifications are gauge-equivalent representatives of OI's $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ — therefore does not hold for matrix models as currently understood. This section discusses the implications of this negative result, the partial recoveries identified in §7.6, and the open directions left by the analysis.

7.1 OI and string theory as independent programs

The most direct implication of §6's analysis is that OI and string theory are independent research programs targeting the same observational target — the Standard Model and its quantum-gravity completion — through structurally distinct substratum descriptions. They are not the same theory in different gauge representations, and they are not different stages of a single derivation in which OI provides the foundational layer and string theory the technical implementation.

This independence has methodological consequences. A successful prediction of OI (e.g., the Cabibbo angle $1/(\pi\sqrt{2})$, the Koide ratio $2/3$) is not automatically a prediction of string theory; conversely, the swampland program's identification of consistent UV completions of the SM does not automatically constrain OI's substratum. Each program is responsible for its own empirical match, with no cross-attribution beyond what shared empirical inputs (E1–E4 in OI's notation) provide.

The independence is also informative about the foundational landscape of fundamental physics. OI and string theory are two of several active programs (alongside loop quantum gravity, causal sets, asymptotic safety, and others) that propose substratum descriptions for the SM. The structural divergence between OI and string theory at A2 and A5 indicates that these are deep ontological choices, not technical details: a substratum framework either commits to deterministic-bijection / linear-wave-equation structure (OI's choice, with the foundational economy this implies) or to path-integral / gauge-Yang-Mills structure (string theory's choice, with the gravitational-unification this provides). The choice cannot be made on technical grounds alone; it requires foundational arguments.

A consequence of the structural divergence is that both programs cannot be foundational descriptions of the same physics simultaneously. If both are taken as candidate substratum descriptions of the Standard Model, then at most one of the two is correct as currently formulated; the other requires modification. Three resolutions are possible: (a) OI is correct as described and string theory needs modification toward classical-substratum dynamics; (b) string theory is correct as described and OI's reconstruction theorem captures features of a spe-

cific compactification rather than foundational structure; (c) both require modification toward a deeper framework from which both emerge as different limits or different aspects. The current empirical and theoretical evidence does not select definitively among these resolutions. The independence claim is therefore not a stable end-state but a provisional characterization of the current state of evidence; the productive direction for future work is to identify which resolution obtains, through continued empirical testing of each framework's predictions, peer-review validation of internal consistency, and investigation of whether the partial recoveries identified in §7.6 can rescue equivalence-class structure.

7.2 The string landscape multiplicity problem is not dissolved by OI's gauge group

The first of the three substantive consequences flagged in §1.3 — dissolution of the string landscape multiplicity problem through OI's $\mathcal{G}_{\text{sub}}$ — was contingent on the equivalence-class reading. Since §6 establishes that the reading does not hold for matrix models, the dissolution does not follow.

The landscape multiplicity problem therefore remains an open question for string theory, to be addressed through string-theoretic methods (swampland conjectures, vacuum-selection mechanisms, anthropic considerations). OI's gauge group $\mathcal{G}_{\text{sub}}$ does not, on the analysis of §6, provide a structural reduction of the landscape; it classifies equivalence among substrata satisfying OI's A1–A6, which the matrix-model formulations of string theory do not.

This is a substantive limitation on OI's reach. A reading of OI as "physics modulo gauge, where the gauge group classifies all UV-completion choices" was natural but turns out to be too strong. The accurate reading is that OI's gauge group classifies equivalence within OI's structural class — the class of finite deterministic bijection systems with bounded coupling and linear wave equations on graphs of statistical isotropy. Substratum descriptions outside this class (matrix models, continuum string theory, perhaps loop quantum gravity, perhaps causal sets) are not classified by $\mathcal{G}_{\text{sub}}$ and may belong to different equivalence classes (or no equivalence class, if they fail to reproduce the SM).

7.3 Cross-attribution of predictions does not follow

The second consequence flagged in §1.3 — that OI's predictions become string-theory predictions through gauge equivalence — likewise does not follow. OI's prediction set (twenty-two classified quantitative SM relations, the sharpest matching observation at sub-percent precision) is a feature of OI's specific substratum derivation chain, which uses linearity of the wave equation, the alphabet-size hypothesis, and cubic-group decomposition. None of these is present in matrix models in the form OI requires, so the prediction chain does not transport.

This means the empirical match of OI's predictions is OI-specific evidence, neither confirmed nor refuted by string-theoretic considerations. If OI's predictions

hold up under further experimental scrutiny (DESI Year 5 for the cosmological prediction; JUNO for $\sin^2\theta_{12}$; lattice tests for additional Layer-2(a) predictions), this is evidence for OI's substratum description specifically. If they fail, it's evidence against OI specifically. String theory is not implicated either way through the gauge-equivalence route examined here.

7.4 Computational gateway via OI's lattice does not extend to string theory

The third consequence — OI's computationally-tractable lattice as a tool for string-theoretic computations — also does not follow. Without gauge equivalence, OI's lattice computations produce predictions of OI's substratum, not of string compactifications. They cannot substitute for direct string-theoretic computation (Calabi-Yau geometry, brane dynamics, flux integration) within a specific compactification.

This limitation does not, however, diminish OI's own computational tractability. OI's predictions remain computable from the cubic-lattice substratum without needing to engage string-theoretic structures. The framework's empirical successes are independent of string-theoretic computation, and that independence is preserved by the negative result.

7.5 Connection to swampland conjectures

The swampland program [Vafa2005, OoguriVafa2007, Palti2019, Lehnert2025] aims to identify which low-energy effective field theories admit consistent UV completion in quantum gravity, by formulating conjectures (Weak Gravity, de Sitter, AdS Distance, Cobordism, etc.) that consistent EFTs must satisfy. The program's success would be a constructive characterization of the SM-reproducing subset of the landscape.

OI's reconstruction theorem provides an alternative characterization of the same SM-reproducing target, but through different methodology: OI derives the SM from the embedded-observer ontology, whereas the swampland program identifies which UV completions are compatible with consistent quantum gravity. The two characterizations are independent in the sense established by §6, but they may agree on which low-energy effective theories are admissible. This section examines OI's compatibility with the major swampland conjectures, identifying both the consistency results and the cross-validation pathways the conjectures provide for OI's specific predictions.

7.5.1 Weak Gravity Conjecture The Weak Gravity Conjecture [ArkaniHamedMotlNicolisVafa2007] states that any consistent quantum-gravity EFT containing a $U(1)$ gauge field must contain a particle whose charge-to-mass ratio in Planck units is ≥ 1 — equivalently, gravity is the weakest force.

OI's emergent QFT contains the SM gauge structure $SU(3) \times SU(2) \times U(1)$ with the standard charged-particle spectrum. The lightest electrically charged

particle is the electron, with charge-to-mass ratio $q/m \sim 10^{21}$ in Planck units — vastly exceeding the WGC threshold. The conjecture is satisfied robustly by OI's emergent QFT for the standard $U(1)$ gauge sector.

The neutrino sector provides a sharper test. [IbañezMartinLozanoValenzuela2017] uses the WGC applied to a $U(1)_{B-L}$ gauge field to constrain neutrino-sector physics: if neutrinos are Dirac, the WGC requires a tower of charged states with masses below the electroweak scale, which has not been observed; if neutrinos are Majorana, the WGC constraint is satisfied via the Majorana mass term itself. OI's [SM §8.4] predicts Majorana neutrinos, putting OI's prediction in the WGC-allowed region. The cross-validation pathway is direct: WGC + observed neutrino phenomenology → Majorana neutrinos; OI → Majorana neutrinos. Both routes converge on the same prediction through structurally distinct reasoning.

7.5.2 de Sitter Conjecture The de Sitter Conjecture [ObiedOoguriSpodyneikoVafa2018] in its refined form [OoguriPaltiShiuVafa2019] states that the scalar potential V of a consistent quantum-gravity EFT satisfies $|\nabla V|/V \geq c \sim O(1)$ at any de Sitter critical point, OR $\min(\mathrm{Hess}\, V) \leq -c'/M_P^2 V$, where c, c' are positive $O(1)$ constants. The conjecture forbids stable de Sitter vacua and is in tension with the standard ΛCDM cosmology with a positive cosmological constant.

OI's [GR §7] carries dark energy in Type II running-vacuum (RVM) form rather than as a fixed cosmological constant — the functional form the local expansion shared by every smooth model, the content the Type II channel, a structural reading rather than a derived law — with the parameter ν encoding the running and the framework's expected magnitude $|\nu| \sim 10^{-32}$. Type II RVM is parametrically away from a static de Sitter vacuum: the cosmological "constant" runs with the Hubble rate, and the late-universe phase is asymptotically de Sitter in the limit $H \to 0$ but not strictly so at any finite time. The dynamical character is consistent with the de Sitter Conjecture's structural content; the small predicted $|\nu|$ value is consistent with the empirical near-ΛCDM behavior. Strict static de Sitter would violate the conjecture; dynamical near-de Sitter satisfies it.

The recent DESI DR2 results, rejecting the cosmological constant at 2.8σ–4.2σ in extended cosmological models with potential phantom crossing [DESI_DR2_2025], provide independent empirical pressure on static dark-energy models — the same direction the de Sitter Conjecture pushes. OI's position relative to this data must be stated carefully, consistent with [GR §6.2, §7.1]: the framework's predicted running $|\nu| \sim 10^{-32}$ is observationally indistinguishable from a cosmological constant, so the DESI phantom-crossing *magnitude* is in 3–4σ tension with OI's prediction even though the running-vacuum *form* is favoured qualitatively. The de Sitter Conjecture consistency claim rests on the strict-static distinction — OI is nominally dynamical but observationally Λ-like — and should be read at that level of strength.

7.5.3 Distance Conjecture The Distance Conjecture [OoguriVafa2007] states that as one moves to large distance in the moduli space of a consistent quantum-gravity EFT, an infinite tower of states becomes light, with masses scaling exponentially with the distance traveled.

OI's framework has no fundamental moduli: the substratum bijection is fixed, and the emergent QFT's parameters are derived from the cubic-lattice structure rather than being moduli to be selected. The Distance Conjecture is therefore satisfied trivially in a degenerate sense — there is no moduli space to traverse.

This is a structural difference between OI and string-theoretic frameworks. In string theory, the landscape's structure is the moduli space of compactifications; the Distance Conjecture provides a non-trivial constraint on EFTs derived from that moduli space. In OI's framework, the analog of the moduli space is the universality-class structure of Part II — different substrata in different universality classes are not connected by continuous moduli but by structural-class membership. The conjecture's content does not directly transport, but a weakened analog might apply: substrata "far" from OI's class in the universality-class space might be connected to OI's class only through transitions involving infinite towers of states. This is speculative and open.

7.5.4 AdS Distance Conjecture The AdS Distance Conjecture [LustPaltiVafa2019] states that AdS minima of consistent quantum-gravity EFTs are accompanied by an infinite tower of states becoming light, with masses scaling with the AdS radius. The conjecture excludes scale-separated AdS vacua in the strict sense.

OI's emergent QFT does not predict an AdS minimum: the cosmological structure is positive-cosmological-constant (de Sitter-like), with the small $|\nu|$ deviation per §7.5.2. The AdS Distance Conjecture is therefore not directly applicable to OI's framework. If a sub-cluster of observer-admitting compactifications reproduced OI's predictions but admitted an AdS minimum, the conjecture would constrain that sub-cluster — but this would be a constraint within the universality-class space, not on OI itself.

7.5.5 Cross-validation pathways The swampland conjectures collectively define a region of EFT parameter space compatible with consistent quantum gravity. OI's emergent QFT lives within this region across the conjectures examined (WGC, dS, Distance, AdS Distance). This is non-trivial cross-validation: OI's structural predictions (Majorana neutrinos, dynamical dark energy via Type II RVM, no fundamental moduli, no AdS minimum) align with the swampland-allowed region without being designed for it.

The cross-validation operates at the level of structural prediction, not numerical agreement. OI predicts Majorana neutrinos because the substratum's structural conditions force lepton-number violation; the WGC predicts Majorana neutrinos because Dirac neutrinos require an unobserved tower of states. Both routes

converge on the same prediction through different reasoning. Similar structural convergence operates for dark energy: OI predicts running cosmological constant from the GR cluster; the de Sitter Conjecture forbids static de Sitter vacua. Both favor dynamical dark energy.

The pathway provides empirical leverage. If LEGEND-1000 [LEGEND2024] confirms Majorana neutrinos, both OI and the WGC-derived prediction are confirmed. If DESI Year 5 strengthens the rejection of the cosmological constant, both OI's Type II running-vacuum channel and the de Sitter Conjecture are favored. The empirical tests are joint tests of OI and the swampland-favored EFT structure — confirmation supports both; refutation challenges both.

This section does not resolve whether OI's specific predictions can be derived from swampland reasoning, or whether swampland-consistent EFTs uniquely determine OI's universality class. Those are open questions. What this section establishes is that OI's framework is consistent with the major swampland conjectures examined, and that cross-validation pathways exist for OI's structural predictions through swampland-derived constraints.

7.6 Partial recoveries and what would be required to revisit the equivalence-class reading

§6.8 identified five partial recoveries, ordered roughly from most to least consequential. *Emergent string-theoretic features from OI's substratum* (recovery (ii)) is the most consequential if it holds: OI's GR §5 already produces the Bekenstein-Hawking $A/4$ entropy through thermal self-consistency [StromingerVafa1996 has the microscopic derivation]; whether OI similarly produces holographic-duality structure analogous to AdS/CFT, graviton modes from long-distance dynamics, and T/S-dualities at appropriate emergent scales is an open technical question. The structural barrier identified in §6 does not preclude this recovery: emergent-scale physics can have nonlinear and stochastic features arising from underlying linear-deterministic dynamics through coarse-graining. *Shared observational invariants* (recovery (v)) is structurally distinct: it asks whether OI and string theory agree at the observable level because both respect invariants of the observation process, identifying invariants across structural classes rather than within OI's class. *Alternative bridges* (AdS/CFT with discrete boundary, tensor networks, causal dynamical triangulations) is tractable but would constitute substantial follow-up work. *Approximate-equivalence via empirical agreement* is testable: computing OI's predictions (Cabibbo, Koide) within a specific SM-reproducing compactification and comparing. *A2/A5 reformulation* is the most speculative — it would require substantial reworking of either OI's conditions or string-theoretic structure.

The two most consequential — emergent features and shared invariants — are not in competition. They address structurally distinct questions: the first treats OI as a candidate foundational description from which string-theoretic content emerges; the second treats observation itself as the locus where different substra-

tum descriptions are forced into agreement. Both extend OI's structural-realism stance — that what is real is invariant structure, not specific representatives — to the comparison with other unification programs.

7.7 Recovery (ii) developed: emergent string-theoretic features from OI's substratum

Recovery (ii) of §7.6 — that OI's substratum produces emergent string-theoretic features at appropriate emergent scales — is the most consequential of the five partial recoveries if it holds. This section develops what is currently tractable in this direction, distinguishing structural features of string theory that plausibly emerge from OI's substratum from features that do not, with honest scoping of which features are open research questions.

The structural barrier identified in §6 (A2 and A5 violations in matrix-model formulations) does not preclude this recovery. The barrier operates at the level of substratum descriptions: OI's substratum is deterministic-bijection-on-cubic-lattice; matrix models' substrata are Hermitian-matrix-path-integral. These differ as substratum descriptions. However, emergent-scale physics derived from each substratum can have shared features, in the same way that two different microscopic theories can share long-distance behavior through RG universality. The question is which string-theoretic features are robust at the emergent scale and which are tied specifically to string-theoretic substratum content.

7.7.1 Horizon holography vs full AdS/CFT The strongest emergence claim is for *horizon holography* — the Bekenstein-Hawking $A/4$ area law and the holographic information bound. OI's [GR §5] derives the $A/4$ coefficient from the substratum's thermal self-consistency at the cosmological-horizon scale, with the entropy density emerging as a partial-trace invariant of the embedded-observer marginal. The derivation does not appeal to specific string-theoretic content; it invokes only the framework's structural features (the trace-out, the cosmological-horizon information capacity, the thermal equilibrium of the hidden sector). The $A/4$ coefficient transports to OI from the partial-trace structure, not from string-theoretic content.

This produces a structural feature shared with string-theoretic horizon physics: any horizon entropy in OI's framework is bounded by $A/4$, and the holographic principle (information content of a region bounded by area, not volume) holds. The shared feature is at the level of the partial-trace observational structure, consistent with §14.2's transport assessment for channel-level unitarity. Recent confirmation comes from GW250114 [GW250114], where Hawking's area theorem was confirmed at 99.999% confidence — a classical-horizon-dynamics result that both OI and string theory preserve; the 1/4 coefficient itself (in OI by mode counting under the horizon conditions of [GR §2, §8.5], in string theory by microscopic state counting [StromingerVafa1996]) has no direct empirical test.

The harder question is whether OI produces *full AdS/CFT-style holography* — bulk-boundary correspondence with specific dictionary between bulk fields and boundary operators. Here the answer is more nuanced. OI's cosmological structure is positive-Λ (de Sitter-like), not anti-de Sitter; the cosmological horizon is a de Sitter horizon, not an AdS boundary. The bulk-boundary correspondence in standard AdS/CFT relies on the asymptotic AdS structure to define boundary operators uniquely, and this asymptotic structure is absent in OI's framework.

Two possible directions for emergence remain open:

- *de Sitter holography.* Work on de Sitter holography [StromingerDS2001] develops bulk-boundary correspondence for de Sitter spacetime with a CFT defined on the asymptotic past or future de Sitter boundary. OI's cosmological structure is asymptotically de Sitter (under the framework's $|\nu| \sim 10^{-32}$ Type II RVM), so de Sitter holography would be the natural framework. Whether OI's substratum produces a specific de Sitter holographic dictionary — bulk graviton fields $\leftrightarrow$ boundary CFT operators with specific scaling dimensions — is open. The structural features (asymptotic de Sitter, finite horizon entropy, holographic bound) are present; the specific dictionary requires further analysis.
- *Approximate AdS in deep IR.* In some limits, de Sitter spacetime admits approximate AdS structure (e.g., near the horizon, in certain coordinate patches). Whether OI's framework produces an effective AdS region with corresponding effective AdS/CFT structure is similarly open.

The honest assessment: OI naturally produces horizon holography (the $A/4$ entropy law and the holographic bound) but does not naturally produce full AdS/CFT-style correspondence as an emergent feature. A is a structural consequence of the partial-trace observational structure; the latter requires specific compactification-dependent content that OI's substratum does not directly provide.

7.7.2 Graviton modes from long-distance dynamics OI's [GR §3–§5] derives linearized Einstein gravity from the partial-trace structure of the substratum bijection at the cosmological-horizon scale. The mechanism: the substratum's coherent excitations include long-wavelength fluctuations of the visible-sector metric, which are the emergent graviton modes. The framework reproduces the standard GR structure at the macroscopic level with Type II RVM corrections, including the linearized-graviton sector with the right propagator structure (massless spin-2 modes, transverse-traceless polarization, lightlike propagation).

This emergence is structurally analogous to the string-theoretic graviton emergence: in both frameworks, gravitons arise as collective modes of the substratum dynamics. In string theory, the graviton is a closed-string vibrational mode with worldsheet σ-model structure. In OI, the graviton is an emergent collective mode

of the substratum's coherent excitations on the cubic lattice, derived through the partial-trace-and-coarse-grain procedure.

The *structural similarity* is at the level of the IR effective theory: both frameworks produce massless spin-2 modes coupled to the stress-energy tensor in the linearized limit, satisfying the equivalence principle and gauge-invariance under linearized diffeomorphisms. This is what physicists mean by "the emergent low-energy theory contains gravitons."

The *structural difference* is at the UV completion. In string theory, the graviton is a fundamental string mode with worldsheet origin; in OI, the graviton is an emergent collective mode with no string-mode interpretation. The differences become important at high energies — string-scale corrections to graviton propagators in string theory vs Planck-scale corrections from substratum-level lattice structure in OI; the existence of an infinite Regge tower of massive string modes vs OI's finite-but-large substratum modes. These UV differences are testable in principle but typically inaccessible at current and near-future energy scales.

The recovery (ii) claim is supported for graviton-mode emergence: OI produces emergent graviton modes with the structural features that string-theoretic graviton modes share — namely the IR effective-theory content. The framework does not produce the specifically string-theoretic UV completion (worldsheet structure, infinite massive tower) but produces the universally-required IR structure that any successful emergent-graviton framework must reproduce.

7.7.3 T-duality and S-duality analogs T-duality in string theory is the equivalence of closed-string theories on a circle of radius R and radius α'/R, with momentum modes $\leftrightarrow$ winding modes. The duality requires a string-mode structure with both momentum and winding quantum numbers, neither of which has a direct analog in OI's substratum.

OI's substratum has lattice-momentum modes (excitations with definite lattice wavevector), but no winding modes (no extended objects winding around compact dimensions). The cubic lattice is the physical spatial structure, not a compactification. T-duality therefore has no clean analog in OI's framework: the duality-pair structure (momentum $\leftrightarrow$ winding) is absent because winding modes are absent.

A weakened analog might exist at the universality-class level. If two SM-reproducing universality classes have algebra-channel pairs related by a specific symmetry — say, an automorphism that exchanges high-momentum and low-momentum content — this would be a T-duality-like feature at the universality-class level rather than the substratum level. Whether such a structure exists in OI's universality-class framework is open. The honest assessment: T-duality at the substratum level has no analog; T-duality-like features at the universality-class level are speculative.

S-duality in string theory is the equivalence of strong-coupling and weak-

coupling descriptions in certain string theories (most cleanly in type IIB). The duality maps the string coupling $g_s \to 1/g_s$, exchanging fundamental strings with D1-branes (and other extended objects) under the duality.

OI's framework has emergent gauge couplings derived from the substratum structure. Whether there is a structural S-duality-like feature — e.g., a duality between a strong-coupling regime where the visible-sector EFT description breaks down and a weak-coupling regime in a different visible-sector EFT description — is an open question. The substratum-level dynamics is fixed (the bijection on the cubic lattice), so there is no direct analog of varying the string coupling. The visible-sector EFT couplings can be varied continuously, but whether the strong-coupling limit maps onto a different weak-coupling EFT description is not established.

The honest assessment: S-duality analogs in OI are speculative. Some structural features of S-duality (the existence of dual descriptions of the same observable physics through different regime variables) might transport to OI's universality-class framework, but no specific S-duality-like map has been identified.

7.7.4 Synthesis: what emerges and what does not Recovery (ii) develops with the following pattern:

Feature	Emerges from OI?	Caveat
Horizon holography ($A/4$ law, holographic bound)	Yes	Structural consequence of partial-trace observation
Bulk-boundary correspondence (full AdS/CFT)	No	OI is dS, not AdS; de Sitter holography open
Linearized graviton modes	Yes	IR effective theory shared with string theory
String UV completion (worldsheet, Regge tower)	No	OI's UV is lattice substratum, structurally different
T-duality	No	No winding modes in OI substratum
S-duality	Speculative	Universality-class-level analogs open

The pattern: OI naturally produces the *IR effective theory* and the *structural observational features* of string-theoretic frameworks (horizon holography, graviton modes, holographic bound), but does not produce the *specifically string-theoretic UV content* (string worldsheet, Regge tower, T-duality, AdS/CFT bulk-boundary dictionary). The IR content transports because it is a feature of the

partial-trace observational structure; the UV content does not transport because it is specific to string-theoretic substratum content that OI's substratum does not produce.

This refines the §6 negative result. OI and string theory share the IR effective theory at the partial-trace observational level (recovery (v)), and emergent structural features that follow from the partial-trace structure (the horizon-holography component of recovery (ii)), but they differ on string-theoretic UV content. The partial recovery via (ii) succeeds for the IR features and the horizon-holography component; it does not succeed for the full string-theoretic UV content.

The implication for the §1.4 equivalence-class proposal: the proposal does not hold at the substratum level (per §6), and it does not hold at the level of string-theoretic UV content (per the negative parts of this section), but it does hold at the level of IR effective theory and partial-trace observational features (per the positive parts of this section and §14). Recovery (ii) is therefore a partial recovery in the precise sense: it identifies which features transport — the structural features of partial-trace observation (Born rule, channel-level unitarity, non-Markovian marginal, horizon holography, linearized graviton modes) and the IR effective theory — but not the substratum-specific UV content.

The work that remains: a rigorous derivation of de Sitter holographic dictionary from OI's substratum (§7.7.1); a quantitative comparison of OI's and string-theoretic graviton-propagator UV corrections (§7.7.2); identification or exclusion of universality-class-level T- and S-duality analogs (§7.7.3). These are the natural follow-up directions to this recovery's development.

7.8 What this paper has established

This paper articulates the equivalence-class proposal precisely (§1.4), resolves it in the negative for matrix models (§6 establishes A2 and A5 fail in ways reflecting structural commitments rather than discretization artifacts), and identifies five partial recoveries as research directions (§6.8 + §7.6). The negative result is informative: OI and string theory are not gauge-equivalent at the substratum level, not in a foundational-versus-effective hierarchy at any obvious level, and not in a derivable relationship as currently formulated. Their relationship must be sought either at the level of emergent-scale physics (recovery (ii)) or at the level of observational invariants both frameworks must respect (recovery (v)). The §14 four-feature audit develops the second direction: three of four observational features transport directly between frameworks, characterizing the partial-trace observational level as where the intersection lives.

7.9 Observer admission and the structure of the landscape

Part II §9.6 formalizes a structural distinction between *observer-admitting* substrata — those that admit a partition satisfying C1–C4 — and *observer-lacking* substrata, where no such partition exists. The distinction operates at the level

of the substratum's bijection structure, prior to any observer being instantiated. This section examines how the distinction reframes the relationship between OI and the string landscape.

The landscape, viewed through observer admission. The string landscape comprises approximately 10^{500} distinct compactifications. The standard reading [Vafa2005, Susskind2003] treats the landscape as a multiplicity of candidates for "the universe," with anthropic selection narrowing the candidate set to those compatible with our observations. Under the observer-admission reading, the landscape's multiplicity decomposes structurally:

- *Observer-lacking compactifications.* Compactifications producing four-dimensional vacua without chiral matter, with rapid hidden-sector mixing, with insufficient horizon capacity, or with broken cosmological structure cannot satisfy C1–C4. They correspond to observer-lacking substrata. These compactifications are mathematical objects in string theory's space of vacua, but they are not candidate physical universes in OI's framework — observers cannot be embedded in them.
- *Observer-admitting compactifications.* Compactifications producing four-dimensional vacua with chiral matter, stable cosmological structure (de Sitter or near-de Sitter with horizon timescales $\tau_B \sim H^{-1}$), and sufficient information capacity (the cosmological horizon's entropy $\sim 10^{122}$) satisfy C1–C4 and are observer-admitting. The SM-reproducing compactifications surveyed in §5 — heterotic on Calabi-Yau with appropriate bundles, Type II / D-brane Pati-Salam constructions, F-theory $SU(5)$ with hypercharge flux — fall in this class.

The observer-admission criterion distinguishes the two classes structurally. Most compactifications in the landscape are likely observer-lacking; the SM-reproducing subset is a small fraction. The numerical multiplicity of the landscape (10^{500}) is largely a measure of observer-lacking substrata, not of competing candidates for our universe.

The translation of C1–C4 to string-theoretic structure. The structural conditions for observer admission [Main §3] are formulated for finite deterministic substrata. Translating them to string-theoretic compactifications requires identifying analogs in string-theoretic structure:

- *C1 (coupling)* requires substantial coupling between visible-sector and hidden-sector dynamics. In string-theoretic terms: the four-dimensional EFT must have non-trivial interaction between observable matter (chiral fermions, gauge fields, Higgs sector) and the gravitational / Kaluza-Klein / hidden gauge sectors. Compactifications with completely decoupled hidden sectors fail C1.
- *C2 (memory persistence)* requires the written record to persist to readback, the slow bath one realization: the hidden-sector correlation time much longer than the observer's measurement timescale. In string-theoretic

terms: the cosmological structure must have a stable phase with horizon timescales $\tau_B \sim H^{-1}$ much longer than laboratory timescales τ_S. AdS compactifications without de Sitter structure, or compactifications with rapidly mixing cosmological structure, fail C2.

- *C3 (sufficient capacity)* requires the hidden sector to have sufficient information capacity to store the interaction history. In string-theoretic terms: the cosmological horizon's information capacity (set by the holographic bound at $\sim 10^{122}$ for our observed universe) must be sufficient. Compactifications with horizons of insufficient capacity fail C3.

These translations are intuitively clear: a compactification reproducing our universe's cosmological structure satisfies all three. A compactification producing AdS, anti-de Sitter, or rapidly-mixing dS-with-very-short-horizon-time fails one or more. The translation is plausible structurally, though a rigorous version would require detailed analysis specific to compactification classes.

The OI-string relationship reframed. Under the observer-admission reading, the relationship between OI and string theory is structurally distinct from the standard "two competing programs" framing. OI operates entirely within observer-admitting substrata — the framework's foundational axiom "observation occurs" is taken to be structurally specific to observer-admitting cases (per §9.6). String theory's landscape encompasses both observer-admitting and observer-lacking substrata, with the observer-admitting subset being where SM-reproducing physics resides.

The relationship therefore takes this form:

- OI's universality class lives in the observer-admitting subset.
- Some SM-reproducing string compactifications also live in the observer-admitting subset.
- The intersection between OI's universality class and the observer-admitting compactification subset is what the partial-trace-features analysis of §14 characterizes.
- Compactifications outside the observer-admitting subset are not candidates for "our universe" in OI's framework — they are observer-lacking substrata, mathematical objects but not physical universes.

This reframing has substantive consequences for the landscape problem. The standard landscape problem treats 10^{500} vacua as competitors; selecting "the right one" appears to require either anthropic reasoning or unprincipled choice. The observer-admission reading dissolves this: most of the landscape is observer-lacking and not in competition for "our universe." The candidate space is the observer-admitting subset, which is much smaller. Within the candidate space, our universe is in some specific universality class, distinguished from other observer-admitting universality classes by specific predictions.

Connection to anthropic reasoning. The standard anthropic argument explains the landscape's empirical match to our universe through selection: we

observe a universe compatible with observers because we couldn't observe a universe incompatible with observers. The observer-admission reading sharpens this. Anthropic selection separates into two distinct kinds of work:

- *Structural exclusion of observer-lacking substrata.* Substrata that lack the structural features for observer-admission are not in the candidate space at all. There is no anthropic selection within them — they are not competitors. This is structural exclusion, not selection.
- *Within-candidate-space selection of universality class.* Among observer-admitting substrata, our universe is in some specific universality class. Standard anthropic reasoning applies here: we are in the universality class that produces our observations, and the framework's predictions distinguish OI's class from other observer-admitting classes within the candidate space.

The combination dissolves the "fine-tuning" worry to a substantial degree. Fine-tuning intuitively arises from the question "why is our universe one specific compactification out of 10^{500}?" The observer-admission reading reframes: most of the 10^{500} are not candidate compactifications for "our universe" at all (they are observer-lacking); within the candidate space, our universe is in a specific universality class characterized by specific structural features (per §14's four-feature audit and the universality-class structure of Part II).

Implications for the relationship. The "two competing programs" framing of OI vs. string theory is not the right characterization under the observer-admission reading. The structurally accurate framing is:

- *String theory* provides a broad framework encompassing both observer-admitting and observer-lacking substrata, with the SM-reproducing subset (observer-admitting) being where physical universes reside.
- *OI* operates within the observer-admitting subset, providing structural conditions (A1–A6) that pick out a specific universality class.
- *The intersection* between OI's universality class and the SM-reproducing observer-admitting compactifications is what §14's partial-trace-features analysis establishes.
- *Outside the intersection,* OI and string theory diverge on universality-class membership, with each framework potentially identifying its own universality class within the observer-admitting subset.

This reframing makes the OI-string relationship structurally clean: not competition, but complementary roles in characterizing the structure of the candidate space for physical universes. The framework's content remains: OI's specific predictions distinguish OI's universality class from other observer-admitting universality classes; the predictions are tests of which class our universe is in.

Open structural questions. The observer-admission framing of the landscape is a structural reading, not a rigorous result. Several questions remain open:

- *Rigorous translation of C1–C4 to string-theoretic structure.* The translations above are plausible but not formal. Working out a rigorous version for specific compactification classes (heterotic, Type II, F-theory) would require detailed analysis specific to each class.
- *Numerical estimate of the observer-admitting subset.* The standard 10^{500} figure is for the full landscape including observer-lacking compactifications. Estimating the observer-admitting subset's size — even to within several orders of magnitude — would substantially clarify the landscape problem's empirical content.
- *Observer-admitting compactifications outside OI's universality class.* If observer-admitting compactifications exist that are not in OI's universality class, they would be alternative candidates for "the universe" — distinct from OI in their predictions, with their own internal structure. Identifying such compactifications would be a substantial result.
- *Connection to swampland criteria.* The swampland program identifies criteria for compactifications consistent with quantum gravity (Weak Gravity Conjecture, de Sitter Conjecture, Distance Conjecture). The observer-admission criterion is structurally distinct but may overlap with swampland criteria — both restrict the candidate compactification space, possibly with different overlap regions. Working out the relationship would clarify both frameworks.

These questions are not addressed in this paper; they are flagged as research directions opened by the observer-admission reframing. The reframing itself — reading the landscape's multiplicity as containing predominantly observer-lacking substrata, with the observer-admitting subset being structurally smaller — is a substantive contribution that the framework's structural-realism stance makes available to the landscape discussion.

Part II: Universality Classes of Embedded Observers (Question 2)

8. Formalizing embedded observation

The notion of "embedded observer" in OI's framework is informal: an observer with finite distinguishable states, embedded in a substratum, with non-trivial coupling to a hidden sector that has slow dynamics relative to observation, and high information capacity in the hidden sector. For the universality-class question to admit precise formulation, we need a formalization sufficient to define equivalence relations on substratum-with-observer systems. Three candidate formalizations are available.

8.1 Categorical formalization

The categorical approach treats embedded observers as functors $F : \mathbf{Sub} \to \mathbf{Obs}$ from a category of substratum descriptions to a category of observable descriptions, with the universality-class equivalence given by natural equivalence of functors (substrata producing isomorphic observable descriptions, possibly through different observation functors). The formalization is well-defined but requires substantial machinery (explicit construction of **Sub** and **Obs** with their respective morphism structures) that has not been worked out in the OI literature. We do not pursue it further here.

8.2 Operational formalization

The operational approach treats embedded observers as operational protocols $(\mathcal{P}, \mathcal{M}, p)$ — preparations, measurements, and outcome probabilities — without committing to a specific underlying substratum, in the spirit of Mackey's quantum logic [Mackey1963] and the generalized probabilistic theories program [Hardy2001, Barrett2007]. The universality class is the equivalence class under operational indistinguishability. This formalization is closest to empiricist intuition and methodologically clear, but it is empiricist — it does not commit to substratum-level realism — and may sit uneasily with OI's structural-realism stance, which treats the equivalence class itself as ontologically real.

8.3 Algebraic formalization

The algebraic approach treats embedded observers as CPTP channels with specific structural features. An embedded observer is characterized by: - An algebra $\mathcal{A}_V$ of visible-sector observables (typically a von Neumann algebra acting on a Hilbert space $\mathcal{H}_V$) - A CPTP channel $\Phi : \mathcal{B}(\mathcal{H}_V) \to \mathcal{B}(\mathcal{H}_V)$ representing the dynamics of visible-sector states under the substratum's evolution traced over the hidden sector - A unitary realization: a hidden sector $\mathcal{H}_H$, a reference state ρ_H, and a unitary U on $\mathcal{H}_V \otimes \mathcal{H}_H$ lifting Φ to a unitary dynamics (Definition 9.2; when ρ_H is pure the associated isometry is the Stinespring dilation)

The substratum is captured by the realization; the observer is captured by the visible-sector algebra and the CPTP channel. Equivalence between embedded observers is equivalence of the channel and algebra structure, with the realization being gauge relative to the fixed reference preparation.

The universality-class proposal in algebraic terms: two substrata are in the same class if they produce equivalent $(\mathcal{A}_V, \Phi)$ pairs, where equivalence is *-isomorphism of the algebras and conjugacy of the channels (channels related by an algebra isomorphism).

Status. This formalization is intermediate in heaviness between categorical and operational. It uses standard machinery from operator-algebra and CPTP-channel theory, both well-developed. It captures OI's framework directly: OI's reconstruction theorem produces an algebra-channel pair from the substratum,

and $\mathcal{G}_{\text{sub}}$ is characterized by transformations preserving this pair. Extending to string theory's emergence requires identifying the corresponding algebra-channel pair from string-theoretic dynamics, which is non-trivial but tractable for matrix-model formulations and AdS/CFT.

8.4 Choosing a formalization

The three formalizations are related but not equivalent. The operational formalization is weakest — it characterizes observers by their data without committing to underlying structure. The algebraic formalization adds structure (algebra of observables, channel dynamics) that operational data alone does not specify. The categorical formalization is heaviest — it requires explicit category-theoretic machinery that the others do not.

For the purposes of this paper, we adopt the **algebraic formalization** as the principal framework. It is well-developed in the operator-algebra and quantum-information literatures; it captures OI's framework and string-theoretic emergence directly; and it provides sufficient structure to define universality classes precisely without requiring the heavy categorical machinery.

We will return to operational and categorical perspectives where they illuminate features the algebraic formalization does not capture, but the principal definitions and arguments use the algebra-channel structure of §8.3.

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9. Universality classes of embedded observers

9.1 Formal definitions

We adopt the algebraic formalization of §8.4. The following definitions make precise the universality-class structure used throughout the rest of the paper.

Definition 9.1 (Embedded observer). *An embedded observer is a triple $(\mathcal{H}_V, \mathcal{A}_V, \Phi)$ where: - $\mathcal{H}_V$ is a separable Hilbert space (the visible-sector state space); - $\mathcal{A}_V \subseteq \mathcal{B}(\mathcal{H}_V)$ is a von Neumann algebra (the algebra of visible-sector observables); - $\Phi : \mathcal{B}(\mathcal{H}_V) \to \mathcal{B}(\mathcal{H}_V)$ is a CPTP channel (the visible-sector dynamics generator).*

Definition 9.2 (Substratum-with-observer system). *A substratum-with-observer system is a quintuple $(\mathcal{H}_V, \mathcal{A}_V, \Phi, \mathcal{H}_H, U)$ where $(\mathcal{H}_V, \mathcal{A}_V, \Phi)$ is an embedded observer, $\mathcal{H}_H$ is a separable Hilbert space (the hidden-sector state space), and U is a unitary on $\mathcal{H}_V \otimes \mathcal{H}_H$ such that*

$$\Phi(\rho) = \mathrm{Tr}_H[U(\rho \otimes \rho_H)U^\dagger]$$

for some hidden-sector reference state ρ_H. The triple $(\mathcal{H}_H, \rho_H, U)$ is a unitary realization *of Φ. When $\rho_H = |0\rangle\langle 0|$ is pure, the map $V_{U,0} : \psi \mapsto U(\psi \otimes |0\rangle)$ is an isometry $\mathcal{H}_V \to \mathcal{H}_V \otimes \mathcal{H}_H$ satisfying $\Phi(\rho) = \mathrm{Tr}_H[V_{U,0}\, \rho\, V_{U,0}^\dagger]$, and it is this isometry — not U — that is a Stinespring dilation of Φ.*

Remark (why the distinction is load-bearing). A unitary realization carries strictly more data than the channel constrains, in two separate ways, and both matter below. First, when ρ_H is pure the channel sees U only through its action on $\mathcal{H}_V \otimes \mathbb{C}|0\rangle$; the action of U on the rest of $\mathcal{H}_V \otimes \mathcal{H}_H$ is entirely free. Second, when ρ_H is mixed the realization remains a realization rather than a Stinespring dilation, and its hidden-sector dimension is not the Kraus rank — it can be strictly *smaller*. The completely depolarizing qubit channel has Kraus rank 4, yet $\rho_H = \mathbb{1}/2$ with $U = \mathrm{SWAP}$ realizes it on a two-dimensional hidden sector. This is not an edge case for OI: §10.4's construction uses the uniform mixed reference state, so OI's own $(\mathcal{H}_H, \rho_H, U_\varphi)$ is a unitary realization and not a minimal Stinespring dilation, and the uniqueness statements below must not be read as applying to it directly.

Definition 9.3 (Observational equivalence). *Two embedded observers* $(\mathcal{H}_V^{(1)}, \mathcal{A}_V^{(1)}, \Phi^{(1)})$ *and* $(\mathcal{H}_V^{(2)}, \mathcal{A}_V^{(2)}, \Phi^{(2)})$ *are observationally equivalent, written* $(\mathcal{H}_V^{(1)}, \mathcal{A}_V^{(1)}, \Phi^{(1)}) \sim_{\mathrm{obs}} (\mathcal{H}_V^{(2)}, \mathcal{A}_V^{(2)}, \Phi^{(2)})$*, if there exists a* -isomorphism $\theta : \mathcal{A}_V^{(1)} \to \mathcal{A}_V^{(2)}$ such that*

$$\theta \circ \Phi^{(1)}|_{\mathcal{A}_V^{(1)}} = \Phi^{(2)}|_{\mathcal{A}_V^{(2)}} \circ \theta.$$

Definition 9.4 (Universality class). *A universality class is an equivalence class of embedded observers under* $\sim_{\mathrm{obs}}$*. Two substratum-with-observer systems are in the same universality class if their embedded observers (the first three components of the quintuple) are observationally equivalent — irrespective of the specific Stinespring dilations producing them.*

The definitions are deliberately framework-neutral. They invoke standard machinery from operator-algebra theory (von Neumann algebras, CPTP channels, *-isomorphisms) without requiring framework-specific structural commitments. This is essential for the universality-class framework's purpose: characterizing equivalence at the level of observation, independent of the specific substratum machinery producing observable physics.

9.2 Relation to OI's gauge group: a proposition

OI's substratum gauge group $\mathcal{G}_{\mathrm{sub}}$ is a special case of observational equivalence. We can state this precisely.

Proposition 9.5 (OI gauge group as observational sub-equivalence). *Let* (S_1, φ_1) *and* (S_2, φ_2) *be OI-class substrata (satisfying A1-A6) related by a transformation* $g \in \mathcal{G}_{\mathrm{sub}}$*. Let* $(\mathcal{H}_V^{(i)}, \mathcal{A}_V^{(i)}, \Phi^{(i)})$ *be the embedded observer derived from* (S_i, φ_i) *via the Main paper's trace-out construction. Then* $(\mathcal{H}_V^{(1)}, \mathcal{A}_V^{(1)}, \Phi^{(1)}) \sim_{\mathrm{obs}} (\mathcal{H}_V^{(2)}, \mathcal{A}_V^{(2)}, \Phi^{(2)})$.

Proof sketch. By construction, $\mathcal{G}_{\mathrm{sub}}$ is the set of substratum-level transformations preserving observational data (transition probabilities $T_{ij}(t)$, emergent Hamiltonian up to D-gauge, and $\hbar$). The trace-out construction maps

observational-data-preserving substratum transformations to *-isomorphisms on the visible-sector algebra commuting with the channel. Hence $g \in \mathcal{G}_{\mathrm{sub}}$ induces θ_g satisfying Definition 9.3. □

The converse — that all observationally-equivalent OI-class substrata are related by some $g \in \mathcal{G}_{\mathrm{sub}}$ — is the completeness statement of [Substratum] Theorem 24, established for OI-class substrata via the Stinespring uniqueness + classical-substratum descent argument [Substratum §4].

Corollary 9.6 (Universality classes are coarser). *Let $\mathcal{C}$ be the universality class containing an OI-class substratum (S, φ). Then $\mathcal{C}$ contains every $\mathcal{G}_{\mathrm{sub}}$-related OI-class substratum, and possibly substrata not in OI's structural class that produce observationally-equivalent embedded observers.*

The corollary makes precise the "universality classes are coarser than $\mathcal{G}_{\mathrm{sub}}$" claim: the universality class extends OI's gauge equivalence by potentially including non-OI-class substrata whose embedded-observer triples are observationally equivalent through different dilation mechanisms.

9.3 Many-to-one structure of dilations

A structural feature important for the universality-class framework is the *many-to-one* relationship between substratum-with-observer systems and their embedded observers. Many distinct substratum-with-observer quintuples produce the same embedded-observer triple; the additional data is the unitary realization $(\mathcal{H}_H, \rho_H, U)$. The uniqueness available here is at the level of the Stinespring isometry, and it is worth separating the algebraic content from what it does and does not say about U.

Proposition 9.7a (Minimal Kraus uniqueness). *Let $\mathcal{H}_V$ be finite-dimensional and let*

$$\Phi(\rho) = \sum_{i=1}^{r} A_i \rho A_i^\dagger = \sum_{j=1}^{s} B_j \rho B_j^\dagger$$

with $\{A_i\}_{i \le r}$ and $\{B_j\}_{j \le s}$ each linearly independent. Then $r = s$ and there is a unique unitary $W \in \mathrm{U}(r)$ with

$$B_j = \sum_i W_{ji} A_i.$$

Proof. Vectorize: the map $\rho \mapsto \sum_i A_i \rho A_i^\dagger$ determines, and is determined by, the positive semidefinite operator $M = \sum_i |a_i\rangle\langle a_i|$ where $|a_i\rangle = \mathrm{vec}(A_i)$. Linear independence of the family says the $|a_i\rangle$ are linearly independent, so $r = \operatorname{rank} M = s$ and both families are bases of $\operatorname{ran} M$. Writing X and Y for the matrices with those columns, $XX^\dagger = M = YY^\dagger$ with X, Y of full column rank forces $Y = XW^{\mathsf{T}}$ for a unique unitary W. □

Linear independence is exactly minimality, and it is not removable: without it the family size is not even determined. For any 3×2 isometry M the family

$C_k = \sum_i M_{ki} A_i$ realizes the same channel with three members rather than two, and no unitary relates families of different sizes.

Proposition 9.7b (Minimal pure-reference dilation uniqueness). *Let* $(\mathcal{H}_H^{(1)}, |0_1\rangle\langle 0_1|, U^{(1)})$ *and* $(\mathcal{H}_H^{(2)}, |0_2\rangle\langle 0_2|, U^{(2)})$ *be pure-reference unitary realizations of the same channel* Φ*, and let* $V_a\psi = U^{(a)}(\psi \otimes |0_a\rangle)$ *be the associated Stinespring isometries. If both are minimal —* $\overline{\mathrm{span}}\{(A \otimes \mathbb{1})V_a\psi \,:\, A \in \mathcal{B}(\mathcal{H}_V),\ \psi \in \mathcal{H}_V\} = \mathcal{H}_V \otimes \mathcal{H}_H^{(a)}$ *— then* $\dim \mathcal{H}_H^{(1)} = \dim \mathcal{H}_H^{(2)}$ *and there is a unique unitary* $W : \mathcal{H}_H^{(1)} \to \mathcal{H}_H^{(2)}$ *with*

$$V_2 = (\mathbb{1}_V \otimes W)\, V_1, \qquad \text{equivalently} \qquad U^{(2)}(\psi \otimes |0_2\rangle) = (\mathbb{1}_V \otimes W)\, U^{(1)}(\psi \otimes |0_1\rangle)\ \ \forall \psi.$$

No equality of the full unitaries away from the reference subspaces follows.

Proof. Minimality of a Stinespring isometry is linear independence of its Kraus family, so this is Proposition 9.7a transported through $A_i = (\mathbb{1}_V \otimes \langle i|)V_a$. □

The final clause is not a hedge. The channel constrains $U^{(a)}$ only on $\mathcal{H}_V \otimes \mathbb{C}|0_a\rangle$, and the action elsewhere is free: with $\mathcal{H}_V = \mathcal{H}_H = \mathbb{C}^2$, the realizations $U^{(1)} = \mathrm{CNOT}$ and $U^{(2)} = \mathrm{CNOT}\,(\mathbb{1} \otimes Z)$ produce the same channel, share the same hidden sector, and are both minimal — yet minimality forces $W = \mathbb{1}$, so no unitary conjugation carries one to the other. Any statement of the form $U^{(2)} = (\mathbb{1}_V \otimes W)U^{(1)}(\mathbb{1}_V \otimes W^\dagger)$ is therefore false as an identity of operators.

Remark (the conjugation form). After additionally choosing W so that $W|0_1\rangle = |0_2\rangle$ — an admissible normalization, not a consequence — the relation of Proposition 9.7b can be written as a conjugation on the reference subspace. It still does not extend to an equality of the full operators.

Remark (why [Substratum] Lemma 24.1 can say more). That lemma does assert a conjugation of the dilating unitaries, $\hat{\varphi}' = (\mathbb{1}_V \otimes W_B)\,\hat{\varphi}\,(\mathbb{1}_V \otimes W_B)^{-1}$, and this is consistent with the restriction here rather than in tension with it. Proposition 9.7b fixes a single channel and a single reference preparation, so nothing constrains U off $\mathcal{H}_V \otimes \mathbb{C}|0\rangle$. Lemma 24.1 instead fixes the whole family $\{\Phi_t\}_{t\geq 0}$ generated by one unitary and works on the cyclic boundary sector $\mathcal{C}_B = \overline{\mathrm{span}}\{\hat{\varphi}^{\,t}(\mathcal{C}_V \otimes \mathrm{supp}\,\omega_H)\}$ — the subspace the dynamics reaches from the visible factor. There is no off-reference remainder there by construction: cyclicity is exactly the statement that the reference subspace generates everything, which is what upgrades an isometry statement to an operator statement. The price is that the conclusion holds on $\mathcal{C}_V \otimes \mathcal{C}_B$ and not on a larger hidden sector, which is the deep-sector remainder that lemma isolates as generator (iii). Single-channel uniqueness of the kind stated here cannot substitute for it, and the completeness argument of [Substratum §4] correctly invokes the semigroup form.

These propositions give the universality-class framework's many-to-one structure its precise form: at the level of minimal pure-reference realizations, what the embedded observer determines is the Stinespring *isometry*, up to hidden-sector unitary. That unitary is the algebraic counterpart of OI's $\mathcal{G}_{\mathrm{sub}}$ generator

(i) (hidden-sector relabeling); non-minimality is the counterpart of generator (iii) (deep-sector enlargement); and the extension of the isometry to a full tensor-product unitary is a third freedom with no generator, because it is invisible to the channel.

Corollary 9.8 (Substratum description as gauge). *The embedded-observer triple $(\mathcal{H}_V, \mathcal{A}_V, \Phi)$ does not determine a unique substratum realization. For a minimal pure-reference representative it determines the Stinespring isometry only up to hidden-sector unitary; the extension of that isometry to a full tensor-product unitary is additionally unconstrained off the prepared reference subspace; and non-minimal realizations introduce further enlargement freedom. The universality-class data is the embedded-observer triple.*

So the gauge freedom in the substratum description has at least three pieces — hidden-sector unitary, unitary-extension freedom off the prepared subspace, and non-minimal enlargement — where the earlier formulation named only the first and third.

Scope (relative to the fixed-reference observational interface). The second piece is gauge only because Definition 9.2 always prepares the same ρ_H. The off-reference action of U is invisible to Φ for that reason and no other. If the hidden sector became independently preparable, that action would become observable and would cease to be gauge. The claim is therefore not that the substratum has less structure than the channel, but that the channel, at a fixed reference preparation, does not see it.

This sharpens the structural-realism claim. What is invariant under the universality-class framework's gauge structure is the embedded-observer triple; the substratum description is gauge data relative to the fixed-reference interface. OI's framework treats the equivalence class modulo $\mathcal{G}_{\text{sub}}$ as ontologically real; the universality-class framework treats the embedded-observer triple as ontologically real.

9.4 Relation to OI's gauge group, refined

With the formal definitions in place, the relationship between universality classes and OI's gauge group can be stated more precisely than in §9.2.

OI's $\mathcal{G}_{\text{sub}}$ classifies substratum-with-observer systems within OI's structural class (substrata satisfying A1-A6, with the trace-out construction of [Main]) producing the same embedded observer. Universality classes (Definition 9.4) classify substratum-with-observer systems across all structural classes producing the same embedded observer. The inclusion is strict if there exist non-OI-class substrata producing observationally-equivalent embedded observers; this is the open question of §9.8 below.

The framework's structural-realism stance refines correspondingly. OI's claim is that the equivalence class modulo $\mathcal{G}_{\text{sub}}$ is real; the universality-class claim is that the embedded-observer triple $(\mathcal{H}_V, \mathcal{A}_V, \Phi)$ is real, with the substratum

description (any specific dilation $(\mathcal{H}_H, U)$ producing the triple) being gauge. The two claims agree within OI's structural class but differ on the question of cross-class membership.

9.5 Structure of universality classes

Universality classes have internal structure beyond bare equivalence. Three structural features:

Sub-classes by structural features. Two embedded observers in the same universality class produce the same algebra-channel pair, hence the same observational features (Born rule, unitarity, non-Markovian marginal, commutant gauge-invariance pattern). They may differ on which structural features the substratum producing them respects — OI's substratum satisfies A1-A6, string-theoretic substrata may not. Sub-classes within a universality class can be characterized by the structural conditions the substratum representatives satisfy.

Sub-classes by gauge content. Two observers in the same universality class produce the same gauge-invariance pattern (commutant restriction) but may differ on the specific gauge group. OI's substratum produces $SU(3) \times SU(2) \times U(1)$ uniquely; string-theoretic substrata may produce the same group through compactification choices, but the same universality class might also include substrata producing different gauge groups (other compactifications, non-OI structural-class substrata) provided they share the algebra-channel structure.

Refinement by substratum-symmetry. Per §14.4, the *pattern* of gauge-invariance via commutant transports across frameworks while the *specific* gauge group does not. The universality class containing OI's substratum and the SM-reproducing string compactifications may also contain substrata producing different gauge groups; the SM-reproducing class is a sub-class characterized by the observable target (specifically $SU(3) \times SU(2) \times U(1)$).

These structural features mean universality classes are not flat — they have hierarchical sub-structure. The OI-string relationship in particular involves multiple levels: a coarse universality class characterized by partial-trace observational features; a finer sub-class characterized by SM-reproducing observable target; a still finer sub-class characterized by specific predictive content (Cabibbo, Koide, etc.).

9.6 Observer-admitting and observer-lacking substrata

The framework's universality classes are populated by *observer-admitting* substrata — pairs (S, φ) that admit a partition $V \subset S$ satisfying the framework's structural conditions C1–C4 (coupling, record persistence, sufficient capacity, history readback) [Main §3]. This restriction defines the scope of the framework's discourse and is structurally distinct from the absence of an observer in

any specific substratum. We make the distinction explicit here because it bears on the relationship between OI and other unification programs in §12.

Definition 9.9 (Observer-admitting substratum). *A substratum (S, φ) is observer-admitting if there exists a partition V such that the triple (S, φ, V) satisfies C1–C4. A substratum is observer-lacking if no such partition exists.*

Three structural conditions on φ characterize observer-admission:

- *Coupled dynamics:* φ does not decompose as $\varphi_V \times \varphi_H$ for any candidate partition. Substrata where visible and hidden sector dynamics factorize cannot satisfy C1.
- *Non-equilibrium phase:* φ admits phases where local mixing time is much shorter than overall recurrence time. Substrata in perpetual equilibrium cannot satisfy C2 [Main §4.6].
- *Sufficient hidden capacity:* potential hidden sectors have $|\mathcal{C}_H|$ large enough to store interaction history without overflow. Substrata with insufficient hidden-sector capacity cannot satisfy C3.

These are structural features of the substratum's bijection φ itself, prior to any observer being instantiated. An observer-admitting substratum has the structural capacity for partial-trace observation; an observer-lacking substratum does not.

Proposition 9.10 (Observer-admitting and observer-lacking substrata are structurally distinct). *Observer-admission is determined by structural features of φ, not by whether an observer is "present" or "absent" in any specific substratum. The classes are mutually exclusive partitions of the space of substrata.*

The proposition is a clarification rather than a deep result: it states that the observer-admission criterion operates at the level of the bijection structure, separating substrata into two structurally-distinct classes. An observer-admitting substratum has the structural features that allow observation to occur; an observer-lacking substratum does not. The classes are not "the same substratum with observation added or removed" — they are different substrata characterized by different structural features of φ.

Corollary 9.11 (Universality classes are populated by observer-admitting substrata only). *The universality classes of Definition 9.4 are equivalence classes within the observer-admitting subset. Observer-lacking substrata are outside the framework's universality-class structure.*

This corollary clarifies the framework's scope. Universality classes characterize substratum-with-observer systems by their algebra-channel structure under partial-trace observation; observer-lacking substrata do not produce algebra-channel pairs through partial trace and are therefore not in the universality-class structure. They are mathematical objects (finite sets with bijections) with dynamics, but not with extractable observational features.

The "physics" of observer-lacking substrata. Observer-lacking substrata have φ-dynamics — they evolve, they have cycle structure, they have all the formal features of permutations on finite sets. But they do not produce *physics* in the framework's sense: no quantum mechanics, no observable record, no measurement outcomes. The framework's foundational claim — that physics is the structural features of partial-trace observation — implies that physics is a feature of observer-admitting substrata, not of all substrata. This is consistent with the main paper's "observation occurs" axiom: the axiom is taken to be a structural feature of our universe (which is observer-admitting) rather than a contingent fact about whether someone happens to be looking.

Anthropic implication. The observer-admission distinction has structural-anthropic content. Most arbitrary (S, φ) pairs do not admit observers — random permutations on finite sets typically lack the coupled-dynamics, non-equilibrium-phase, and high-capacity features required for C1–C4. Observer-admitting substrata are a structurally-characterized subset. Our universe is in this subset, structurally; we are observers in an observer-admitting substratum. This is structural anthropic selection rather than contingent anthropic selection: the criterion for "physical universe" is observer-admission, with observer-lacking substrata being abstract mathematical objects rather than candidate physical universes.

The structural-anthropic reading is consistent with but more precise than the standard anthropic position. Standard anthropics says we observe a universe with parameters compatible with observers because we couldn't observe a universe without those parameters. The observer-admission reading says: substrata with observers and substrata without observers are structurally distinct kinds of objects; observer-admitting substrata are the candidate space for "physical universe"; we are in one of them.

9.7 Worked examples

We illustrate the universality-class definition with three examples.

Example 1: OI's class. OI's substratum is a finite deterministic bijection on a $d = 3$ cubic lattice with $K = 6$ internal components. The substratum enters as a unitary realization of the visible-sector CPTP channel (Definition 9.2), with a mixed reference state; the visible-sector algebra is the emergent QFT operator algebra. The universality class containing OI's substratum is the class of substrata producing the same algebra-channel pair — by OI's reconstruction theorem, this includes all $\mathcal{G}_{\text{sub}}$-related substrata (relabelings, alphabet changes, deep-sector enlargements, graph statistical-class equivalents). Whether the class extends beyond $\mathcal{G}_{\text{sub}}$ depends on whether non-OI-class substrata produce the same algebra-channel pair. All members of this class are observer-admitting per §9.6.

Example 2: String-theoretic class. A string-theoretic substratum (e.g., type IIA on a specific Calabi-Yau threefold with specific fluxes) produces a low-

energy effective field theory. The dilation is the full string-theoretic structure (worldsheet plus target space, or matrix-model structure plus large-N limit); the visible-sector algebra is the four-dimensional effective field theory operator algebra. The universality class is the class of substrata producing the same algebra-channel pair through string-theoretic emergence. Observer-admission is a non-trivial constraint: not all string compactifications are observer-admitting, as developed in §7.9.

Example 3: Cross-class candidate. A class candidate is the union: substrata that produce the SM gauge group + three generations + SM-like Yukawa structure through any mechanism (OI's cubic-lattice derivation, string-theoretic compactification, possibly LQG-style structures, causal-set structures, or asymptotic-safety–derived structures with appropriate matter-emergence). This class is characterized by the observable target (the SM at low energies) rather than by the substratum mechanism. All members are observer-admitting (the SM-reproducing constraint forces observer-admission, since reproducing the SM requires producing the observed cosmological structure within which we are observers). Part I (§§3-7) establishes that, at the level of partial-trace observational features, OI and string theory are members of this candidate cross-class; §13.1.4 extends the per-condition analysis to LQG, causal sets, and asymptotic safety, finding that all three fail OI's A1–A6 (so are outside OI's specific class) but may share the partial-trace observational features class with OI and string theory. The question is whether the cross-class is well-defined and what additional features characterize it.

9.8 Open: does the OI-string cross-class exist?

The candidate cross-class of Example 3 is well-defined as a set: substrata producing the SM at low energies. The question is whether it is a *universality class* in the sense defined — whether all members produce equivalent algebra-channel pairs.

Part III §14 establishes partial transport: Born rule, channel-level unitarity, P-indivisibility transport directly; commutant gauge-invariance pattern transports but not the specific gauge group. Specific Yukawa couplings, CP-violating phases, and other detailed predictions have not been demonstrated to transport.

If only the structural features at §14.1-14.3 transport, the OI-string cross-class is universal at the level of these features but not at the level of detailed predictions. The class might be subdivided further: the partial-trace-features class is broader; the SM-reproducing class is narrower; the OI-Cabibbo-and-Koide-predictions class is narrower still.

Establishing definitively whether OI's specific predictions (Cabibbo, Koide, etc.) transport to the string-theoretic class would require either (a) finding a string-theoretic compactification reproducing OI's predictions, (b) showing that any SM-reproducing compactification automatically produces these predictions, or

(c) showing that no string-theoretic compactification can reproduce these predictions exactly. Each of (a)-(c) is a substantial research question.

The provisional conclusion: the OI-string cross-class is well-defined at the level of partial-trace observational features (per §14) and includes specific gauge-group-and-content sub-classes, but the full structure of the cross-class — including whether OI's specific predictions are universal to the SM-reproducing sub-class or specific to OI's representative — is open.

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10. OI's framework as a worked example

This section uses OI's framework to illustrate how universality classes interact with substratum-level structure. The aim is to show that universality classes are not abstract mathematical curiosities but capture the structural content of OI's reconstruction theorem in a more general form.

10.1 OI's reconstruction theorem in universality-class language

OI's reconstruction theorem (Theorem 23 of [Substratum]) states that the empirical/structural inputs uniquely fix, under the theorem's stated conditions, the **local propagating lattice/gauge residue** modulo $\mathcal{G}_{\text{sub}}$. In universality-class language, this is a uniqueness result for the local OI residue characterized by the relevant partial-trace features plus A1-A6; it is not a theorem that one Bell-inclusive full substratum uniquely populates the class.

The reconstruction theorem's content is that the stated inputs select one **local lattice/gauge residue** modulo $\mathcal{G}_{\text{sub}}$ within OI's structural class. H-Bell separately conditions whether a measurement-independent Bell-violating composite can inhabit the same ontology, and neither H-Bell nor Lemma 24.1 makes that Bell-composite structure unique.

10.2 What OI does and does not establish at the universality-class level

OI's reconstruction theorem establishes uniqueness of the **local reconstructed residue** within the OI structural class. It does not establish Bell-inclusive uniqueness even within that class, and it does not establish anything about *other* universality classes. Part I (§§3-7) is explicit on this: the equivalence-class extension to string-theoretic substrata fails because string-theoretic substrata are in different universality classes (substrata violating A2 and A5 are not in OI's class).

The universality-class framework therefore provides a precise statement of OI's reach: OI's reconstruction theorem operates within a specific universality class, and the framework's predictions (Cabibbo, Koide, etc.) are features of OI's class. Whether other universality classes exist that produce the same observable

physics is an open question; if they do, OI's predictions are class-specific rather than substratum-level-universal.

10.3 The structural-realism implication

OI's structural-realism claim (per [Substratum §1.1]) is scoped to the **reconstructed local residue modulo** $\mathcal{G}_{\text{sub}}$. The universality-class framework refines this: what is class-specific is that local residue within OI's class; what is universality-class-level is the partial-trace observational pattern. Bell-inclusive completion data remain an additional underdetermined layer.

The strongest structural-realism claim consistent with both OI's framework and the universality-class framing is: what's real at the universality-class level is the algebra-channel structure of the visible-sector observation; what's class-specific is the substratum representative producing this structure. This is a hierarchical structural realism: features at multiple levels are real, with finer features being more class-specific and coarser features being more universal.

This is consistent with OI's framework but more nuanced. OI's specific predictions (Cabibbo, Koide) are real features of OI's universality class; they are not universal across all classes producing the SM. A different universality class containing the SM at low energies might have different specific predictions or might lack predictive content at the corresponding level.

10.4 Constructing OI's algebra-channel pair

The universality-class formalism of §8.3 / §9.1 requires an explicit algebra-channel pair to be constructive. We now exhibit the triple $(\mathcal{H}_V^{\text{OI}}, \mathcal{A}_V^{\text{OI}}, \Phi^{\text{OI}})$ at the EFT scale, completing Step 1 of the four-step recipe of §13.1.1 and providing the concrete object that the comparison work of §11.2 acts on.

The Hilbert space. OI's substratum is a $d = 3$ cubic lattice with $K = 2d = 6$ internal degrees of freedom per site ([SM §3.2 Theorem 6]). Under the trace-out construction of [Main §3.4], the visible sector is the sublattice accessible to the embedded observer; the hidden sector is the complementary substratum. The visible-sector Hilbert space at the substratum scale is $\ell^2(\mathcal{C}_V)$ where $\mathcal{C}_V$ is the visible sublattice's configuration space. At wavelengths long compared to the lattice spacing $\epsilon = 2\sqrt{\sigma}\, l_p$, $\ell^2(\mathcal{C}_V)$ resolves to the Fock space $\mathcal{H}_V^{\text{OI}}$ of the emergent QFT — the Standard Model with gauge group $SU(3) \times SU(2) \times U(1)$, three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]), and one Higgs doublet ([SM §4]).

The algebra. Let $\mathcal{A}_V^{\text{OI}}$ be the von Neumann algebra of gauge-invariant local observables on $\mathcal{H}_V^{\text{OI}}$:

$$\mathcal{A}_V^{\text{OI}} = \langle \mathcal{O}(x) : \mathcal{O} \text{ a gauge-invariant local SM operator} \rangle''$$

where $\langle \cdot \rangle''$ denotes the bicommutant (von Neumann algebra closure). Generators include:

- *Gauge-field invariants.* The Yang-Mills curvature combinations $\operatorname{tr} F^a_{\mu\nu} F^{a,\mu\nu}$ for each factor of $SU(3) \times SU(2) \times U(1)$, plus topological terms $\operatorname{tr} F\tilde{F}$.
- *Fermion bilinears.* $\bar{\psi}_i \psi_j$, $\bar{\psi}_i \gamma^\mu \psi_j$, $\bar{\psi}_i \gamma^\mu \gamma^5 \psi_j$ for the SM fermion content (six quarks, six leptons, three generations), with appropriate gauge-index contractions.
- *Higgs invariants.* $\phi^\dagger \phi$, $|D_\mu \phi|^2$, $(\phi^\dagger \phi)^2$, and the Yukawa contractions $\bar{\psi}_L \phi \psi_R$ structurally fixed by the cubic-group representation theory of [SM §7].
- *Cubic-lattice-forced higher-dimensional operators.* Wilson coefficients of dimension-5 and dimension-6 operators are not free parameters in OI but are determined by the cubic-lattice substratum geometry; the specific structural relations $\lambda = 1/(\pi\sqrt{2})$ (Cabibbo, [SM §7.1]), $A = \sqrt{2/3}$ (Wolfenstein, [SM §7.1]), and the Koide 2/3 relation are encoded as algebraic identities among Yukawa-sector generators.

The gauge-invariance restriction is a commutant condition: $\mathcal{A}_V^{\mathrm{OI}}$ is the subalgebra of the full local algebra that commutes with the action of $SU(3) \times SU(2) \times U(1)$ on $\mathcal{H}_V^{\mathrm{OI}}$.

The channel. The CPTP channel encoding visible-sector dynamics is generated by the substratum bijection φ traced over the hidden sector. As a unitary realization (Definition 9.2):

$$\Phi^{\mathrm{OI}}(\rho) = \mathrm{Tr}_H[U_\varphi \, (\rho \otimes \rho_H^{\mathrm{OI}}) \, U_\varphi^\dagger]$$

where U_φ is the unitary lift of the bijection φ on $\mathcal{H}_V^{\mathrm{OI}} \otimes \mathcal{H}_H^{\mathrm{OI}}$ ([Substratum §3.1]), and ρ_H^{OI} is the uniform reference state on the hidden sector (the OI-canonical choice from the maximal-entropy invariant-measure lemma of [Main §1.2]). That reference state is *mixed*, so this triple is a unitary realization rather than a Stinespring dilation, and Propositions 9.7a–9.7b do not apply to it directly: obtaining a Stinespring isometry from it requires first purifying ρ_H^{OI}, which enlarges the hidden sector. At the EFT scale, Φ^{OI} encodes:

- *SM RG flow* from the cubic-lattice scale ($1/\alpha_0 = 23.25$ structurally, [SM §6.1]) down to the electroweak scale ($1/\alpha_1 = 59.00$, $1/\alpha_2 = 29.57$, $1/\alpha_3 = 8.47$ at M_Z, [SM §6.3]).
- *Non-perturbative structural features.* Instanton-sum closure consistent with T-invariance and detailed balance.
- *Anomaly structure.* Anomaly cancellation forced by the hypercharge structure derived from cubic-group representation theory ([SM §4.6]).
- *Higher-dimensional operator coefficients* forced by cubic-lattice geometry, including the Yukawa structural relations producing the CKM and PMNS matrices and the fermion mass ratios ([SM §7]).

The pair as universality-class invariant. The algebra-channel pair $(\mathcal{A}_V^{\mathrm{OI}}, \Phi^{\mathrm{OI}})$ is invariant under $\mathcal{G}_{\mathrm{sub}}$: by Proposition 9.5, any $g \in \mathcal{G}_{\mathrm{sub}}$ induces a *-isomorphism on the algebra commuting with the channel up to conjugacy.

The pair is therefore a universality-class invariant of OI's structural class, encoding precisely the observational content the framework licenses.

What the pair captures, and what it does not. The pair captures the gauge structure (the $SU(3) \times SU(2) \times U(1)$ commutant), the renormalizable couplings (gauge couplings at M_Z, Yukawa structure producing CKM/PMNS), and the higher-dimensional operator structure forced by the cubic lattice. It does *not* capture the substratum bijection's specific values (those are gauged away by $\mathcal{G}_{\text{sub}}$), the substratum-level dynamics directly (only its trace-out image), or trans-Planckian corrections where the EFT description breaks down.

The construction uses standard machinery from algebraic quantum field theory: nets of local von Neumann algebras with isotony and microcausality [BrunettiFredenhagen2024], with type-III factor structure [Yngvason2005] inherited from the relativistic-QFT setting; the channel via Stinespring dilation [Stinespring1955]. The framework's contribution is not new operator-algebraic machinery but the identification of the channel's specific structure as the trace-out of a deterministic substratum bijection, with the resulting algebra-channel pair as the universality-class invariant of OI's structural class. See §15.2 for further discussion of the algebra-channel formalism in independent literature.

10.5 Status as Step 1 of §13.1.1's recipe

The construction of §10.4 completes Step 1 of the four-step *-isomorphism recipe of §13.1.1. The remaining three steps remain open and are deferred to follow-up work:

- *Step 2.* Construct a specific string compactification's pair $(\mathcal{H}_V^{\text{string}}, \mathcal{A}_V^{\text{string}}, \Phi^{\text{string}})$ at the same EFT scale. Natural candidates are MSSM-like F-theory or heterotic compactifications producing the exact observed gauge group and chiral content. The algebra is determined by the compactification's particle content and gauge group; the channel encodes perturbative dynamics plus non-perturbative effects (instanton contributions, threshold corrections from heavy compactification modes, RG running from the GUT/string scale to M_Z).
- *Step 3.* Test *-isomorphism $\mathcal{A}_V^{\text{OI}} \cong \mathcal{A}_V^{\text{string}}$. The renormalizable-sector check is plausibly affirmative (both reproduce SM gauge structure and chiral content); the higher-dimensional-operator check is where divergence is expected, since OI's Wilson coefficients are forced by cubic-lattice geometry while string-compactification Wilson coefficients are determined by the compactification's specific topology and flux choices.
- *Step 4.* Test channel conjugacy $\theta \circ \Phi^{\text{OI}} = \Phi^{\text{string}} \circ \theta$ for the *-isomorphism θ (if Step 3 succeeds). Even if the algebras are* -isomorphic, the channels may differ on specific RG flow, anomaly content, or instanton effects, placing the two systems in the same algebra class but different channel class.

Steps 2-4 require detailed comparison with a chosen compactification and are at the boundary of in-chat tractability — likely Layer 2(b) in the framework's

offline-tractability classification (substantial setup, multi-session). The construction here makes the comparison concrete by exhibiting OI's side and reduces the open work in §11.2 from "algebra-channel comparison" to "execute Steps 2-4 against a specified compactification."

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11. The OI-string universality class

Part III §14 establishes that OI and string theory share structural features at the partial-trace observation level. This section examines what that establishes about cross-class membership and what additional work is needed.

11.1 What §14's findings establish

The §14 four-feature audit shows that the Born rule, channel-level unitarity, and P-indivisibility transport directly between OI and string-theoretic emergence mechanisms. The commutant gauge-invariance *pattern* transports but the specific gauge group does not.

In universality-class language: OI's class and the string-theoretic class share the partial-trace observational features. This is necessary for the two classes to be in the same broader universality class, but not sufficient — they would also need to share the algebra-channel structure of the visible-sector observation, including the specific algebra of observables and the specific channel dynamics.

The §14.4 finding is that the specific gauge group transports across frameworks under the observable-target constraint (both produce $SU(3) \times SU(2) \times U(1)$ when reproducing the SM), but through different substratum-symmetry mechanisms. This is consistent with same-universality-class membership at the level of the SM gauge group, but the specific dynamics on the SM gauge group (Yukawa couplings, CP-violating phases, scale of supersymmetry breaking if any) may differ.

11.2 What additional work is needed

§10.4 completes the explicit construction of OI's algebra-channel pair $(\mathcal{H}_V^{\mathrm{OI}}, \mathcal{A}_V^{\mathrm{OI}}, \Phi^{\mathrm{OI}})$ at the EFT scale — Step 1 of the four-step recipe of §13.1.1. To establish definitively whether OI and string theory are in the same universality class, the remaining three steps must be addressed:

Question 1: Do OI's specific predictions transport? OI predicts the Cabibbo angle as $1/(\pi\sqrt{2})$ from cubic-group representation theory. Does any SM-reproducing string compactification reproduce this exact value through its compactification mechanism, or are string-theoretic predictions for the Cabibbo angle compactification-specific (and presumably not exactly $1/(\pi\sqrt{2})$ in general)? If the latter, OI's predictions are not universality-class-level features but OI-specific representative features.

Question 2: Do the algebras agree? OI's emergent QFT has a specific algebra of observables determined by its cubic-lattice structure (constructed in §10.4). String-theoretic compactifications produce algebras determined by the compactification specifics. Are these algebras *-isomorphic for SM-reproducing constructions, or merely structurally similar (sharing partial-trace features but differing on specific algebraic content)?

Question 3: Do the channels agree? Beyond the algebra structure, the channel Φ encodes the dynamics on visible-sector states. Does OI's Φ^{OI} (as constructed in §10.4) agree with the string-theoretic Φ^{string} on identical SM gauge group plus identical particle content, or does it differ on detailed dynamical features (e.g., the specific RG flow, specific anomaly content, specific instanton effects)?

These are substantial technical questions corresponding to Steps 2-4 of the §13.1.1 recipe (construct the comparison pair, check *-isomorphism, check channel conjugacy). Each would require detailed comparison of OI's framework predictions, now made concrete in §10.4, with specific string-theoretic compactifications producing the same observable target. Part III §14.5 establishes the structural-features-of-trace-out level; the universality-class question requires going further to algebra-channel-level comparison against a specified compactification.

11.3 Provisional characterization

Pending the technical work of §11.2, the OI-string relationship at the universality-class level is provisionally:

- **Same partial-trace-features class:** Confirmed by §14.1-14.3. Both frameworks produce Born rule, channel-level unitarity, and non-Markovian marginal observation.
- **Same gauge-group-target class:** Confirmed by both frameworks producing $SU(3) \times SU(2) \times U(1)$ at low energies, though through different substratum-symmetry mechanisms.
- **Same algebra-channel class:** Open. Requires technical work to establish.
- **Same specific-predictions class:** Open. OI's Cabibbo, Koide, and related predictions may or may not transport to string-theoretic compactifications.

The provisional characterization is that OI and string theory are likely in the same partial-trace-features and gauge-group-target classes but possibly in different algebra-channel and specific-predictions classes. Definitive characterization requires the algebra-channel comparison work flagged in §11.2. The conceptual move — taking invariants across substratum descriptions as the proper target of structural-realism commitment — has independent corroboration in the recent philosophy-of-physics literature on common-core realism for QM-string-theory dualities [DawidFranzmann2025]; see §15.3 for the connection.

12. Implications

12.1 Refined structural realism

The universality-class framework refines OI's structural-realism stance. OI's claim is that what's real is the equivalence class modulo $\mathcal{G}_{\text{sub}}$; the universality-class refinement is that there is structure at multiple levels: partial-trace observational features (most universal, shared across structural classes), specific gauge group (intermediate, depends on substratum-symmetry but realizable by multiple substrata), specific predictions (most class-specific, features of the specific universality-class representative).

What is "real" at each level differs. At the most universal level, the structural features of partial-trace observation are real — Born rule, unitarity, non-Markovian marginal are features of any embedded-observer system, not specific to OI or string theory. At the gauge-group level, $SU(3) \times SU(2) \times U(1)$ is real — features any SM-reproducing universality class produces. At the specific-predictions level, OI's Cabibbo angle $1/(\pi\sqrt{2})$ is real *for OI's universality class* but not necessarily for other classes.

This is a hierarchical structural realism: realism at multiple levels, with class-specificity decreasing as level becomes more universal. It is consistent with OI's structural-realism stance while extending it to characterize what is shared with other unification programs and what is class-specific.

12.2 Methodology of fundamental physics

The universality-class framework suggests a methodological reframing of fundamental physics. Rather than asking "what is the substratum?" — which assumes a unique answer that committing to a specific framework presupposes — we can ask "what universality class are we in?" — which allows for multiple substratum-level descriptions consistent with the same universality class.

This is methodologically more agnostic about substratum specifics but more committed about substratum-class membership. It treats fundamental physics as a research program of identifying the universality class of our universe through observational features, with substratum descriptions being framework-specific tools for representing the class rather than competing claims about ultimate reality.

The methodological move connects to operationalist and empiricist positions in foundations of physics, but with structural-realism preserved at the universality-class level: what's real is the universality class, not just the operational data, and not the specific substratum representative.

12.3 Connection to OI's main paper

OI's main paper takes "observation occurs" as the foundational axiom. The universality-class framework formalizes this axiom's content: observation occurs, and the structure of observation produces universality classes characterized by partial-trace observational features. The features that any embedded-observer system must respect (Born rule, unitarity, non-Markovian marginal, commutant gauge-invariance pattern) are forced by the structure of observation, not by any specific substratum.

This is consistent with OI's main paper's foundational stance but extends it. Where the main paper identifies QM with embedded observation — "observation occurs" plus structural conditions — the universality-class framework derives partial-trace observational features from "observation occurs" alone. The structural conditions A1-A6 are then class-specific refinements that yield OI's specific predictions; without them, observation still produces the universal features, but specific predictions remain framework-dependent.

This places OI's main paper in a broader context. The "observation occurs" axiom has implications beyond OI's framework — it implies the partial-trace observational features in any framework respecting the axiom. OI's specific predictions are features of OI's specific universality-class representative, with the universal features being shared with any other framework respecting the observation axiom.

12.4 Predictions stratified by hierarchy level

The framework's predictions live at distinct levels of the hierarchy developed in §2 and characterized through Parts I–III. Empirical confirmation or refutation has different meaning depending on which level a prediction belongs to: a partial-trace-level prediction confirmed supports any framework with embedded observers; a representative-specific prediction confirmed supports the specific substratum representative. This section consolidates the framework's empirical content, stratified by level, with the empirical status as of this writing.

Level 1: Partial-trace observational features. Universal across frameworks producing observation through partial-trace structure on quantum states. Predictions of any embedded-observer framework respecting the structural features identified in §14.

Feature	Empirical status	Reference
Born rule (probability $\|\langle a\|\rho\rangle\|^2$)	Confirmed at 10^{-3}–10^{-4}	[Sinha2010, Söllner2012, Kanthak2025]
Channel-level unitarity (Stinespring dilation)	Standard QM, theorem-level	[Stinespring1955, Barandes2026]

Feature	Empirical status	Reference
Non-Markovian marginal (P-indivisibility)	Confirmed in colloidal, micromechanical, metrology	[Mehl2012, Groeblacher2015, QEC-NonMarkovian2025]
Holographic bound ($S \leq A/4$)	GW250114 confirms the classical area theorem; the entropy bound itself is untested directly	[GW250114]
Bekenstein-Hawking 1/4 coefficient	Conditional on H-slope with the horizon and frame conditions; no direct empirical test (GW250114 constrains the area theorem)	[GR §2, §8.5, GW250114]

Level 1 confirmations support OI but do not distinguish OI from alternative frameworks producing the same partial-trace structure (string theory, holographic models, hidden-degree-of-freedom open quantum systems). All such confirmations are joint confirmations of the universal level.

Level 2: Gauge-group-target features. Shared by any framework producing SM gauge structure. OI shares this level with SM-reproducing string compactifications and any other program reaching the SM target.

Feature	Empirical status	Reference
$SU(3) \times SU(2) \times U(1)$ gauge structure	Confirmed (full SM)	[PDG]
Three chiral generations	Confirmed	[PDG]
Anomaly cancellation across generations	Confirmed	[PDG]
Renormalizable Yang-Mills + Higgs sector	Confirmed	[PDG]

Level 2 confirmations are confirmations of the gauge-group-target level but do not distinguish OI from other SM-reproducing frameworks. They establish shared structure without identifying the specific universality class.

Level 3: OI universality class. Predictions that follow from OI's algebra-channel pair (§10.4) and would hold for any substratum in OI's universality class. Other substrata producing the same algebra-channel pair would share these predictions.

Feature	Empirical status	Reference
Cubic-lattice-forced higher-dim operators	Indirect (via specific predictions below)	[SM §3.1]
Type II RVM dynamical dark energy	running-vacuum *form* favoured over plain ΛCDM; $\nu \approx 0$ magnitude consistent with Bertini (2025, itself consistent with zero), in mild tension with DESI DR2's evolving-DE preference	[Bertini2025, DESI_DR2_2025, GR §7]
MOND scale $a_0 = cH/6$	Tully-Fisher relation; ongoing tests	[GR §7.3]
Holographic $A/4$ at cosmological horizon	Structural (theory-level transport; no direct test)	[GR §5]
No GUT-mechanism proton decay ($\tau_p \sim 10^{45}$ yr)	Super-K bounds at $\tau_p > 2.4 \times 10^{34}$ yr already exclude minimal SU(5); favors no-GUT class	[SuperK2020_proton, SM §6.7, §8.7]
No GUT-mechanism magnetic monopoles	GUT-scale monopoles undetected cosmologically; consistent with OI without inflation invocation	[MoEDAL2024, SM §6.7]
No GUT-chain cosmic strings	NANOGrav 15-year data disfavors stable cosmic-string interpretation	[NANOGravStrings2023, SM §6.7]
Non-unification of $\alpha_3, \alpha_2, \alpha_1$ below M_{Pl}	Pure SM running confirmed; SUSY-GUT scenarios under LHC pressure ($M_{\mathrm{SUSY}} \gtrsim 3$ TeV with split-SUSY tuning)	[LHC_SUSY_2025, SM §6.4, §6.7]

Level 3 confirmations are confirmations of OI's universality class but not of OI's specific representative. A substratum in the same universality class through different mechanism (e.g., a string compactification realizing the same algebra-channel pair) would share these predictions. Note that the no-GUT predictions (proton decay, monopoles, cosmic strings) are shared with the broader "SM

all the way to Planck" universality class containing OI; what distinguishes OI structurally is the *reason* the framework belongs to that class — the cubic-commutant gauge-emergence mechanism (per [SM §6.7]).

Level 4: OI representative-specific. Predictions specific to OI's substratum representative. Confirmation supports OI's specific substratum; refutation challenges OI but not necessarily the broader universality class.

Feature	Empirical status	Reference
Cabibbo angle $\sin\theta_C = 1/(\pi\sqrt{2}) \approx 0.2251$	Confirmed at 0.04σ	[SM §7.2]
Koide ratio $Q = 2/3$	Confirmed at 0.02%	[SM §7.3]
JUNO $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$	Retrodictive match at 0.07σ; exact value gives a pre-registered forward test at JUNO design precision	[Juno]
Higgs structural ratio $A = \sqrt{2/3}$	Consistent with current data	[SM §7.4]
Higgs quartic $\lambda(M_{\rm Pl}) = 0$ (composite-scalar prediction)	Confirmed at 0.6σ	[SM §7.4]
22 SM observables, classified (seven parameter-free structural on the empirically selected cubic substratum)	Sharpest entries sub-percent; classes per [SM §7.6]	[SM §7.5]
Majorana neutrinos, normal ordering	Awaiting LEGEND-1000 sensitivity	[SM §8.4, LEGEND2024]
$\lvert\nu\rvert \sim 10^{-32}$ in Type II RVM	Consistent with DESI DR2 $\nu \approx 0$ at 2σ	[GR §7, Bertini2025]
Universal $1/\alpha_0 = 23.25$ shared origin of three gauge couplings	Retrodictive match at few-percent (chain includes the empirical threshold δ_0; SM §6)	[SM §6.1, §6.7]

Level 4 confirmations are confirmations of OI's specific representative — the strongest support the framework can receive. The framework's predictions are all consistent with current data; the awaiting tests (Majorana confirmation at LEGEND-1000, JUNO measurement, DESI Year 5 ν tightening, Hyper-Kamiokande proton-decay bounds) will discriminate further.

Empirical strategy. The stratification clarifies what each test does. A Born rule test at 10^{-5} precision confirms Level 1 (any embedded-observer framework respecting the partial-trace structure). A LEGEND-1000 Majorana confirmation supports Level 4 (OI's specific neutrino-sector prediction) but also supports broader frameworks predicting Majorana (e.g., type-I seesaw extensions, swampland-derived predictions per §7.5.1). A precise JUNO measurement of $\sin^2\theta_{12}$ tests Level 4 directly: confirmation at OI's value $1/3 - 1/(4\pi^2)$ is OI-specific. A DESI Year 5 measurement of ν tests Level 3: confirmation of small running supports OI's universality class without necessarily distinguishing OI from other Type II RVM frameworks. Hyper-Kamiokande proton-decay sensitivity at $\tau_p \sim 10^{35}$ years tests Level 3: non-observation strengthens the broader no-GUT class containing OI; observation at this level would falsify OI.

The most discriminating tests for OI's specific predictions live at Level 4. The strongest confirmations of the framework's foundational stance live at Level 1. The intermediate levels — gauge-group target and universality class — are where shared content with other frameworks lives, with confirmations supporting structural features that any framework reaching the same level shares.

This stratification refines the §12.1 structural-realism position: *what is real is invariant structure*, with realist commitment appropriately calibrated to the level of structure. Level 1 confirmations support universal structural commitments; Level 4 confirmations support specific-representative commitments. The framework's empirical content lives across all four levels, with each level's confirmations carrying different weight for different aspects of the framework's claims.

12.5 The no-GUT structural prediction

The framework predicts that no grand-unified gauge structure exists at any energy scale. This prediction has three layers, established jointly by [SM §4.6], [SM §6.4], and [SM §6.1] and brought together in [SM §6.7]:

(A) *Group-theoretic.* The SM gauge group emerges directly from cubic-commutant decomposition; no intermediate large gauge group exists at any scale.

(B) *Coupling-level.* Gauge couplings do not unify. They are split at the matching scale $\mu = M_{\mathrm{Pl}}$ by C_2-dependent thresholds (induced-gauge-theory structure), with pure SM running below M_{Pl} continuing the splitting.

(C) *Mechanistic.* OI's gauge couplings emerge from the fermion determinant, with no fundamental gauge action and no moduli connecting the predicted values to other values.

This prediction is structurally distinct from string-theoretic frameworks, which traverse intermediate large gauge groups (heterotic E_8 chains, F-theory SU(5), type IIA Pati-Salam) en route to the SM. The OI-distinguishing content is layer (C) plus the universal $1/\alpha_0 = 23.25$ value (ambient; native for hypercharge under H-observer-bundle and H-Y-vertex, [SM §6.5]): the three SM gauge couplings

emerge from a *single* substratum quantity, not from independent compactification moduli.

The no-GUT prediction's empirical consequences are catalogued at Level 3 in §12.4: proton lifetime $\tau_p \sim 10^{45}$ years, no GUT-mechanism monopoles, no GUT-chain cosmic strings. None of these uniquely identify OI; they identify the broader "SM all the way to Planck" universality class to which OI belongs. The unique-to-OI content is the structural reason the framework belongs to this class (the cubic-commutant gauge-emergence mechanism plus the induced-gauge-theory structure forcing the universal $1/\alpha_0$).

Empirical-lean assessment. The current empirical state moderately favors the no-GUT universality class over standard GUT scenarios across multiple discriminators. Super-Kamiokande proton-decay bounds [SuperK2020_proton, SuperK2024_eta] at $\tau_p(p \to e^+\pi^0) > 2.4 \times 10^{34}$ years and $\tau_p(p \to \nu K^+) > 5.9 \times 10^{33}$ years already exclude minimal $SU(5)$ GUT and constrain the broader supersymmetric SU(5) parameter space [SUSY_GUT_Bourilkov2015]. LHC searches have pushed natural-SUSY parameter space to $M_{\rm SUSY} \gtrsim 3$ TeV [LHC_SUSY_2025], with the surviving SUSY-GUT scenarios requiring split-SUSY-style tuning that sacrifices the original hierarchy-problem motivation. NANOGrav 15-year pulsar timing data [NANOGravStrings2023] disfavors stable cosmic-string interpretations of the observed stochastic gravitational-wave background, with the signal preferentially explained by supermassive black-hole binary populations. Cosmological searches have found no GUT-scale magnetic monopoles, consistent with OI's prediction without requiring inflation as a dilution mechanism (inflation may still be motivated by horizon and flatness considerations, but the monopole problem is structurally absent in OI).

No single observation sharply distinguishes OI from all GUT-class scenarios — split-SUSY and topologically-suppressed F-theory models retain partial viability against current data. But the cumulative pattern across discriminators is a real empirical lean toward the no-GUT class containing OI, not merely consistency. The decisive future tests are Hyper-Kamiokande's projected $\tau_p \sim 10^{35}$ year sensitivity for $p \to e^+\pi^0$ [HyperK_TDR2018] and DUNE's $\sim 6.5 \times 10^{34}$ year sensitivity for $p \to K^+\nu$, plus continued LHC SUSY exclusion. Both will further sharpen the empirical position over the next decade.

Connection to the unified naturalness pattern. The no-GUT prediction is an instance of a broader pattern in the framework: substratum-level structural symmetries dissolve naturalness problems by forcing emergent parameter values that would otherwise be free. The instances catalogued in [SM §8.8] are: (i) *strong CP* — an open instance: substratum T-invariance narrows θ to $\{0, \pi\}$ and gives reciprocal transition counts but does not fix $\bar{\theta}$ ([SM §5], H-top and H-det); (ii) *hierarchy problem* — singlet-taste chiral symmetry provides composite-Higgs-style protection of the Higgs mass against cutoff-scale corrections ([SM §8.8], partial resolution); (iii) *cosmological constant* — trace-out's information

compression ratio dissolves the 10^{122} discrepancy ([GR §6]). The no-GUT structural prediction extends this pattern: rather than requiring a GUT mechanism to explain gauge-coupling unification (and inheriting all the GUT scenarios' associated empirical commitments), OI's direct cubic-commutant gauge emergence sidesteps the unification puzzle structurally. Three of the four — hierarchy, cosmological constant, gauge structure — are dissolved by the same kind of substratum-level structural mechanism, and strong CP is narrowed but not dissolved by it, giving the framework a unified treatment of what standard physics treats as four independent open problems requiring four independent solutions.

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13. Open questions and limits

13.1 Technical questions

The main technical open questions:

Q1. Are the algebra-channel structures of OI and string theory *-isomorphic at the SM-reproducing target? (§11.2 Question 2.)

Q2. Do OI's specific predictions (Cabibbo, Koide, etc.) hold across all substrata in the same SM-reproducing universality class, or are they OI-specific features? (§11.2 Question 1.)

Q3. What is the precise definition of the "partial-trace observational features" class? Are the four features identified in §14 necessary and sufficient to characterize this class, or are there additional features?

Q4. Are there universality classes other than OI's and string theory's that produce the SM at low energies? (LQG-derived classes, causal-set-derived classes, asymptotic-safety classes, etc.) If so, what features characterize them?

13.1.1 Q1 developed: what *-isomorphism would entail

To establish whether OI's and a specific string-theoretic compactification's algebra-channel pairs are -*isomorphic at the SM-reproducing target, four steps would be required: (1) construct OI's algebra-channel pair at the EFT scale (gauge-fields + fermion-bilinears + Higgs-operators algebra restricted to commutant of $SU(3)\times SU(2)\times U(1)$, with channel from the substratum's bijection traced over the hidden sector); (2) construct the compactification's pair at the same scale (algebra from particle content + gauge group; channel from perturbative + non-perturbative effective dynamics including instantons, threshold corrections, RG running); (3) check* -isomorphism (whether algebraic relations agree on gauge-coupling structure, Yukawa-coupling structure, anomaly content, topological terms — both renormalizable and higher-dimensional operators); (4) check channel conjugacy ($\theta \circ \Phi^{(1)} = \Phi^{(2)} \circ \theta$ for the *-isomorphism θ). **Status:** Step 1 is executed in §10.4 (construction of $(\mathcal{A}_V^{\mathrm{OI}}, \Phi^{\mathrm{OI}})$); Steps 2-4 remain open and require a specified compactification.

Three possible outcomes: (a) full isomorphism + conjugacy — same universality class, OI's specific predictions transport to the compactification; (b) algebras isomorphic, channels differ — different classes despite leading-order SM agreement, predictions class-specific; (c) algebras not isomorphic — different higher-dimensional structure, the most likely outcome based on the landscape's UV-completion multiplicity. The expected outcome is (b) or (c): different SM-reproducing compactifications generically produce different higher-dimensional operators with different Wilson coefficients, so OI's specific predictions are likely class-specific rather than universal across the SM-reproducing landscape — consistent with the structural-realism stance of §12.1.

13.1.1.1 Converged comparison protocol A specific protocol for executing Steps 2-4 against a chosen compactification has been worked out as preparation for offline computational follow-up. The protocol is:

Compactification choice: heterotic line-bundle Calabi-Yau in the Anderson-Gray-Lukas-Palti class. Specifically, the Schoen-quotient model $X = \tilde{X}/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ with five-line-bundle SM model [Anderson-Gray-Lukas-Palti 2011-2014]. This compactification was selected after considering three candidates (heterotic line-bundle on Schoen quotient; type IIA Pati-Salam on intersecting D6-branes; F-theory $SU(5)$ with hypercharge flux) on multiple criteria: 4D EFT characterization completeness (heterotic AGLP has multi-loop Yukawa structure characterized in the literature); three-generation match (all three candidates produce three generations, but through structurally distinct mechanisms — see below); pure SM gauge group at IR (all three reach SM after appropriate breaking); bundle-cohomology computational tractability; phenomenological literature density (heterotic AGLP has the most extensive literature). The Schoen-$\mathbb{Z}_3 \times \mathbb{Z}_3$ model dominates on most criteria and was selected over its alternatives.

Comparison scale: below the Stueckelberg $U(1)$ mass (~10^{16} GeV). Heterotic line-bundle compactifications produce an extra Stueckelberg-massive $U(1)$ via the Green-Schwarz mechanism (anomaly cancellation in heterotic with line bundles); typical Stueckelberg masses in AGLP models are $\sim 10^{16}$ GeV, set by the compactification scale. Below this mass, the heterotic effective theory has only the SM gauge group, matching OI's structure at the IR. The comparison must therefore be conducted at scales below the Stueckelberg mass to ensure both sides have matching gauge structure. This restricts the universality-class claim to the IR EFT (scales $M_Z \lesssim \mu \lesssim 10^{16}$ GeV) but does not prevent the comparison from proceeding.

Comparison test: Yukawa textures. Both frameworks reach three generations by structurally distinct mechanisms (heterotic uses bundle-cohomology index of the relevant cohomology groups on the CY, an entirely topological invariant; OI uses the cubic group's T_1 irrep with multiplicity 3 from the spatial nearest-neighbor structure). Three generations is therefore not a discriminator; both frameworks pass this test by different routes. The natural quantitative

test is whether the *Yukawa textures* arising from the cubic-group reduction in OI ([SM §7]) match the textures arising from bundle-cohomology computations on the Schoen-$\mathbb{Z}_3 \times \mathbb{Z}_3$ heterotic model. OI's Yukawa textures are characterized in [SM §§7.1-7.3] (CKM matrix, charged-lepton masses, PMNS angles); the heterotic Yukawa textures are characterized in [Anderson-Gray-Lukas-Palti 2013-2014]. The comparison reduces to: do these textures agree at the scale below the Stueckelberg mass?

Verdict criteria. The comparison produces one of three verdicts: (i) Yukawa textures agree at the comparison scale (suggesting OI and Schoen-$\mathbb{Z}_3 \times \mathbb{Z}_3$ are in the same algebra-channel-pair universality class at the IR EFT level); (ii) textures disagree on dimension-4 (renormalizable) operators (frameworks not in same class even at IR); (iii) textures agree on renormalizable operators but disagree on higher-dimensional Wilson coefficients (frameworks share leading-order SM but differ on subleading structure — the expected outcome per §13.1.1's three-outcome decomposition).

A subtle point about scale-choice. The comparison-scale specification matters for the verdict. Below the Stueckelberg mass, both frameworks have just SM, and the comparison tests Yukawa textures. Between the Stueckelberg mass and M_{Pl}, the heterotic theory has an extra $U(1)$ + GUT-related structure that OI does not have; the algebras differ at this scale. A comparison at the *higher* scale would yield a *different-classes* verdict not because of fundamental disagreement but because of the integrating-out structure. The honest comparison protocol must specify the scale, and the IR EFT scale (below Stueckelberg mass) is the natural choice for testing whether OI and the heterotic compactification can coexist in the same universality class at all.

Status of execution. The protocol is fully specified at the structural level. Executing it requires: (i) extracting OI's Yukawa-texture predictions in a form directly comparable to bundle-cohomology output; (ii) extracting AGLP's Yukawa-texture predictions for the Schoen-$\mathbb{Z}_3 \times \mathbb{Z}_3$ model in matching form; (iii) running the *-isomorphism check at the IR EFT level; (iv) running the channel-conjugacy check on the dilation structures (substratum-to-EFT for OI; heterotic-UV-with-Stueckelberg-integrated-out for the compactification). Steps (i) and (ii) are literature-extraction work; (iii) and (iv) are the operator-algebra computations. All four steps are deferred to offline computational follow-up.

Connection to the no-GUT structural finding. The Stueckelberg $U(1)$ mass at $\sim 10^{16}$ GeV is the heterotic compactification's natural intermediate scale, set by the compactification structure. This is the string-theoretic analog of the seesaw scale (per Q8 in §13.3). The comparison scale's choice — at or below this Stueckelberg mass — implicitly settles the comparison's relationship to Q8: at the IR EFT level, OI and heterotic may share universality class even though OI lacks a structural intermediate scale (per Q10's negative resolution in §13.3). Universality-class matching at the IR is consistent with structurally-distinct intermediate-scale physics, with the structural difference being absorbed into the channel rather than being a barrier to algebra-isomorphism. This re-

fines the §13.1.1 expected-outcome decomposition: outcome (b) is the working expectation (algebras agree at IR EFT; channels differ at intermediate scales).

13.1.2 Q2 developed: criteria for prediction transport

Q2 asks whether OI's specific predictions are universal at the universality-class level or specific to OI's representative within the class. Q1's outcome decomposition implicitly resolves much of Q2 — under outcome (a), predictions transport; under (b) and (c), they do not. Direct treatment refines the picture by stratifying predictions across the universality-class hierarchy of §9.5: predictions at the *partial-trace-features level* (Born rule, channel positivity, non-Markovianity index, commutant pattern) transport trivially; predictions at the *gauge-group level* (anomaly conditions, no flavor-changing neutral currents at tree level) transport across all members sharing the gauge group; predictions at the *algebra-channel level* (Yukawa values, CKM structure, RG trajectories) transport only under outcome (a)'s isomorphism + conjugacy; predictions at the *substratum level* (combinatorial counts on the cubic lattice, specific spectral properties) do not transport in general.

OI's specific predictions sit at varying levels: the Cabibbo angle $\theta_C = 1/(\pi\sqrt{2})$ from cubic-group representation theory is substratum-specific; the Koide ratio $Q = 2/3$ is intermediate (gauge-group structure $\times$ cubic-group multiplicity, likely substratum-specific); the Bekenstein-Hawking $A/4$ coefficient is more universal — string theory derives the same coefficient through Strominger-Vafa's microscopic D-brane count, suggesting common universality-class membership or substratum-independent thermodynamic forcing. The general answer: OI's specific predictions are mostly substratum-level features of OI's specific universality-class representative, with exceptions that follow from gauge-group structure or partial-trace observational features. The practical upshot: validating OI's predictions supports OI's specific representative; falsifying them would refute that representative without necessarily refuting the broader universality-class framework.

13.1.3 Q3 developed: necessary and sufficient features for the partial-trace observational class

Q3 asks whether the four features of §14 (Born rule, channel-level unitarity, P-indivisibility, commutant gauge-invariance pattern) are necessary and sufficient to characterize the partial-trace observational features class. *Necessary:* yes, modulo qualifications. The first two follow directly from the embedded-observer triple definition (Born rule from partial-trace structure; channel-level unitarity from Stinespring's theorem). P-indivisibility is necessary under hidden-sector entanglement; commutant gauge-invariance under substratum symmetry. *Sufficient:* no. Two embedded-observer triples sharing all four features can be observationally inequivalent by differing on specific algebraic relations (Yukawa couplings, anomaly content, central extensions), specific channel dynamics (RG flow, threshold corrections, non-perturbative effects), or Hilbert-space struc-

ture (multiplicity of irreducible representations, Type classification). The four features characterize a *meta-class* containing many universality classes as subclasses — consistent with §9.5's hierarchical structure.

Refinement candidates that would tighten the characterization: (α) information-theoretic structure of the marginal (Holevo bound, mutual-information capacity); (β) spectral properties of the channel (eigenvalues, decay times, ergodic structure); (γ) multi-time correlation structure (three-time, four-time non-Markovian features beyond P-indivisibility). These would add necessary conditions but are not expected to be jointly sufficient: Definition 9.3's *-isomorphism criterion depends on specific algebraic content that no finite list of generic features can fully capture. Provisional resolution: the four features are necessary at the partial-trace level (under stated qualifications) but not sufficient to determine universality classes uniquely; full identification requires the algebra-channel-level work of §13.1.1 and §13.1.2.

13.1.4 Q4 developed: other universality classes that may produce the SM

Q4 asks whether the SM-reproducing universality class includes substrata other than OI's and string theory's. Principal candidates are loop quantum gravity (LQG)–derived structures, causal-set–derived structures, and asymptotic-safety–derived structures. The per-condition A1–A6 analysis applied to these programs in the manner of §6 yields preliminary structural conclusions. We give compact treatments of the three principal candidates; the analysis is more telegraphic than the string-theoretic case in §§3–7, since the goal is to characterize structural compatibility at the level of OI's substratum conditions, not to reproduce a full per-program audit.

Loop Quantum Gravity. LQG's substratum [Rovelli2004; Ashtekar-Lewandowski2004] is the space of spin networks — graphs labeled by SU(2) representations and intertwiners — modulo spatial diffeomorphisms. Dynamics is defined either by Thiemann's Hamiltonian constraint on the kinematical Hilbert space or by spin-foam amplitudes summing over histories. Per-condition: A1 (finiteness) **fails** in the full theory — even at fixed graph, the Hilbert space is infinite-dimensional through unbounded spin labels, and the sum-over-graphs structure of the full theory makes $|S| = \infty$ generically; finite-graph and bounded-spin variants admit a finite reading but are not standard LQG. A2 (determinism / bijection) **fails** — the dynamics is quantum, with spin-foam amplitudes constructed as Feynman-like sums rather than deterministic bijections on configuration space. A3 (bounded coupling degree) holds at fixed graph valence (typically ≤ 4 in EPRL spin foams) but the bounded-coupling reading depends on the graph being fixed, which is in tension with sum-over-graphs. A4 (center independence) **holds** — diffeomorphism invariance is built in. A5 (linearity) **holds** at the operator level (the Hamiltonian constraint is a linear operator on Hilbert space) but is complicated by the constraint structure: solutions to the constraint are highly nonlinear in the spin-network labels even

when the constraint operator itself is linear. A6 (background independence) **holds** — this is LQG's defining feature. Net: A1 and A2 fail; A4 and A6 hold robustly; A3 and A5 are reframing-dependent. The structural pattern of failure is similar to string theory's matrix-model formulation in §6, with the key structural mismatches at A1 and A2 being deeper than the discretization-bridge issue of §6.1 — LQG's quantum dynamics and infinite Hilbert spaces are central to the program rather than artifacts of a particular formulation.

Causal sets. The causal-set program [BombelliLeeMeyerSorkin1987; Rideout-Sorkin1999; Surya2019] takes the substratum to be a locally finite partially-ordered set, with dynamics typically modeled by Classical Sequential Growth (CSG) — a stochastic generative process that adds events according to a probability measure on histories. Per-condition: A1 (finiteness) **holds at any finite stage** (finite causal sets are genuinely finite) but **fails** in the limit of unbounded growth. A2 (determinism / bijection) **fails** — CSG is explicitly stochastic, with each event-addition governed by a probability rather than a deterministic update rule. This is the most direct structural divergence from OI's substratum: where OI takes the substratum dynamics to be deterministic with stochasticity emerging through partial trace, causal sets take the dynamics to be intrinsically stochastic at the substratum level. A3 (bounded coupling) holds: each event has a finite number of immediate predecessors and successors. A4 (center independence) **holds** — the formalism is intrinsically relational. A5 (linearity) does not translate cleanly — CSG is described by transition probabilities rather than a wave equation, so OI's linearity-of-wave-equation framing has no direct analog. A6 (background independence) **holds**. Net: A2 fails outright, A1 fails in the limit, A5 lacks a clean translation. Causal sets and OI agree on relational structure (A4, A6) but diverge sharply at the determinism axis (A2). The structural-realist reading: OI's substratum and the causal-set substratum represent different commitments at the foundational level, with OI placing the stochasticity at the partial-trace observational layer while causal sets place it at the substratum layer.

Asymptotic safety. The asymptotic-safety program [Weinberg1979; Reuter1998; Eichhorn2018] takes gravity to be defined by a non-Gaussian UV fixed point of the renormalization group, with the metric as the dynamical variable in a continuum quantum field theory framework. Per-condition: A1 (finiteness) **fails** — continuum QFT. A2 (determinism / bijection) **fails** — quantum, path integral. A3 (bounded coupling) reframing-dependent in the same way as for matrix models; the natural reading at the spacetime level has unbounded coupling. A4 (center independence) **holds** — general covariance. A5 (linearity) **fails** — Einstein-Hilbert action plus higher-derivative terms is nonlinear in the metric, with the nonlinearity essential to producing gravitational dynamics. A6 (background independence) **holds**, modulo standard subtleties about expanding around fixed backgrounds. Net: similar pattern to string-theoretic matrix models — A2, A5, A1 fail; A4 and A6 hold; A3 reframing-dependent. The structural pattern resembles string theory closely.

Cross-program pattern. All three candidate programs fail OI's A2 (determinism / bijection) for the same structural reason — they are quantum or stochastic theories at the substratum level, with substratum dynamics defined by path integrals or probability measures rather than by deterministic bijections. All three fail A1 (finiteness) in their standard formulation, though variants admit finite readings. A4 and A6 hold robustly across all three. A3 and A5 vary by program. Net: none of the three candidates are in OI's specific universality class as that class is delimited by A1–A6. The deeper reading is that determinism (A2) is the most discriminating structural condition — it cleanly separates OI's class from all four candidates considered (string theory in §6 plus LQG, causal sets, asymptotic safety here). This is consistent with the framework's foundational commitment per [Substratum §1.1]: the determinism of φ is what allows the trace-out to produce the reduced observable law — universally admitting the quantum representation, explicitly non-unique — with stochasticity emerging from observer-ignorance rather than substratum-intrinsic randomness.

Partial-trace-features class membership. Although none of the candidate programs are in OI's specific universality class (per A1–A6 failure), all three may share the partial-trace observational features class. Each program is intended to reproduce SM-compatible low-energy physics, which forces Born-rule structure at the observational level (the theories are quantum); channel-level unitarity follows from Stinespring's theorem applied to any embedded-observer reading; P-indivisibility is more program-specific (LQG's spin-foam dynamics and causal-set CSG dynamics may or may not produce non-Markovian marginal observation in the OI sense); commutant gauge-invariance pattern depends on the gauge structure of each program's emergent QFT. For each candidate, partial-trace-class membership is open and would require the algebra-channel comparison work flagged in §13.1.1 for the OI–string case. The pattern of expected outcomes mirrors §13.1.1's three-outcome decomposition: most likely the algebras differ at higher-dimensional operator structure even where partial-trace features agree, placing each candidate in the partial-trace-features meta-class but in a different sub-class within it.

Implications for §11 and Limit 1. The Q4 development extends the §11 OI-string class analysis from one comparison program to four. §11.3's provisional characterization of the OI-string relationship as "same partial-trace-features class, possibly different algebra-channel class" extends to the OI-LQG, OI-causal-set, and OI-asymptotic-safety relationships under the same provisional reading. Limit 1's "one concrete data point" claim is updated by §13.1.4 to a partial census across four programs (per §13.3 update below). The natural research direction left open is the per-program partial-trace-feature audit: for each candidate, does the program's emergence machinery produce all four features of §14, and do the algebras agree at SM-reproducing target?

13.2 Conceptual questions

Q5. Is the universality-class framework the right level of abstraction, or is it too coarse to capture features relevant for fundamental physics? (E.g., quantum gravity might depend on substratum-level features that universality classes wash out.)

Q6. Does the universality-class framing weaken or strengthen OI's predictive content? On the one hand, OI's predictions are class-specific rather than universal, which weakens their status. On the other hand, the framework establishes that class membership is what's real, which strengthens the claim that finding the right class is the right research target.

Q7. What is the relationship between universality classes of embedded observers and other equivalence classes in fundamental physics (e.g., RG universality classes, equivalent QFTs related by dualities, holographic correspondence between bulk and boundary descriptions)?

13.2.1 Q5 developed: is universality-class abstraction the right level?

Q5 asks whether the universality-class framework is too coarse to capture features relevant for fundamental physics — particularly quantum gravity, which might depend on substratum-level features that universality classes wash out. The Q1–Q4 developments give a partial answer: the framework operates at multiple levels of abstraction, with the partial-trace-features class as the most universal, the SM-reproducing class as intermediate, the OI–A1–A6 class narrower, and substratum-level features narrowest. Quantum-gravity content lives at varying levels: holographic entropy laws ($S = A/4$) are at the partial-trace-features level (per §11) since the same coefficient is reached by different substrata through different mechanisms; specific Planck-scale corrections to dispersion relations or graviton propagators are substratum-level (different programs predict different corrections). The framework's claim is therefore not that universality-class abstraction is the *only* relevant level, but that different physical predictions live at different levels, with the abstraction level matching the prediction's structural status. The honest concern remains: if QG predictions of phenomenological interest (specific signatures of substratum discreteness in cosmic-ray observations, specific corrections to graviton dispersion, specific non-perturbative effects at Planck scale) are intrinsically substratum-level, then universality-class framing offers limited purchase on them — these are framework-specific predictions, not class-level ones. Whether this is a limit or a feature depends on whether the prediction class one cares about is universal or class-specific; the framework itself does not adjudicate. The methodological answer is that universality-class abstraction is the appropriate level *for those predictions that live at it*, with framework-specific predictions retaining their status as predictions of the specific representative — in OI's case, of OI's universality-class representative within the broader meta-class.

13.2.2 Q6 developed: does universality-class framing weaken or strengthen OI's predictive content?

Q6 asks whether the universality-class framing weakens or strengthens OI's predictive content — class-specificity reduces the absolute reach of OI's predictions, but it sharpens what the predictions *are* predictions of. The dual reading: *weakened* in absolute terms — OI's Cabibbo-angle prediction $\theta_C = 1/(\pi\sqrt{2})$ is a feature of OI's specific universality-class representative, not a universal feature of any SM-reproducing framework, so confirming the prediction does not falsify all alternatives but only those in different universality classes; *strengthened* in framework-relative terms — the prediction is structurally forced *within* OI's class, and class membership is itself an empirically meaningful claim rather than a framework-internal stipulation. The methodological reframing follows §12.2: empirical confirmations support OI's specific-class membership, while the broader question of whether other classes also reproduce these predictions becomes the next-order empirical inquiry. Practically, OI's predictive content is most sharply expressed as conditional: *given OI's class membership*, $\theta_C = 1/(\pi\sqrt{2})$; the unconditional reading "$\theta_C = 1/(\pi\sqrt{2})$ is a universal feature of SM-reproducing physics" is not what the framework establishes. Confirmation under either reading is empirical support, but the conditional form is structurally honest about what the framework demonstrates. The Q4 development above sharpens this by adding three programs (LQG, causal sets, asymptotic safety) that share OI's SM-reproducing target but lie outside OI's class; for each, whether OI's specific predictions transport is open, and the answer is empirically meaningful in the same sense as for the OI–string comparison. The strengthening reading dominates in research-program terms: identifying the right universality class is a sharper target than identifying "the" theory, and the framework's predictions become diagnostic for class membership rather than absolute claims about reality.

13.2.3 Q7 developed: relationship to other equivalence classes in fundamental physics

Q7 asks how universality classes of embedded observers relate to other equivalence classes prominent in fundamental physics — RG universality classes (different microscopic theories with the same long-distance behavior), QFT dualities (different descriptions of the same theory), and holographic correspondences (bulk and boundary descriptions of the same physics). The relationships differ in structure. *RG universality classes* are the closest mathematical analog: they identify substrates as equivalent if they produce the same long-distance / coarse-grained behavior, with the equivalence relation defined by RG flow rather than by partial-trace observation. The frameworks are compatible — RG-universality-equivalent substrates would typically also be partial-trace-observation-equivalent, since the partial-trace observation depends on long-distance content — but partial-trace observational equivalence is in general a *weaker* equivalence: it can fail to distinguish substrates that RG flow

distinguishes if the trace-out integrates over precisely those microstructure features that RG preserves. *QFT dualities* (S-duality, T-duality, mirror symmetry, AdS/CFT) identify two theory descriptions as equivalent — the duality is an isomorphism of the operator algebras and dynamics, not a partial-trace relationship. Dual descriptions are automatically in the same universality class in §9's sense (they have *-isomorphic algebras and conjugate channels).* Holographic correspondence* is a specific kind of duality: bulk and boundary descriptions are dual, with the boundary acting as the "observable" side of an embedded-observer reading. The universality-class framework's relationship to holography is closer than to RG: holographic correspondence is consistent with viewing the boundary as visible-sector and the bulk as hidden-sector, with dualities of bulk descriptions being automorphisms of the channel rather than separate classes. Practically, the universality-class framing is *broader* than RG (capturing structural features that RG washes out, such as gauge-group structure and observer-admission status) and *broader* than dualities (capturing equivalences across non-dual frameworks like OI and string theory at the partial-trace level), and is *consistent with* holography in the sense that holographic dualities operate within universality classes rather than across them. Whether universality-class abstraction *adds* something beyond the union of RG, dualities, and holography is the essential question; the present framing's claim is yes — it captures structural agreement between programs (like OI and string theory) that no RG, duality, or holographic relationship establishes, since those relationships operate within frameworks while universality-class observational equivalence operates across them.

13.3 Open questions on intermediate-scale physics

The structural investigation of OI's no-GUT prediction (developed in [SM §6.7] and [SM §8.7]) surfaced several substantive open questions that bear on the framework's predictive content but require further work to develop. They are catalogued here as research directions; their development is deferred to future work.

Q8. *The neutrino mass scale puzzle.* The framework's [SM §8.4] predicts Majorana neutrinos via the dimension-5 Weinberg operator $(LH)(LH)/\Lambda_{\text{seesaw}}$, with Λ_{seesaw} treated as a Tier 3 (solution-specific) parameter. The observed neutrino mass scale $m_\nu \sim 10^{-3}$ eV requires $\Lambda_{\text{seesaw}} \sim 10^{16}$ GeV — coincidentally the GUT scale. In standard GUT scenarios, this scale is structurally explained by gauge-coupling unification. In OI's no-GUT framework, no GUT scale exists structurally; the natural cutoff is the lattice scale $M_{\text{Pl}}/(2\sqrt{\sigma})$, of order $M_{\text{Pl}} \sim 10^{19}$ GeV when σ is of order one, which would give $m_\nu \sim 10^{-6}$ eV — three orders of magnitude too small. The framework currently treats Λ_{seesaw} as solution-specific input rather than deriving it. *Open question:* does the framework have an intermediate scale between M_Z and M_{Pl} that arises structurally from the substratum's staggered-taste structure or other lattice-level features, and that could naturally produce $\Lambda_{\text{seesaw}} \sim 10^{16}$ GeV? Speculative candidates in-

clude scales associated with the staggered-taste mass spectrum (where different tastes have different couplings $|\mu_\Gamma| = 1, |\mu_X| = 1/3$ per [SM §4.7] Theorem 8b), or scales associated with the boundary-layer thickness in the visible-vs-hidden sector partition.

Q9. *The hierarchy problem in OI's composite-Higgs structure.* The framework's [SM §7.4] predicts $\lambda(M_{\rm Pl}) = 0$ as a structural consequence of the Higgs being a composite scalar (the singlet taste). This prediction is empirically supported at 0.6σ and identifies the Higgs sector as composite at the substratum level. Standard composite-Higgs scenarios in BSM phenomenology dissolve the hierarchy problem via chiral-symmetry protection (the Higgs as a pseudo-Nambu-Goldstone boson of an approximate chiral symmetry). *Partial resolution* (developed in [SM §8.8]): the framework structurally implements an analog of this protection through the singlet-taste chiral symmetry. The $\mu = +1$ and $\mu = -1$ singlet tastes are paired under chirality at the substratum level (exact discrete symmetry); the Higgs vev spontaneously breaks this pairing in the emergent EFT. The Higgs is the order parameter of the broken chirality, with mass protected against cutoff-scale corrections by Goldstone's theorem.

The framework *derives* cutoff insensitivity — no $O(M_{\rm Pl}^2)$ corrections to m_H^2, only explicit-breaking corrections of order $g^2 f^2/(16\pi^2)$ — together with the compositeness scale $f \sim M_{\rm Pl}$ being structurally forced. The framework *does not derive* the Higgs vev value $v \sim 246$ GeV, which remains Tier 3 (solution-specific). The boundary: the framework derives why m_H is not at the cutoff (chiral protection); it does not derive why v is at the electroweak scale rather than at the compositeness scale.

Quantitatively, the framework's partial resolution differs from "no resolution" by a factor of $\sim 10^{34}$ in the required fine-tuning. Standard SM without chiral protection requires fine-tuning m_H^2 to 34 decimal places; the framework's mechanism requires only that the Tier 3 input v be at the electroweak scale, with m_H following from $\lambda(M_{\rm Pl}) = 0$ + RG running.

Open question: is there a structural mechanism in OI that derives $v \ll f$, or is v genuinely solution-specific? Two candidate paths: (i) deriving $v \ll f$ from substratum structure through a mechanism currently not in the framework; (ii) accepting v as solution-specific, with the structural-vs-solution-specific partition being part of the framework's content. The framework's current commitment is (ii): v sits in the solution-specific category alongside CP phases, generation-specific Yukawa magnitudes, and other parameters whose values are not forced by the structural derivation chain.

Q10. *Possible unified resolution of Q8 and Q9.* Both Q8 (neutrino mass scale) and Q9 (Higgs hierarchy) point to a single underlying question: does OI's substratum produce a structural intermediate scale between M_Z and $M_{\rm Pl}$ that's not a GUT scale but plays the roles that GUT scales play in standard scenarios — setting the seesaw scale for neutrino masses and protecting the Higgs mass via chiral mechanisms? **Tentative resolution:** negative on the analysis

conducted here. The framework's substratum-level scales are essentially the lattice scale $M_{\rm Pl}/(2\sqrt{\sigma})$, of order $M_{\rm Pl}$ when σ is of order one, and emergent scales below M_Z; candidates for intermediate scales (staggered-taste mass gap, boundary-layer thickness, the $\delta_0 = \mathrm{Tr}[Y^2]$ coincidence at the matching convention $\mu = M_{\rm Pl}$, C_2-dependent threshold effective scales, cosmological-horizon-induced scales, emergent dimensional transmutation) all yield scales at $M_{\rm Pl}$ modulo $O(1)$ factors or scales below M_Z, but none naturally at $\sim 10^{16}$ GeV. The seesaw scale and the Higgs mass therefore remain as Tier 3 (solution-specific) parameters in the framework, set by the specific bijection φ rather than by structural derivation. *Implication:* OI's predictions on intermediate-scale physics are limited to what can be derived from $M_{\rm Pl}$ + EW-scale physics, with no claim of structural intermediate-scale derivation. This is consistent with the framework's existing treatment of $\Lambda_{\rm seesaw}$ as Tier 3 (per [SM §8.4]) and represents an honest scoping rather than a hidden deficiency. The negative resolution sharpens the structural-vs-solution-specific distinction: many parameters that GUT-style frameworks treat as derivable from unification structure are, in OI, genuinely solution-specific. This *may* be revisited if a structural mechanism overlooked in the chat-tractable analysis is identified by future work, but the chat-tractable conclusion is that no such mechanism is currently apparent in the framework's existing structure.

Q11. *Quantitative baryogenesis in a spectrum with no right-handed neutrino.* The framework's substratum-level baryon-number conservation (per [SM §8.7]) coexists with sphaleron-mediated $B+L$ violation in the emergent EFT. But the OI spectrum contains no right-handed neutrino — anomaly cancellation forces exactly five representations per generation and the singlet taste is the Higgs ([SM §4.9, §8.4]) — so standard type-I thermal leptogenesis, which requires heavy right-handed Majorana neutrinos, does not directly apply. *Open question:* the full quantitative calculation — connecting whatever ultraviolet physics generates the dimension-5 Weinberg operator (the $\Lambda_{\rm seesaw}$ scale of Q8) to the resulting baryon asymmetry $\eta = n_B/n_\gamma \approx 6\times 10^{-10}$ — is open, and the mechanism is not fixed by the framework. Resolution depends partly on Q8: without a structural derivation of the Weinberg-operator scale, the baryogenesis prediction inherits the same status (Tier 3 / solution-specific).

Q12. *Integrated treatment of non-perturbative emergence.* [SM §8.7]'s table of substratum-level structural precursors mapping to emergent-level non-perturbative SM features (T-invariance $\to \theta = 0$; topology of induced gauge action $\to$ sphalerons; bipartite checkerboard $\to$ chiral fermions and anomalies; block-diagonal $M \to$ substratum B conservation) provides a unified pattern that organizes the framework's treatment of non-perturbative SM physics. *Open question:* is this mapping exhaustive, or are there non-perturbative SM features (e.g., specific instanton-density distributions, specific non-perturbative QCD vacuum structure, specific anomaly-inflow mechanisms) that lack identified substratum-level precursors? A complete cataloging would either confirm the framework's structural completeness on this dimension or identify gaps requiring further development.

These five open questions (Q8-Q12) represent the framework's substantive research frontier as of the no-GUT investigation. Each is a candidate research direction for future development; together they form a coherent set focused on whether OI's substratum has the structural richness needed to resolve standard SM open problems (neutrino masses, hierarchy, baryogenesis, non-perturbative completeness) without invoking new physics beyond what the framework already commits to.

13.4 Limits of the approach

The universality-class framework has limits worth flagging.

Limit 1: Programmatic with a partial census. The framework articulates the universality-class question and provides candidate formalizations. Per §13.1.4, four programs (string theory, LQG, causal sets, asymptotic safety) are now positioned within the universality-class hierarchy: all four fail OI's A1–A6 and are therefore outside OI's specific class, but all four may be in the broader partial-trace-features class (the per-program partial-trace audit is open). The framework remains programmatic in that systematic enumeration of universality classes is incomplete and the partial-trace-features class membership question for each candidate is unresolved; however, the cross-program structural pattern (A2 determinism is the most discriminating condition; A4 and A6 robustly hold across all candidates) provides constructive content beyond the bare existence of the universality-class question.

Limit 2: Depends on operator-algebra technology. The algebraic formalization adopted in §8.4 and used throughout §9 depends on operator-algebra and CPTP-channel theory. These are well-developed but place the framework in a specific mathematical setting. Frameworks that do not naturally admit operator-algebra description (e.g., theories with no clear Hilbert-space structure) may not fit the formalization without modification.

Limit 3: Depends on continuum / discrete bridge. The formalization assumes that substratum descriptions, regardless of discreteness, yield visible-sector reduced laws admitting CPTP-channel representation with operator-algebra structure — representability is universal for finite laws ([Main §3.4]); operational closure beyond it is a model hypothesis, not a framework property. For OI's discrete substratum and string theory's continuum substratum this is plausible (both yield quantum-representable observable-level descriptions). For substrata whose visible-sector observable structure is not of this form, the framework may not apply.

Limit 4: The foundational thesis is unfalsifiable on its own; the content lives in the model of the incompleteness. The framework's name and starting point — that what is known is reached through observation, which is necessarily incomplete (an embedded observer cannot access the total state) — is, read as a bare thesis, not a falsifiable physical claim. It is close to a restatement of the empirical stance common to all of science: every theory is

a description constructed from inside, from partial access. So the bare thesis cannot be wrong, and therefore cannot by itself predict anything. All of the framework's predictive content lives one level down, in the *specific model* of the incompleteness: the C1–C4 architecture (a fast visible sector reading and writing to a slow, high-capacity hidden register), the cubic lattice, the wave-equation dynamics, and the ETH-genericity of the hidden sector. Each of these is either an empirically-anchored selection or a structural stipulation (catalogued for the SM sector in [SM §8.3, *Derivation status: the three tiers*]; for the reconstruction in [Substratum §3, Refinement]). The framework's power and its circularity are accordingly the same structure seen from two sides: it is a maximally tight compression of known physics onto a small set of commitments — establishing that the Standard Model's discrete structure, the gravitational thermodynamics, and the Born-rule statistics are facets of one object rather than independent inputs (a genuine claim about the unity of physics) — while those commitments are themselves chosen to recover the physics already observed. Reading "observation is incomplete" as the *source* of the predictions overstates the framework; reading it as the *organizing principle* under which a large body of physics compresses to one structure is the defensible claim, and the one this paper's universality-class machinery is built to make precise. This limit is not a defect to be removed by further work — it is intrinsic to any framework whose foundation is the empirical stance itself — and stating it plainly is what distinguishes a structural-realist compression from a claimed derivation of the world from nothing.

13.5 Relationship to the methodology paper

The universality-class framework is closely related to a planned methodology paper articulating OI's overall posture for philosophy-of-physics venues. The methodology paper would cover the broader posture (constructive structural realism, single-foundational-commitment derivation, the architecture-level engagement with the no-go results of quantum foundations, "physics modulo gauge" reframing); the universality-class framework provides the technical machinery for the structural-realism component when applied to comparison with other unification programs. The two contributions are companions: the methodology paper situates OI's posture in the philosophy-of-physics literature; the universality-class framework provides one technical articulation of what that posture entails.

Part III: Cross-Framework Synthesis

14. Cross-framework audit of observational structures

This section develops the constructive content of recovery (v) — shared observational invariants — by examining whether structural features of OI's trace-out also follow from string theory's emergence mechanisms. The framing is that observation is a structurally privileged operation: OI's substratum and string theory's substrata are different, but the *act of observation* on each substratum produces observational features that may be substratum-independent.

We proceed feature by feature. For each, we characterize the structural feature in OI's framework, identify the analogous mechanism in string-theoretic emergence, and assess whether the feature transports across frameworks. Four feature classes are examined: Born-form quantum probability structure, channel-level unitary dilation, memory/non-Markovian marginal dynamics with divisibility tracked separately, and commutant gauge-invariance.

The discussion is technical but exploratory. The aim is not to prove cross-framework equivalence at the observational level — that would require substantial work beyond chat-track scope — but to establish whether the proposed shared invariants are structurally well-defined in both frameworks. If they are, recovery (v) admits constructive sharpening; if some don't transport, the framework's universality claims must be refined.

14.1 The Born rule

OI's framing. At the finite observable-law layer, OI supplies an exact fixed-basis Born-form representation rather than a parameter-free derivation of the quadratic probability functional [Main §3]. A finite reversible realization lifts to a permutation unitary, and with Q_{fb} defined by the dictionary $p = |U|^2$, tracing over the hidden register reproduces the visible transition law exactly. The important scope distinction is that the construction does **not select** the exponent: permutation-unitary entries are 0 or 1, so $|U|^p$ gives the same transition table for every $p > 0$. For a general quantum representation the visible density matrix $\rho_V = \mathrm{Tr}_H|\psi\rangle\langle\psi|$ gives $p(a) = \mathrm{Tr}(\Pi_a \rho_V)$ in the usual way; identifying that full coherent observable/instrument structure simultaneously across all interventions is part of the operational-lifting frontier. OI's proved contribution at this layer is therefore the exact Born-form representation of classical deterministic substrata, not an independent selection theorem for why nature chooses the quadratic dictionary.

String-theoretic analog. String theory uses standard quantum mechanics throughout — in matrix-model formulations (BFSS, IKKT), in worldsheet conformal field theory, and in the boundary CFT of AdS/CFT correspondence. The Born rule is built in as a feature of QM. When an observer in a string-theoretic context measures an observable, the probability is given by the standard quantum-mechanical trace formula on the boundary CFT or the world-

sheet.

Transport assessment. The **Born-form representation structure** transports straightforwardly: once a framework is represented quantum mechanically and reduced by partial trace, subsystem probabilities use the standard quadratic trace rule. The distinction is that OI constructs an exact fixed-basis quantum representation starting from a classical reversible substratum, whereas string theory assumes the quantum formalism from the outset. This comparison should not be read as an OI derivation of the exponent 2 itself; Main §3.1/§3.4 makes explicit that the permutation construction is exponent-blind and that the dictionary's physical origin is a separate frontier item.

The shared-invariants reading is therefore representation-theoretic rather than a new Born-rule uniqueness theorem. The quadratic Born rule itself is empirically tested at the 10^{-3}–10^{-4} level via multi-path interferometry [Sinha2010, Söllner2012], with ongoing precision improvements through ultracold-atom platforms [Kanthak2025] (further discussion in §15.5).

14.2 Channel-level unitarity

OI's framing. OI's visible-sector dynamics is a CPTP channel Φ generated by the trace-out of substratum bijection dynamics over the hidden sector [Main §3.2]. Stinespring's theorem [Stinespring1955] identifies Φ as the partial trace of a unitary U acting on the dilated Hilbert space $\mathcal{H}_V \otimes \mathcal{H}_H$. Channel-level unitarity is the statement that the dilation U is unitary; the visible-sector channel Φ is generally non-unitary (because of the trace-out's loss of information to the hidden sector) but lifts to unitary dynamics on the dilation.

String-theoretic analog. AdS/CFT bulk reconstruction has a closely analogous structure. Boundary CFT operators are encoded in bulk operators with redundancy, and the encoding is a quantum error correcting code [HarlowBulkReconstruction2018, AlmheiriDongHarlow2014]. Bulk reconstruction from boundary subregions ("entanglement wedge reconstruction") shows that bulk operators in the entanglement wedge of a boundary subregion can be reconstructed from operators on that subregion. The reconstruction is unitary at the level of the full bulk-boundary system, with effective non-unitarity arising when restricting to a subregion.

The structural parallel is direct: in AdS/CFT, the "trace-out" is restriction to a boundary subregion, and the "visible-sector channel" is the effective dynamics on that subregion after tracing over the complement. The bulk-boundary system as a whole is unitary; the subregion-level channel is generally non-unitary; the channel admits a unitary dilation given by the full bulk reconstruction.

Transport assessment. Channel-level unitarity transports across frameworks. Both OI and AdS/CFT have the same structural feature: a CPTP channel at the observable level lifting to unitary dynamics on a dilation. The dilation in OI is the Stinespring lift to $\mathcal{H}_V \otimes \mathcal{H}_H$; in AdS/CFT, the dilation is the full

bulk-boundary system before restricting to a subregion.

A subtle but important point: the *mechanism* by which dilation produces unitarity is different. OI's dilation is mathematically the Stinespring construction applied to a classical-substratum bijection; AdS/CFT's dilation is the holographic encoding of bulk operators in boundary degrees of freedom via the gravitational path integral. Both produce unitary dilations of CPTP visible-sector channels, but through different mechanisms. The shared structural feature is the dilation pattern itself, not the specific machinery producing it.

14.3 Memory, non-Markovianity, and recurrence-scale indivisibility

OI's framing. OI's C4 readback condition is first a statement of *conditional memory*: in general $P(a_3 \mid a_2, a_1) \neq P(a_3 \mid a_2)$ because hidden-sector correlations retain information about visible history [Main §2.3, §3.4]. That property is not identical to P-indivisibility. In Main's divisibility sense, P-indivisibility means that for some pair of rooted visible maps there is no stochastic propagator that divides the later map through the earlier one. What OI proves for one fixed finite reversible representative is more specific: C4 supplies a non-permutation rooted map, finite recurrence later returns the visible map to the identity, and a non-permutation stochastic map has no stochastic inverse; hence an indivisibility witness occurs somewhere in the recurrence cycle. Bell is a separate constraint: a response-complete deterministic completion that is both measurement-independent and ontically parameter-independent obeys $|S_{\mathrm{CHSH}}| \leq 2$, so Bell-violating OI composites retain measurement independence and relinquish ontic parameter independence while preserving operational no-signaling [Main §3.3]. Thus C4 memory is the temporal mechanism, recurrence supplies the proved indivisibility witness, and neither memory nor indivisibility is a Bell selector.

String-theoretic analog. Holographic effective field theories explicitly classify modes by their Markovianity. The sound scalar in compressible relativistic fluids dual to AdS planar black holes is a non-Markovian field with Markovianity index $3 - d$ for a d-dimensional fluid. The holographic Schwinger-Keldysh formalism [HolographicSK2024, HolographicNonHydro2024] derives effective field theories that include non-Gaussian noise and non-Markovian dynamics as features of the boundary effective description, arising from integrating out bulk degrees of freedom. This supplies a close analog of the *memory/non-Markovian* part of OI's mechanism.

The structural parallel is therefore at the memory level: in both frameworks, restricting attention to a subset of degrees of freedom can produce history-dependent effective dynamics when the integrated-out degrees of freedom retain correlations across time. Partial trace or coarse-graining by itself does not force this outcome; a memoryless hidden sector can produce a Markovian reduced law.

Transport assessment. The memory/non-Markovian mechanism transports

across frameworks. The stronger P-indivisibility property does **not** follow merely from the cited holographic non-Markovianity statements, and is not claimed here for string theory without a separate divisibility analysis. OI's recurrence-scale witness uses additional structure that the cross-framework analogy does not automatically supply: one fixed finite reversible representative plus C4 readback.

This distinction is important for the Bell comparison as well. Holographic non-Markovian effective theories provide an analog of hidden-memory dynamics, not by themselves a proof of the standard quantum Bell correlation set. In OI, a response-complete deterministic completion that combines the pointwise graph cone with a measurement-independent pre-setting ensemble is Bell-local and satisfies $|S_{\mathrm{CHSH}}| \leq 2$; quantum Bell violation therefore requires the separately stated ontic-parameter-dependent branch, with operational no-signaling retained. Hidden memory and stochastic indivisibility alone do not supply that branch.

The memory mechanism has independent theorem-level corroboration. [BrandnerMeyer2025] establishes a rigorous weak-memory regime for autonomous linear evolution equations with hidden-degree-of-freedom coupling, providing a controlled framework for the non-Markovian visible dynamics OI invokes. [TomizukaTakeda2026] constructs a structurally similar hidden-model decoherence picture independently, motivated by classical–quantum dynamics in gravity. [Barandes2026]'s stochastic-quantum theorem concerns indivisible stochastic processes and unitary quantum systems; Main §3.4 retains that result for the compressed form and interpretation under the translation hypothesis, while keeping indivisibility distinct from conditional-memory non-Markovianity and keeping fixed-basis representability internal. The framework's central technical move therefore sits within an established theorem-level neighborhood without identifying these different notions by definition (§15.1 develops this in detail).

14.4 Commutant gauge-invariance

OI's framing. OI's emergent QFT inherits gauge invariance from the substratum's commutant structure [SM §3.1]. Cubic-group decomposition of the $K = 6$ internal components produces multiplicities $(3, 2, 1)$, with the commutant of the cubic-group action on the $K = 6$ space being $SU(3)\times SU(2)\times U(1)$. Background independence (A6) promotes the global commutant symmetry to local gauge invariance. The trace-out preserves only commutant operators (those commuting with the gauge group action), so observable correlations are gauge-invariant by construction.

String-theoretic analog. Gauge invariance in string-theoretic emergence has multiple sources, but the closest analog to OI's commutant structure is the *open-string sector* of D-brane stacks. A stack of N D-branes has $U(N)$ gauge symmetry generated by the open-string excitations on the brane worldvolume;

the commutant of the worldvolume action gives the gauge group. Restricting to closed strings (graviton sector) gives the singlets under the gauge group, analogous to OI's trace-out preserving only commutant operators.

For AdS/CFT specifically, the boundary CFT has a global symmetry that becomes a bulk gauge symmetry under the duality. Boundary operators charged under the global symmetry correspond to bulk fields charged under the gauge symmetry; gauge-invariant bulk physics corresponds to singlet boundary operators. The commutant structure is preserved across the duality.

Transport assessment. Commutant gauge-invariance transports across frameworks, with the caveat that the specific gauge groups produced differ. In OI's framework, the gauge group is $SU(3) \times SU(2) \times U(1)$ derived from the cubic-group commutant on $K = 6$ components. In string-theoretic compactifications, the gauge group depends on the specific compactification — D-brane configurations, fluxes, Calabi-Yau topology — and the SM gauge group is realized through specific arrangements rather than forced by a unique commutant structure.

The structural feature that transports is *the gauge-invariance pattern via commutant restriction* — both frameworks have observable physics determined by the commutant of some symmetry action on the substratum. The structural feature that does not transport in the same form is *the specific commutant producing* $SU(3) \times SU(2) \times U(1)$ — OI derives this from the cubic-group; string theory realizes it from specific compactification choices.

This partial transport refines the shared-invariants claim: the commutant gauge-invariance pattern is universal (any framework with substratum-level symmetry produces gauge-invariant observables via commutant restriction), but the specific gauge group obtained depends on the substratum's symmetry structure. The SM gauge group is forced in OI's framework by the cubic-group symmetry; in string theory, it's one of many possible commutants.

14.5 Synthesis

The four-feature audit produces a mixed transport pattern:

Feature	Transports?	Caveat
Born-form quantum probability structure	Yes at the representation/formalism level	OI exhibits an exact fixed-basis representation; string theory starts from quantum formalism; the comparison does not select the exponent 2

Feature	Transports?	Caveat
Channel-level unitary dilation	Yes	Different dilation mechanisms (Stinespring vs holographic)
Memory / divisibility	Memory mechanism: yes; P-indivisibility: not established cross-framework	OI has C4 memory and a recurrence-scale indivisibility theorem; holographic non-Markovianity alone does not prove stochastic non-divisibility
Commutant gauge-invariance	Pattern transports; specific commutant does not	OI's cubic-group commutant produces $SU(3) \times SU(2) \times U(1)$ uniquely; string theory's commutants are compactification-dependent

The first two features transport directly at the representation/dilation level. The third transports at the memory/non-Markovian mechanism level, while OI's recurrence-scale P-indivisibility uses additional finite-recurrence structure and is not inferred for string theory here. The fourth transports as a *pattern* (gauge invariance via commutant) but not as a *specific output*.

This refines recovery (v) substantively. Partial trace is an architectural mechanism, not a theorem that every reduced process is memory-bearing. Once a quantum probability dictionary is in place, reduced probabilities have the usual Born form; once a visible map is CPTP, a unitary dilation is available. By contrast, non-Markovian memory requires retained hidden correlations, and P-indivisibility requires a further divisibility argument. OI supplies those extra ingredients through C4 and finite recurrence for a fixed representative; the cross-framework audit must check them separately elsewhere.

The picture that emerges is therefore narrower than a universal four-feature derivation: **observation by restriction supplies a common interface pattern, while memory, divisibility, and substratum-symmetry outputs depend on additional structure**. OI and string theory share important representation/dilation architecture and can share hidden-memory mechanisms; they do not thereby become equivalent at the full operational level.

This conclusion has implications for the universality-class question developed in Part II below. Embedded-observer universality classes should be characterized

by a vector of properties rather than by "partial trace" alone: which quantum-probability representation is used, which dilation structure exists, whether the discarded sector retains operational memory, whether the reduced dynamics is actually indivisible, and which gauge structures the substratum admits. OI and string theory can intersect on several coordinates while reaching them through different mechanisms.

14.6 Implications for the OI-string relationship

The cross-framework analysis of §14.1–14.5 establishes that recovery (v) — shared observational structure — has substantive content, but it also fixes the scope of that claim. The commonality is strongest at the representation/dilation interface and becomes conditional once memory, divisibility, or substratum symmetry is involved.

This refines the conclusion of §7.8 and §14.5: OI and string theory are not equivalent at the substratum level (per §6), and they do not share *all* observational features. They share a structural pattern of restricting a larger description to an observable sector; within quantum formulations this carries the standard probability formalism and admits unitary dilation. Whether the reduced dynamics is memory-bearing depends on correlations retained in the discarded sector, and whether it is P-indivisible requires a separate divisibility theorem. OI proves the latter at recurrence scale from C4 plus a fixed finite reversible representative; no corresponding conclusion for string theory is inferred merely from holographic non-Markovianity.

This intersection is non-trivial but should not be overstated. A framework that already uses quantum states and partial trace inherits the standard quantum probability rule for its subsystem and the usual dilation architecture. It does **not** automatically inherit non-Markovianity, P-indivisibility, Bell causal locality, or OI's specific commutant output. Those require additional dynamical, statistical, or symmetry structure.

The constructive content of recovery (v) is therefore: **the restriction/partial-trace interface is a locus of shared architecture, not by itself a derivation of every observational feature audited here**. This connects to OI's "observation occurs" principle while preserving Main's distinction between universal representation, C4 memory, recurrence-scale indivisibility, and the still-conditional operational-local lift.

What does not transport — the specific gauge group, the specific particle content, the specific predictions — is what distinguishes substratum descriptions empirically. OI's predictions for the Cabibbo angle and Koide ratio are features of OI's specific substratum-symmetry derivation; string theory's compactifications would need to reproduce these as features of specific compactification choices. The two frameworks agree on the *kind* of physics observation produces; they differ on the *specific content*.

This is a more refined picture than the §7.1 framing of "OI and string theory as independent programs." They are independent at the substratum level, share structural features at the partial-trace level, and may or may not converge at the specific-content level depending on which compactifications string theory can realize as SM-reproducing. The recovery (v) analysis identifies precisely where the intersection lives: at the level of partial-trace structural features, not above and not below.

14.7 Loop quantum gravity: four-feature audit

The §14.1–14.6 audit was conducted for OI vs string theory. The same four features admit similar audit for other quantum-gravity / unification programs identified in §13.1.4. This section and the next two perform the audit for loop quantum gravity, causal sets, and asymptotic safety. The aim is the cross-program characterization of where the partial-trace observational features hold.

Born rule. Loop quantum gravity (LQG) [RovelliSmolin1990, Thiemann2007] uses standard quantum mechanics throughout: spin-network states form a Hilbert space with inner products, observables are operators on this Hilbert space, and probabilities are computed via the standard Born rule on spin-network projection operators. The Born rule transports without modification.

Channel-level unitarity. LQG dynamics is implemented through spin-foam path integrals [EPRL2008, FreidelKrasnov2008] that produce transition amplitudes between spin-network states. The spin-foam amplitude is unitary at the level of the full path integral; effective dynamics on visible-sector observables (e.g., on smeared geometric observables averaging over Planck-scale spin-network structure) is CPTP, with the unitarity recovered at the spin-foam level via Stinespring-style dilation. Channel-level unitarity transports, with the dilation living in the spin-foam path integral rather than the OI substratum bijection.

Memory / divisibility. Spin-foam dynamics on visible-sector observables, after integrating over hidden internal-edge data, plausibly produces non-Markovian effective dynamics. The mechanism is structurally similar to OI's trace-out: the visible-sector marginal can carry memory of substratum-level dynamics through correlations established in earlier spin-foam evolution. This supports a plausible memory/non-Markovian analog, but it does not by itself establish P-indivisibility in Main's stochastic-propagator sense. Both the existence of the effective memory law and any stronger divisibility failure require technical work; this is the most underdeveloped of the four feature classes for LQG.

Commutant gauge-invariance. LQG has natural gauge invariance via the Ashtekar-connection variable's $SU(2)$ gauge structure plus diffeomorphism invariance. Observable physics is determined by the commutant of the diffeomorphism + gauge group action on the spin-network Hilbert space. The com-

mutant gauge-invariance pattern transports; the specific gauge group is $SU(2)$ at the substratum level (different from OI's $SU(3) \times SU(2) \times U(1)$ from cubic-group commutant). LQG must reach the SM gauge structure through additional matter-coupling structure rather than through commutant alone.

LQG transport summary. The quantum-probability formalism, channel-level unitary-dilation pattern, and commutant gauge-invariance pattern are present in the comparison. The memory/divisibility feature is only plausible at the memory level; P-indivisibility remains open. LQG therefore shares several interface-level structures with OI without yet matching OI's recurrence-scale indivisibility result.

14.8 Causal sets: four-feature audit

Causal set theory [BombelliEtAl1987, Sorkin2003] takes the substratum to be a discrete, locally finite partially ordered set with sequential growth dynamics. The structural setup differs from OI in that the substratum dynamics is fundamentally stochastic (sequential growth via classical or quantum sequential growth) rather than deterministic.

Born rule. Classical sequential growth (CSG) [RideoutSorkin1999] is fundamentally classical-stochastic, not quantum. The Born rule does not directly arise. Quantum sequential growth (QSG) [DowkerSurya2020] adapts the framework to produce quantum probabilities through quantum measure theory. Whether QSG produces the standard Born rule or a generalization (with potentially observable higher-order interference terms) is an open technical question. Transport: open / uncertain. The current status is that the framework is structurally compatible with the Born rule but does not exhibit it as an exact internal representation identity, as OI's finite-horizon equivalence does ($S \iff D \iff Q_{\text{fb}}$, [Main §3.4]). The comparison is about exhibiting the Born form, not about deriving it: neither framework selects the quadratic exponent, which in OI is part of the definition of Q_{fb} rather than a derived consequence ([Main §3.1]).

Channel-level unitarity. CSG sequential growth lacks a clean unitary lift at the substratum level — the dynamics is fundamentally stochastic. QSG's quantum-measure formulation can be reformulated through a CPTP channel structure with appropriate dilation, but the Stinespring dilation is structurally different from OI's (the dilation acts on a different Hilbert space, with different microcausality properties). Transport: structurally yes, but the dilation mechanism differs.

Memory / divisibility. Causal-set coarse graining can plausibly produce history-dependent effective dynamics when fine-grained causal structure retains correlations. That is structurally similar to OI's hidden-memory mechanism. The comparison supports a plausible memory/non-Markovian analog; it does not establish P-indivisibility without a separate divisibility analysis.

Commutant gauge-invariance. Causal sets are intrinsically diffeomorphism-invariant: the causal-order structure determines the manifold up to volume measure, and any continuous diffeomorphism preserves the order. Observable physics is determined by order-invariant quantities. The commutant gauge-invariance pattern transports, but the gauge group is the trivial diffeomorphism group rather than a Yang-Mills structure. Causal sets must reach SM gauge structure through additional matter content, similar to LQG.

Causal sets transport summary. The commutant/order-invariance pattern transports most directly. Quantum-probability and unitary-dilation structure depend on the QSG formulation, while memory is plausible and P-indivisibility remains unaudited. Causal sets are therefore related at several structural levels but do not presently match the full OI feature vector.

14.9 Asymptotic safety: four-feature audit

Asymptotic safety [Weinberg1979, Reuter1998, ReuterSaueressig2019] proposes that quantum gravity is non-perturbatively renormalizable through a non-trivial UV fixed point of the renormalization group. The framework operates within standard QFT machinery with non-perturbative RG analysis.

Born rule. Standard QFT machinery is used throughout; the Born rule is built in via the standard quantum-mechanical formalism. Transport: yes, without modification.

Channel-level unitarity. Standard QFT unitarity is inherited at the level of the asymptotic-safety effective action. The visible-sector effective dynamics — derived by integrating out high-momentum modes in the RG flow — is CPTP, with the dilation living at higher RG scales. Channel-level unitarity transports.

Memory / divisibility. Integrating over high-momentum modes in an RG flow can generate non-local-in-time or memory-bearing effective interactions. This gives a strong mechanism-level analog of OI's visible-hidden memory picture, with the split operating in momentum space rather than the spatial substratum. The existence of such non-Markovian effective structure does not by itself certify P-indivisibility in Main's divisibility sense; that stronger property would require a separate propagator analysis.

Commutant gauge-invariance. Asymptotic safety preserves gauge-invariance of the effective action under the RG flow via Ward identities. Observable physics is determined by gauge-invariant effective operators. The commutant gauge-invariance pattern transports; the specific gauge group is whatever Yang-Mills structure is included in the matter content (typically the SM gauge group, in matter-included formulations).

Asymptotic safety transport summary. Standard quantum probability and unitarity are inherited from the QFT framework, the RG split supplies a strong memory-mechanism analog, and gauge-invariance is preserved. The comparison does not establish P-indivisibility from non-Markovianity alone. Mechanistically

this remains one of the closest cross-program parallels to OI, but not an all-four-feature theorem.

14.10 Cross-program synthesis

The four-feature audit across OI, string theory, LQG, causal sets, and asymptotic safety produces the following pattern:

Program	Born-form / quantum probability	Unitary dilation	Memory / indivisibility	Commutant gauge
String theory	Yes (built in)	Yes	Non-Markovian modes; indivisibility not established here	Pattern; specific differs
OI	Yes (represented)	Yes	C4 memory; indivisible at recurrence scale	$SU(3) \times SU(2) \times U(1)$
LQG	Yes (built in)	Yes	Memory plausible; indivisibility open	Pattern; specific differs
Causal sets	Open (QSG)	Structurally	Memory plausible; indivisibility open	Pattern; trivial gauge
Asymptotic safety	Yes (built in)	Yes	Non-Markovian EFT mechanisms; indivisibility not established here	Pattern; specific differs

Three observations:

The restriction-interface features are robust, but at different logical strengths. Quantum formulations carry their probability rule and admit unitary dilation; several programs also exhibit or plausibly generate memory-bearing effective dynamics when degrees of freedom are integrated out. The stronger P-indivisibility

statement does not follow from non-Markovianity alone and is proved in OI here only at recurrence scale from C4 plus a fixed finite reversible representative. Thus §15.3's universality claim should be read as an architectural transport claim, not as a theorem that partial trace forces every feature in the table. For OI specifically the split is sharper than "integrating out generates memory": partial trace alone does not force memory — the memory-free reversible witnesses are exactly the visible-decoupled maps, whose visible dynamics carries no hidden dependence. But in the uniform full-product finite reversible setting, the T-level form of (C1) is a non-permutation one-step visible matrix, and then memory follows in general rather than only by census: [Main §3.4]'s non-permutation-witness lemma gives P-indivisibility at recurrence scale, while every first-order Markov process is P-divisible. Equivalently, the visible factor of a finite deterministic bijection has zero asymptotic entropy rate (its entire history is fixed by one of finitely many initial substratum states), whereas a stationary first-order Markov chain with non-permutation T has $H(X_{t+1} \mid X_t) > 0$. That second route needs less than uniformity and less than $|V| = 2$: it needs a φ-invariant prior — which makes the visible process stationary — giving positive weight to some visible state whose T-row is non-degenerate. Uniform-on-the-full-product is one such prior, so the ensemble censused above is a special case, and the theorem covers the non-uniform invariant priors that [Main §3.4]'s arbitrary-prior extension admits (check 10, `equivalence_recovery_probes.py`, with a prior supported on a visible-decoupled block as the control that correctly does not fire). The sharp form is worth stating on its own: the visible factor of a finite reversible substratum has zero entropy rate under every invariant prior, so whatever randomness an embedded observer records is bounded by $\log|\mathcal{C}_V \times \mathcal{C}_H|$ and is exhausted at recurrence scale — the same scale at which the indivisibility witness appears, and for the same reason. The exhaustive checks are therefore regressions of that theorem, not its proof: at $|V| = 2$ and $|H| = 2, 3, 4$, respectively, all 16/16, 648/648, and 39,168/39,168 T-level-C1 bijections carry visible memory; the $|H| = 4$ complement consists of exactly $1{,}152 = 2(4!)^2$ visible-decoupled bijections (checks 7–9, `equivalence_recovery_probes.py`). The 648 denominator coincides with [Main §3.1]'s entanglement-breaking census but the diagnostic is different: that census classifies channels as measure-and-prepare, while this one classifies visible processes as memory-bearing, and the two must not be merged.

The commutant gauge-invariance pattern transports universally; the specific gauge group does not. All five programs produce gauge-invariant observable physics through commutant restriction or analog mechanism, but each reaches the SM gauge group through different routes (cubic-group commutant in OI; brane-stack arrangements in string theory; matter-content addition in LQG, causal sets, and asymptotic safety). The specific SM gauge group is universality-class-specific within the broader gauge-invariance-pattern level.

Asymptotic safety has the closest mechanism-level similarity to OI. Its visible-hidden split via momentum-space RG flow is structurally closest to OI's spatial-substratum trace-out. Both produce non-Markovian visible dynamics through

the same architectural pattern (unitary on the dilation, CPTP on visible). This suggests asymptotic safety and OI may be more closely related than either is to string theory or LQG, and may share more universality-class structure.

The cross-program audit refines §13.1.4's per-condition A1–A6 analysis by identifying which observational structures recur even where the structural conditions differ. The evidence supports a shared *interface architecture* — coarse graining/restriction, quantum-probability form where quantum formalism is present, dilation structure, and possible hidden-memory effects — but not a blanket derivation of P-indivisibility or of the Born exponent from A1, A3, A4, A6 plus the SM target. Those stronger properties must be established by the program-specific dynamics. Section 14 is therefore a transport audit, not an independent derivation theorem.

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15. Theoretical and empirical corroboration

The framework's claims are of mixed types. Some are directly empirical (the Born rule, channel-level unitarity, non-Markovian marginal observation) and admit precision experimental tests. Others are structural — the partial-trace mechanism, the algebra-channel formalism, the universality-class hypothesis — and are corroborated through theorem-level results in independent literatures and through analog constructions in adjacent fields. This section collects the most consequential corroborations, organized by claim type. The ordering reflects strength of support: §15.1 documents the strongest external corroboration (theorem-level results for the framework's central technical mechanism); §15.5 acknowledges what remains under-supported.

15.1 The partial-trace mechanism

The framework's central technical move — that visible-sector dynamics under trace-out of a hidden sector is generically non-Markovian and admits a unitary quantum description — has theorem-level corroboration in three independent recent results.

Stochastic-quantum correspondence at theorem level. Barandes [Barandes2026], building on [Barandes2023, Barandes2025], establishes a precise correspondence theorem: every indivisible stochastic process on a configuration space corresponds to a unitarily evolving quantum system, and conversely, every quantum system can be represented as such a process. The theorem provides a first-principles explanation for why quantum theory uses complex Hilbert spaces, linear-unitary evolution, and the Born rule. This is the machinery [Main §3.4] retains — for the compressed form and its interpretation under the translation hypothesis, the fixed-basis representation's existence internal — rather than the gate from the deterministic substratum to the emergent description, now established at theorem-level rigor in the foundations literature independent of the OI framework's specific content. [Main]'s reverse-direction lemma (every

P-indivisible process on a finite configuration space admits a deterministic dilation) is consistent with the same correspondence (the lemma carries its own proof in [Main §3.2]).

Weak-memory regime for hidden-degree-of-freedom dynamics. Meyer and Brandner [BrandnerMeyer2025], following [Brandner2025a], prove that for autonomous linear evolution equations describing systems coupled to inaccessible degrees of freedom, integrating out those degrees of freedom yields well-defined non-Markovian visible dynamics in a weak-memory regime, with explicit error bounds and a convergent perturbation scheme. The setting is general open quantum systems with no stake in OI being right, and the conclusion is that visible-sector dynamics with hidden-sector memory is a mathematically clean construction with a rigorous regime of validity. The framework's C2 persistence condition — in its slow-bath realization — and the resulting non-Markovian visible dynamics fall within this regime, so the framework's central technical move sits inside an established theorem.

Hidden-model decoherence producing classical–quantum dynamics. Tomizuka and Takeda [TomizukaTakeda2026] construct an explicit hidden model in which a mediator is coupled to additional unobserved environmental degrees of freedom, and derive the resulting reduced dynamics by tracing them out. The reduced dynamics is generically non-Markovian, reflecting environmental memory effects. Their motivation is the question of whether classical–quantum dynamics in gravity can arise as effective descriptions of fully quantum systems; the construction is structurally identical to OI's mechanism (visible–hidden partition, unitary on the dilation, trace-out producing non-Markovian visible dynamics) developed independently for a different physics question. The structural identity is independent corroboration that the OI mechanism is the standard way the open-systems literature now thinks about hidden-degree-of-freedom dynamics.

Experimental implementation in quantum metrology. The hidden-Markov-model framework underlying the partial-trace mechanism has direct experimental implementation in quantum metrology [QECNonMarkovian2025]: the model is Markovian on the joint system + environment and non-Markovian on the visible-sector reduced state, with information backflow from the environment producing the framework's signature non-Markovian behavior. The framework's mechanism is therefore not only mathematically rigorous but realizable in current experimental quantum-information platforms.

15.2 The algebra-channel formalism

The §8.3 / §9.1 algebra-channel formalization adopted in this paper draws on standard machinery from algebraic quantum field theory (AQFT). The formalism's reliability and expressive power are established in the AQFT literature.

AQFT canonical references. The Brunetti–Fredenhagen / Buchholz–Fredenhagen program [BrunettiFredenhagen2024] develops the algebraic

approach to QFT through nets of local von Neumann algebras with isotony, microcausality, and covariance, with states arising as positive normalized linear functionals on the algebras. The structure used in §10.4's construction of OI's algebra-channel pair — a von Neumann algebra of gauge-invariant local observables, with the channel implementing the trace-out via Stinespring dilation [Stinespring1955] — is standard within this framework. [Yngvason2005] establishes that the local algebras of relativistic QFT are typically type III von Neumann factors, which has consequences for entanglement and the modular structure that §10.4's construction inherits. [LabuschagneMajewski2025] provides recent updates to the algebraic framework on curved spacetime, relevant to OI's GR-emergence cluster.

Stinespring dilation as universal device. The Stinespring representation theorem [Stinespring1955] guarantees that every CPTP channel admits a unitary dilation. §9.2 (Definition 9.2) and §10.4's channel construction invoke this, with the qualification recorded there: OI's own realization uses a mixed reference state, so what it exhibits is a unitary realization rather than a Stinespring dilation, and the uniqueness statements of Propositions 9.7a–9.7b apply to the pure-reference isometry rather than to it. The recent literature has refined Stinespring dilations into a working tool for quantum channel learning and metrology, including channel tomography techniques [RandomStinespring2025] that provide protocols for distinguishing channels via their dilations — the operational analog of the §13.1.1 four-step recipe's channel-conjugacy step.

15.3 Structural realism in operational and dual-theory contexts

The hierarchical structural-realism position developed in §2 and §12 is supported by independent work in philosophy of physics that arrives at compatible positions through different routes.

Operational theories as structural realism. Adlam [Adlam2022] argues that operational axiomatizations of quantum mechanics (Hardy [Hardy2001], Barrett [Barrett2007], and the broader Mackey [Mackey1963] tradition) can be interpreted as a novel form of structural realism: probing the modal structure of possible operational theories, with quantum mechanics as a specific point in this space. The view aligns with Structure's two-dimensional hierarchy (observation × gauge): the observational axis corresponds to Adlam's modal-structure-of-operational-theories framing, and the gauge axis corresponds to Structure's parallel articulation of the gauge-equivalence levels at which structural commitment lives. The framework's structural-realism commitment is therefore situated within an active research program in philosophy of physics, not a novel position in isolation. The reconstruction theorems also play a second, sharper role in the framework: their uniqueness statements supply the necessity partner to the constructive derivation of quantum mechanics, with the hypothesis-mapping stated in [Main §3.2].

Common-core realism in QM-string dualities. Dawid and Franzmann

[DawidFranzmann2025] develop a structural-realism account specifically for dualities in quantum mechanics and string theory, accommodating ontological pluralism while preserving structural realism through what they call "common core realism" — the proposal that invariant structure across dual models is the proper target of realist commitment. This is precisely the conceptual move §12.1 makes in the universality-class framing: the realist commitment is to structure invariant across substratum descriptions, with specific substratum content being framework-relative. Their work cites parallel programs by Matsubara–Johansson, Rickles, Huggett, and De Haro–Butterfield that converge on similar positions. The universality-class proposal of §9 can be read as one technical articulation of common-core realism specialized to embedded-observer systems.

15.4 Universality classes in adjacent literatures

The universality-class framework of §9 is not unique to embedded-observer systems; analogous structures appear in dissipative quantum systems and many-body operator dynamics, where they are working analytical tools rather than conjectural propositions.

Emergent universality classes in dissipative quantum systems. Zhou et al. [ZhouEtAl2025] develop an effective field theory description for the universal dynamics of dissipative quantum systems with dipole moment conservation, identifying a strongly interacting non-equilibrium fixed point that governs equal-time phase fluctuations. The work is independent of OI but uses the same conceptual apparatus — universality classes characterized by specific structural features (dipole conservation in their case; partial-trace observational features in OI's), with framework-specific content (subdiffusive vs diffusive transport in their case; specific gauge structure in OI's). The existence of working analyses of this kind in active research programs is corroboration that the universality-class framework is a productive analytical tool, not merely a programmatic gesture.

Random-matrix universality from fast-slow decomposition. Lunt et al. [Lunt2025] prove the emergence of a universal random-matrix description of fast-mode dynamics in many-body quantum systems, via the memory-function formalism with a careful split into fast and slow modes. The fast-mode universality is direct analog to OI's partial-trace structure: hidden-sector (fast) dynamics produce universal observational features at the visible-sector (slow) level, independent of the specific microscopic content. The result is theorem-level for the random-matrix universality class; the same architecture (fast-slow decomposition producing universality at the slow-mode level) is the architecture OI's framework uses to produce SM-reproducing universality at the visible-sector level.

Holographic Schwinger-Keldysh as analog of P-indivisibility. Recent work on holographic Schwinger-Keldysh contours [AmmonEtAl2025], extending earlier work [HolographicSK2024], establishes that the AdS/CFT correspondence prescribes a specific Schwinger-Keldysh contour structure for real-time

correlation functions, with non-trivial out-of-time-order four-point functions exploring the full four-fold geometry. §14.3 identifies this as the holographic analog of OI's P-indivisibility — both encoding the non-Markovian marginal structure of partial-trace observation. The recent literature provides concrete prescriptions making the analog precise, supporting the §14.3 transport claim.

15.5 Empirical anchors

Direct experimental confirmation applies to the partial-trace-level features identified in §14 (Born rule, channel-level unitarity, non-Markovian marginal) and to the framework's specific predictions developed in [SM], [GR], and [Juno]. The following are the most recent and precise confirmations.

Born rule precision. The cleanest test of higher-order interference is the multi-path interferometry experiment of [Sinha2010], which bounded the magnitude of three-path interference to less than 10^{-2} of the expected two-path interference, ruling out third- and higher-order interference in optical regimes. The bound was tightened to 10^{-3}–10^{-4} by [Söllner2012] with detector-nonlinearity calibration. Recent proposals using ultracold-atom interferometry [Kanthak2025] aim for further precision via Bose–Einstein condensate platforms that avoid systematic errors from material slits. The accumulated bound is consistent with the Born rule as a partial-trace-level invariant.

Non-Markovian dynamics in physical systems. Direct experimental observation of non-Markovian visible dynamics from hidden slow degrees of freedom: [Mehl2012] in colloidal systems and [Groeblacher2015] in macroscopic micromechanical oscillators. These confirm that the trace-out mechanism is realized in physical systems and that the resulting non-Markovian behavior is operationally accessible. Recent experimental implementations in metrology [QECNonMarkovian2025] demonstrate the same mechanism in quantum-information platforms.

Cosmological tests of the framework's GR-emergence cluster. [DESI_DR2_2025] reports rejection of the cosmological constant at 2.8σ–4.2σ depending on the supernovae dataset, with potential phantom crossing at $z \approx 0.5$ and a 2σ+ detection of nonzero neutrino masses in extended-parameter cosmology. The framework's [GR §7] carries dark energy in Type II running-vacuum form — the functional form the local expansion shared by every smooth model, the content the Type II channel and the magnitude; the [Bertini2025] direct fit to DESI DR2 + DES-Y5 + Planck 2018 reports $\nu = -(2.5 \pm 1.3) \times 10^{-4}$, consistent with zero at 2σ; the framework's expected magnitude $|\nu| \sim 10^{-32}$ (conditional on the Type II reading) is itself indistinguishable from zero at this precision, so the two are consistent. DESI Year 5 results, expected within the publication horizon, will be discriminating.

Neutrinoless double-beta decay. [LEGEND2024] reports LEGEND-200 sensitivity $T_{1/2} > 2.8 \times 10^{26}$ yr at 90% CL on $0\nu\beta\beta$ in ^{76}Ge (combined with GERDA + Majorana Demonstrator). The framework's [SM §8.4] predicts Majorana neutrinos, so $0\nu\beta\beta$ should be observed; LEGEND-1000 is designed to

reach the inverse-ordering range. A positive observation would confirm the framework's neutrino-sector prediction; a null result at LEGEND-1000 sensitivity would constrain the framework. The swampland literature [IbañezMartin-LozanoValenzuela2017] independently constrains neutrino-sector parameters via the Weak Gravity Conjecture, providing a cross-validation pathway: the framework's normal-ordering Majorana prediction sits within the WGC-allowed region.

Black-hole entropy. GW250114 (LIGO/Virgo/KAGRA, September 2025) confirmed Hawking's area theorem at 99.999% confidence — classical horizon dynamics the framework preserves; the 1/4 coefficient of [GR §5], conditional on H-slope with the horizon and frame conditions, has no direct empirical test. The observation is a non-discriminating consistency for the cosmological-horizon application of the partial-trace mechanism.

15.6 What remains under-supported

Three classes of claims remain without independent support and would benefit from focused future work.

The universality-class hypothesis qua novel proposal. The proposal that observationally-equivalent embedded observers form a coarser equivalence than $\mathcal{G}_{\text{sub}}$, populated by genuinely different substratum classes, has no direct corroboration in the literature. The closest analog work (§15.4) supports the framework concept but not the specific cross-class population claim. The §10.4 construction of OI's algebra-channel pair, combined with the §11.2 / §13.1.1 four-step recipe applied against a specific string compactification, is the natural pathway to confirmation.

Per-condition A1–A6 audit for LQG, causal sets, and asymptotic safety (§13.1.4). Structure's analysis is the only cross-program per-condition comparison of its kind. Independent corroboration would require another author independently performing the analysis and arriving at compatible conclusions. The framework provides specific testable claims — e.g., A2 (determinism) is the most discriminating condition across candidate programs — that can be checked against each program's foundational documents.

Observer-admission distinction (§9.6). The structural distinction between observer-admitting and observer-lacking substrata is novel; the closest concept in the literature is anthropic selection (Vilenkin, Tegmark), which is statistical/probabilistic rather than structural. The distinction is not directly testable as stated; its empirical content lies in the predictions it underwrites within the SM-reproducing observer-admitting class, which are tested through the [SM §7] prediction set.

Acknowledgments

This is the fifth core paper of the OI framework, alongside [Main, SM, GR, Substratum]. It consolidates content originally developed across three separate documents: a canonical articulation of the framework's two-dimensional hierarchical structure (now §2), a technical analysis of the OI-string substratum-level relationship (now Part I), and a programmatic framework of universality classes of embedded observers (now Part II), with cross-framework synthesis in Part III. The proposal investigated in Part I was raised in discussion with [collaborator/AI-assisted analysis, depending on author preference]. Errors and speculative content are the author's responsibility.

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